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Data Strategy



FEATURING "Democratizing Transformation" *by Marco Iansiti and Satya Nadella*

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What's Your Data Strategy?

by Leandro DalleMule and Thomas H. Davenport

MORE THAN EVER, the ability to manage torrents of data is critical to a company's success. But even with the emergence of data-management functions and chief data officers (CDOs), most companies remain badly behind the curve. Cross-industry studies show that on average, less than half of an organization's structured data is actively used in making decisions—and less than 1% of its unstructured data is analyzed or used at all. More than 70% of employees have access to data they should not, and 80% of analysts' time is spent simply discovering and preparing data. Data breaches are common, rogue data sets propagate in silos, and companies' data technology often isn't up to the demands put on it.

Having a CDO and a data-management function is a start, but neither can be fully effective in the absence of a coherent strategy for organizing, governing, analyzing, and deploying an organization's information assets. Indeed, without such strategic management many companies struggle to protect and leverage their data—and CDOs' tenures are often difficult and short (just

2.4 years on average, according to Gartner). In this article we describe a new framework for building a robust data strategy that can be applied across industries and levels of data maturity. The framework draws on our implementation experience at the global insurer AIG (where DalleMule is the CDO) and our study of half a dozen other large companies where its elements have been applied. The strategy enables superior data management and analytics—essential capabilities that support managerial decision-making and ultimately enhance financial performance.

The "plumbing" aspects of data management may not be as sexy as the predictive models and colorful dashboards they produce, but they're vital to high performance. As such, they're not just the concern of the CIO and the CDO; ensuring smart data management is the responsibility of all C-suite executives, starting with the CEO.

Defense Versus Offense

Our framework addresses two key issues: It helps companies clarify the primary purpose of their data, and it guides them in strategic data management. Unlike other approaches we've seen, ours requires companies to make considered trade-offs between "defensive" and "offensive" uses of data and between control and flexibility in its use, as we describe below. Although information on enterprise data management is abundant, much of it is technical and focused on governance, best practices, tools, and the like. Few if any data-management frameworks are as

business-focused as ours: It not only promotes the efficient use of data and allocation of resources but also helps companies design their data-management activities to support their overall strategy.

Data defense and offense are differentiated by distinct business objectives and the activities designed to address them. Data defense is about minimizing downside risk. Activities include ensuring compliance with regulations (such as rules governing data privacy and the integrity of financial reports), using analytics to detect and limit fraud, and building systems to prevent theft. Defensive efforts also ensure the integrity of data flowing through a company's internal systems by identifying, standardizing, and governing authoritative data sources, such as fundamental customer and supplier information or sales data, in a "single source of truth." Data offense focuses on supporting business objectives such as increasing revenue, profitability, and customer satisfaction. It typically includes activities that generate customer insights (data analysis and modeling, for example) or integrate disparate customer and market data to support managerial decision-making through, for instance, interactive dashboards.

Idea in Brief

The Challenge

To remain competitive, companies must wisely manage quantities of data. But data theft is common, flawed or duplicate data sets exist within organizations, and IT is often behind the curve.

The Solution

Companies need a coherent strategy that strikes the proper balance between two types of data management: *defensive*, such as security and governance, and *offensive*, such as predictive analytics.

The Execution

Regardless of its industry, a company's data strategy is rarely static; typically, a chief data officer is in charge of ensuring that it dynamically adjusts as competitive pressures and overall corporate strategy shift.

Offensive activities tend to be most relevant for customer-focused business functions such as sales and marketing and are often more real-time than is defensive work, with its concentration on legal, financial, compliance, and IT concerns. (An exception would be data fraud protection, in which seconds count and real-time analytics smarts are critical.) Every company needs both offense and defense to succeed, but getting the balance right is tricky. In every organization we've talked with, the two compete fiercely for finite resources, funding, and people. As we shall see, putting equal emphasis on the two is optimal for some companies. But for many others it's wiser to favor one or the other.

Some company or environmental factors may influence the direction of data strategy: Strong regulation in an industry (financial services or health care, for example) would move the organization toward defense; strong competition for customers would shift it toward offense. The challenge for CDOs and the rest of the C-suite is to establish the appropriate trade-offs between

defense and offense and to ensure the best balance in support of the company's overall strategy.

Decisions about these trade-offs are rooted in the fundamental dichotomy between standardizing data and keeping it more flexible. The more uniform data is, the easier it becomes to execute defensive processes, such as complying with regulatory requirements and implementing data-access controls. The more flexible data is—that is, the more readily it can be transformed or interpreted to meet specific business needs—the more useful it is in offense. Balancing offense and defense, then, requires balancing data control and flexibility, as we will describe.

Single Source, Multiple Versions

Before we explore the framework, it's important to distinguish between information and data and to differentiate information architecture from data architecture. According to Peter Drucker, information is "data endowed with relevance and purpose." Raw data, such as customer retention rates, sales figures, and supply costs, is of limited value until it has been integrated with other data and transformed into information that can guide decision-making. Sales figures put into a historical or a market context suddenly have meaning—they may be climbing or falling relative to benchmarks or in response to a specific strategy.

A company's data architecture describes how data is collected, stored, transformed, distributed, and consumed. It includes the rules governing structured formats, such as databases and file

systems, and the systems for connecting data with the business processes that consume it. Information architecture governs the processes and rules that convert data into useful information. For example, data architecture might feed raw daily advertising and sales data into information architecture systems, such as marketing dashboards, where it is integrated and analyzed to reveal relationships between ad spend and sales by channel and region.

Many organizations have attempted to create highly centralized, control-oriented approaches to data and information architectures. Previously known as information engineering and now as master data management, these top-down approaches are often not well suited to supporting a broad data strategy. Although they are effective for standardizing enterprise data, they can inhibit flexibility, making it harder to customize data or transform it into information that can be applied strategically. In our experience, a more flexible and realistic approach to data and information architectures involves both a single source of truth (SSOT) and multiple versions of the truth (MVOTs). The SSOT works at the data level; MVOTs support the management of information.

The elements of data strategy

	Defense	Offense
Key objectives	Ensure data security, privacy, integrity, quality, regulatory compliance, and governance	Improve competitive position and profitability

Core activities	Optimize data extraction, standardization, storage, and access	Optimize data analytics, modeling, visualization, transformation, and enrichment
Data-management orientation	Control	Flexibility
Enabling architecture	SSOT (Single source of truth)	MVOTs (Multiple versions of the truth)

In the organizations we've studied, the concept of a single version of truth—for example, one inviolable primary source of revenue data—is fully grasped and accepted by IT and across the business. However, the idea that a single source can feed multiple versions of the truth (such as revenue figures that differ according to users' needs) is not well understood, commonly articulated, or, in general, properly executed.

The key innovation of our framework is this: It requires flexible data and information architectures that permit both single and multiple versions of the truth to support a defensive-offensive approach to data strategy.

A New Data Architecture Can Pay for Itself

WHEN COMPANIES LACK a robust SSOT-MVOTs data architecture, teams across the organization may create and store the data they need in siloed repositories that vary in depth, breadth, and formatting. Their data management is often done in isolation with inconsistent requirements. The process is inefficient and expensive and can result in the proliferation of multiple uncontrolled versions of the truth that aren't effectively reused. Because SSOTs and MVOTs concentrate,

> standardize, and streamline data-sourcing activities, they can dramatically cut operational costs.
>
> One large financial services company doing business in more than 200 countries consolidated nearly 130 authoritative data sources, with trillions of records, into an SSOT. This allowed the company to rationalize its key data systems; eliminate much supporting IT infrastructure, such as databases and servers; and cut operating expenses by automating previously manual data consolidation. The automation alone yielded a 190% return on investment with a two-year payback time. Many companies will find that they can fund their entire data management programs, including staff salaries and technology costs, from the savings realized by consolidating data sources and decommissioning legacy systems.
>
> The CDO and the data-management function should be fully responsible for building and operating the SSOT structure and using the savings it generates to fund the company's data program. Most important is to ensure at the outset that the SSOT addresses broad, high-priority business needs, such as applications that benefit customers or generate revenue, so that the project quickly yields results and savings—which encourages organization-wide buy-in.

OK. Let's parse that.

The SSOT is a logical, often virtual and cloud-based repository that contains one authoritative copy of all crucial data, such as customer, supplier, and product details. It must have robust data provenance and governance controls to ensure that the data can be relied on in defensive and offensive activities, and it must use a common language—not one that is specific to a particular business unit or function. Thus, for example, revenue is reported, customers are defined, and products are classified in a single, unchanging, agreed-upon way within the SSOT.

Not having an SSOT can lead to chaos. One large industrial company we studied had more than a dozen data sources containing similar supplier information, such as name and address. But the content was slightly different in each source. For example, one source identified a supplier as Acme; another called it Acme, Inc.; and a third labeled it ACME Corp. Meanwhile, various functions within the company were relying on differing data sources; often the functions weren't even aware that alternative sources existed. Human beings might be able to untangle such problems (though it would be labor-intensive), but traditional IT systems can't, so the company couldn't truly understand its relationship with the supplier. Fortunately, artificial intelligence tools that can sift through such data chaos to assemble an SSOT are becoming available. The industrial company ultimately tapped one and saved substantial IT costs by shutting down redundant systems. The SSOT allowed managers to identify suppliers that were selling to multiple business units within the company and to negotiate discounts. In the first year, having an SSOT yielded $75 million in benefits.

An SSOT is the source from which multiple versions of the truth are developed. MVOTs result from the business-specific transformation of data into information—data imbued with "relevance and purpose." Thus, as various groups within units or functions transform, label, and report data, they create distinct, controlled versions of the truth that, when queried, yield

consistent, customized responses according to the groups' predetermined requirements.

Consider how a supplier might classify its clients Bayer and Apple according to industry. At the SSOT level these companies belong, respectively, to chemicals/pharmaceuticals and consumer electronics, and all data about the supplier's relationship with them, such as commercial transactions and market information, would be mapped accordingly. In the absence of MVOTs, the same would be true for all organizational purposes. But such broad industry classifications may be of little use to sales, for example, where a more practical version of the truth would classify Apple as a mobile phone or a laptop company, depending on which division sales was interacting with. Similarly, Bayer might be more usefully classified as a drug or a pesticide company for the purposes of competitive analysis. In short, multiple versions of the truth, derived from a common SSOT, support superior decision-making.

At a global asset management company we studied, the marketing and finance departments both produced monthly reports on television ad spending—MVOTs derived from a common SSOT. Marketing, interested in analyzing advertising effectiveness, reported on spending after ads had aired. Finance, focusing on cash flow, captured spending when invoices were paid. The reports therefore contained different numbers, but each represented an accurate version of the truth.

Procter & Gamble has adopted a similar approach to data management. The company long had a centralized SSOT for all product and customer data, and other versions of data weren't allowed. But CDO Guy Peri and his team realized that the various business units had valid needs for customized interpretations of the data. The units are now permitted to create controlled data transformations for reporting that can be reliably mapped back to the SSOT. Thus the MVOTs diverge from the SSOT in consistent ways, and their provenance is clear.

In its application of the SSOT-MVOTs model, the Canadian Imperial Bank of Commerce (CIBC) automated processes to ensure that enterprise source data and data transformations remained aligned. CIBC's CDO, Jose Ribau, explains that the company's SSOT contains all basic client profile and preference data; MVOTs for loan origination and customer relationship management transform the source data into information that supports regulatory reporting and improved customer experience. Automated synchronization programs connect SSOT and MVOTs data, with nightly "exception handling" to identify and address data-integrity issues such as inconsistent customer profiles.

Although the SSOT-MVOTs model is conceptually straightforward, it requires robust data controls, standards, governance, and technology. Ideally, senior executives will actively participate on data governance boards and committees. But data governance isn't particularly fun. Typically, enterprise

CDOs and CTOs lead data and technology governance processes, and business and technology managers in functions and units are the primary participants. What's critical is that single sources of the truth remain unique and valid, and that multiple versions of the truth diverge from the original source only in carefully controlled ways. (For more on data governance and technology, see the sidebars "Good Governance, Good Data" and "A Lake of Data.")

Good Governance, Good Data

A SOUND DATA STRATEGY REQUIRES that the data contained in a company's single source of truth (SSOT) is of high quality, granular, and standardized, and that multiple versions of the truth (MVOTs) are carefully controlled and derived from the same SSOT. This necessitates good governance for both data and technology. In the absence of proper governance, some common problems arise:

- **Data definitions may be ambiguous and mutable.** With no concrete definition at the outset of what constitutes the "truth" (whether an SSOT or MVOTs), stakeholders will squander time and resources as they try to manage nonstandardized data.
- **Data rules are vague or inconsistently applied.** If rules for aggregating, integrating, and transforming data are unclear, misunderstood, or simply not followed—particularly when data transformation involves multiple poorly defined steps—it's difficult to reliably replicate transformations and leverage information across the organization.

- **Feedback loops for improving data transformation are absent.** Complex data analyses such as predictive modeling may be undertaken by one group but prove useful across an organization. Without mechanisms for making these outputs available to others (by, for example, integrating them into appropriate MVOTs), stakeholders may needlessly duplicate work or miss opportunities.

Strong data governance usually involves standing committees or review boards composed of business and technology executives, but it relies heavily on robust technology oversight. If technology rules prevent a marketing executive from buying a server on his or her corporate purchasing card, it's much less likely that marketing will, for instance, create unregulated "shadow" MVOTs or a marketing analytic that duplicates an existing one.

A Lake of Data

UNTIL A FEW YEARS AGO, technological limitations made it hard to build the SSOT-MVOTs data architecture needed to support a robust data strategy. Companies depended on traditional data warehouses that stored structured enterprise data in hierarchical files and folders, but these were not always suited to managing vast and growing volumes of data and new formats. To meet the need for a cheaper, more agile and scalable architecture, Silicon Valley engineers devised the "data lake," which can store virtually unlimited amounts of structured and unstructured data, from databases to spreadsheets to free text and image files. Data lakes are an ideal platform for SSOT-MVOTs architecture. A lake can house the SSOT, extracting, storing, and providing access to the organization's most granular data down to the level of individual transactions. And it can support the aggregation of SSOT data in nearly infinite ways in MVOTs that also reside in the lake. Data warehouses still have their uses: They store data for production applications (such as general ledger and order-management systems)

that require tight security and access controls, which few data lakes can do. Many companies have both data lakes and warehouses, but the trend is for more and more data to reside in a lake.

Striking a Balance

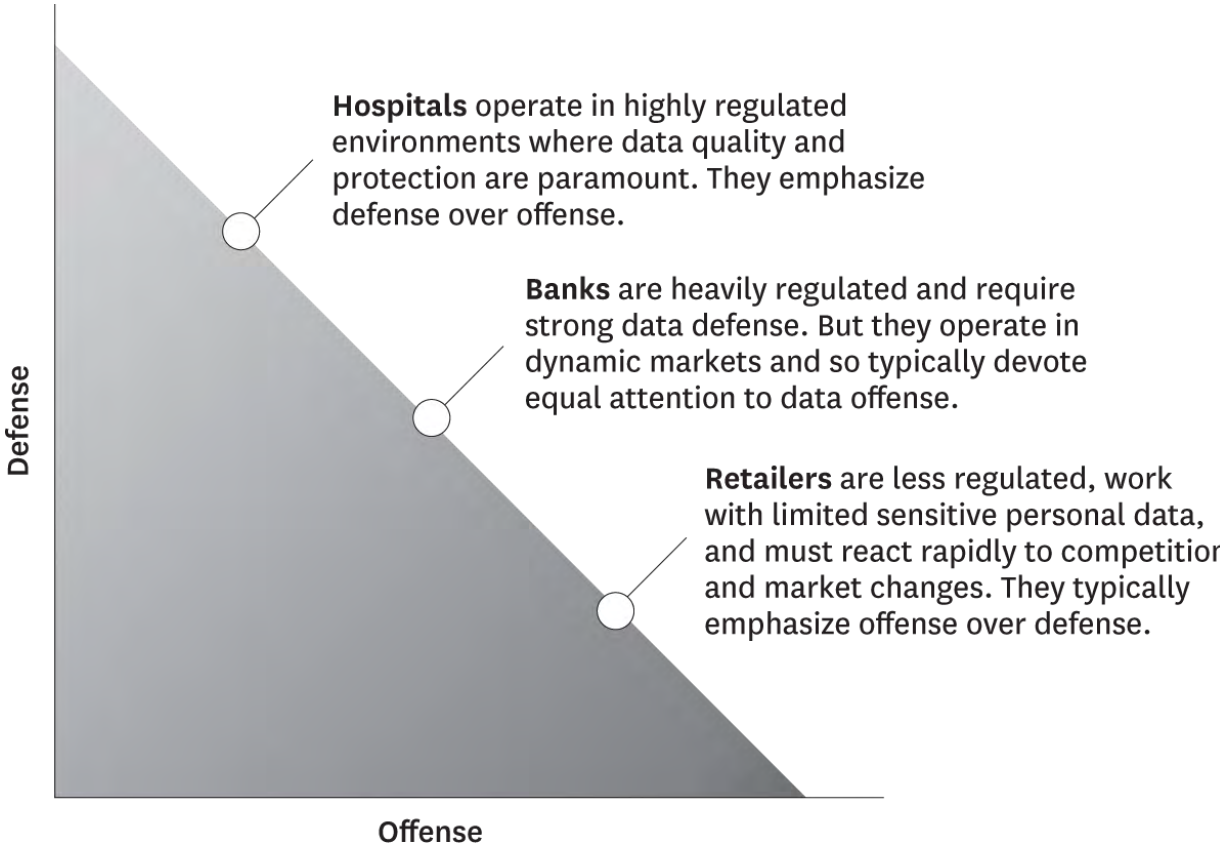
Let's return now to data strategy—striking the best balance between defense and offense and between control and flexibility. Whereas the CEO—often with the CIO—is ultimately responsible for a company's data strategy, the CDO commonly conceives it and leads its development and execution. The CDO must determine the right trade-offs while dynamically adjusting the balance by leveraging the SSOT and MVOTs architectures.

It's rare to find an organization—especially a large, complex one—in which data is both tightly controlled and flexibly used. With few exceptions, CDOs find that their best data strategy emphasizes either defense and control (which depends on a robust SSOT) or offense and flexibility (enabled by MVOTs). Devoting equal attention to offense and defense is sometimes optimal, but in general it's unwise to default to a 50/50 split rather than making considered, strategic trade-offs. To determine a company's current and desired positions on the offense-defense spectrum, the CDO must bear in mind, among other things, the company's overall strategy, its regulatory environment, the data capabilities of its competitors, the maturity of its data-management practices, and the size of its data budget. For example, insurance and financial services companies typically

operate in heavily regulated environments, which argues for an emphasis on data defense. (That is the case at AIG.) Retailers, operating in a less-regulated environment where intense competition requires robust customer analytics, might emphasize offense. (See the exhibit "The data-strategy spectrum.")

The data-strategy spectrum

A company's industry, competitive and regulatory environment, and overall strategy will inform its data strategy.



As Peri points out, defense and offense often require differing approaches from IT and the data-management organization.

Defense, he argues, is day-to-day and operational, and at P&G is largely overseen by permanent IT teams focused on master data management, information security, and so forth. Offense involves partnering with business leaders on tactical and strategic initiatives. Leaders may be reluctant to engage with master data management, but they are happy to collaborate on optimizing marketing and trade promotion spending.

Of course, plenty of cases don't fall neatly into either the offense or the defense category: The CDO of a large hedge fund told us that he was less concerned with data protection than with rapidly gathering and using new data. The most valuable data for his fund is primarily external, publicly or commercially available, captured in real time, and already of good quality, structured, and cleansed. Additionally, although his business is in financial services, it's not heavily regulated. Thus he focuses primarily on data offense. Wells Fargo's CDO, A. Charles Thomas, has enterprise responsibility for customer-related analytics, an offensive activity, and strives to keep the balance between offensive and defensive activities around 50/50, even structuring meeting agendas to focus equally on the two.

The tool "Assess your strategy position" offers diagnostic questions that can help CDOs place their companies on the offense-defense spectrum and gauge whether their data strategy aligns with their corporate strategy. Determining an organization's current and desired positions on the spectrum will force executives to make trade-offs between offensive and

defensive investments. Of course, this tool is not a precise measure. CDOs should use the results to inform data strategy and discussions with other C-level executives.

Assess your strategy position

Choose from among the following 16 objectives the eight that are most important to your business. Selecting that subset will require considered trade-offs that reveal your offense-defense orientation.

		Check the eight that qualify.	Total number of checked boxes
1	Reduce general operating expenses		
2	Meet industry regulatory requirements		
3	Prevent cyberattacks and data breaches		
4	Mitigate operational risks such as poor access controls and data losses		
5	Improve IT infrastructure and reduce data-related costs		
6	Streamline back-office systems and processes		
7	Improve data quality (completeness, accuracy, timeliness)		
8	Rationalize multiple sources of data and information (consolidate and eliminate redundancy)		
9	Improve revenue through cross-selling, strategic pricing, and customer acquisition		
10	Create new products and services		
11	Respond rapidly to competitors and market changes		
12	Use sophisticated customer analytics to drive business results		
13	Leverage new sources of internal and external data		
14	Monetize company data (sell as a product or a service)		
15	Optimize existing strong bench of analysts and data scientists		
16	Generate return on investments in big data and analytics infrastructure		

Data defense
Strong defense is characterized by single source of truth (SSOT) architecture, robust data governance and controls, and a more centralized data-management organization.

Data offense
Strong offense is characterized by multiple versions of the truth (MVOTs) architecture, high data flexibility, and a more decentralized data-management organization.

We find that companies with the most-advanced data strategies started at one point and gradually migrated to a new, stable position. For example, they may have shifted their focus from defense and data control toward offense as their data defense matured or competition heated up. The opposite path—from

offense toward defense, and from flexible toward controlled—is possible but usually more difficult.

Consider how data strategy has shifted at CIBC. The bank established the chief data officer role a few years ago and for the first 18 months maintained a 90% defensive orientation, focusing on governance, data standardization, and building new data-storage capabilities. When Jose Ribau took over as CDO, in 2015, he determined that CIBC's defense was sufficiently solid that he could shift toward offense, including more-advanced data modeling and data science work. Today CIBC's data strategy strikes a 50/50 balance. Ribau expects that the new attention to offense will drive increased ROI from data products and services and nurture analytical talent for the future.

Regardless of what industry a company is in, its position on the offense-defense spectrum is rarely static. As competitive pressure mounts, an insurer may decide to increase its focus on offensive activities. A hedge fund may find itself in a tougher regulatory environment that requires rebalancing its data strategy toward defense. How a company's data strategy changes in direction and velocity will be a function of its overall strategy, culture, competition, and market.

Organizing Data Management

As with most organizational design, data-management functions can be built centrally or decentralized by function or business unit. The optimal design will depend on a company's position on

the offense-defense spectrum. A centralized data function typically has a single CDO with accountability across the entire organization, ensuring that data policies, governance, and standards are applied consistently. This design is most suitable for businesses that focus on data defense.

Conversely, several companies we studied found that data offense can be better executed through decentralized data management, typically with a CDO for each business unit and most corporate functions. “Unit CDOs” tend to report directly to their business but have a matrix reporting relationship to the enterprise CDO. That helps prevent the development of data silos (which can lead to redundant systems and duplicate work) and ensures that best practices are shared and standards are followed. Generally speaking, unit CDOs own their respective versions of the truth, while the enterprise CDO owns the SSOT. A decentralized approach is well suited to offensive strategies because it can increase the agility and customization of data reporting and analytics. In many companies, among them Wells Fargo, CIBC, and P&G, the CDO is responsible for both analytics and data management, facilitating the ability to balance offense and defense.

Finally, in choosing between a centralized and a decentralized data function, it’s important to consider how funding will be determined, allocated, and spent. The budget may appear larger for a centralized function than for a decentralized one simply because it’s concentrated under one CDO. Decentralized budgets

are typically more focused on offensive investments, are closer to the business users, and have more-tangible ROIs, whereas centralized budgets are more often focused on minimizing risk, reducing costs, and providing better data controls and regulatory oversight—activities that are less close to business users and usually have a less-tangible ROI. Thus creating a business case to justify the latter is usually trickier. The importance of investing in data governance and control—even if the payoff is abstract—is more easily understood and accepted if a company has suffered from a major regulatory challenge, a data breach, or some other serious defense-related issue. Absent a traumatic event, enterprise CDOs should spend time educating senior executives and their teams about data-defense principles and how they create value.

Emerging technologies may enable a next generation of data-management capabilities, potentially simplifying the implementation of defensive and offensive strategies. Machine learning, for example, is already facilitating the creation of a single source of truth in many companies we studied. The promise is more-dynamic, less-costly SSOTs and MVOTs. However, no new technology will obviate an effective, well-run data-management function. Our framework will become even more relevant as distributed technology solutions—blockchain, for example—come into play.

Data was once critical to only a few back-office processes, such as payroll and accounting. Today it is central to any business, and the importance of managing it strategically is only growing. In September 2016, according to the technology conglomerate Cisco, global annual internet traffic surpassed one zettabyte (1,021 bytes) —the equivalent, by one calculation, of 150 million years of high-definition video. It took 40 years to get to this point, but in the next four, data traffic will double. There is no avoiding the implications: Companies that have not yet built a data strategy and a strong data-management function need to catch up very fast or start planning for their exit.

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Democratizing Transformation

by Marco Iansiti and Satya Nadella

OVER THE PAST DECADE, Novartis has invested heavily in digital transformation. As the Swiss pharmaceutical giant moved its technology infrastructure to the cloud and invested in data platforms and data integration, it recruited AI specialists and data scientists to build machine-learning models and deploy them throughout the firm. But even as the technical teams grew, managers from across the business—sales, supply chain, HR, finance, and marketing—weren't embracing the newly available information, nor were they thinking much about how data could enhance their teams' work. At the same time, the data scientists had little visibility into the business units and could not easily integrate data into day-to-day operations. As a result, the investments resulted in only occasional successes (in some aspects of the R&D process, for example) while many pilots and projects sputtered.

More recently, however, pilots targeting both R&D and marketing personalization started showing business value and captured the attention and imagination of some of Novartis's

more creative business executives. They became increasingly excited about opportunities to deploy AI in various parts of the company and began to earnestly champion the efforts. (Disclosure: We have both worked with Novartis and other companies mentioned in this article in a variety of ways, including board membership, research, and consulting.) They realized that technologists and data scientists alone couldn't bring about the kind of wholesale innovation the business needed, so they began pairing data scientists with business employees who had insight into where improvements in efficiency and performance were needed.

Novartis also invested in training frontline business employees to use data themselves to drive innovation. A growing number of teams adopted agile methods to address all kinds of opportunities. The intensity and impact of transformation thus accelerated rapidly, driving a range of innovation initiatives, including digitally enabling sales and sales forecasting, reconceiving the order and replenishment system for health-care-services customers, and revamping prescription-fulfillment systems and processes.

The progress in digital transformation became invaluable as the company dealt with the initial chaos of the pandemic. Novartis business teams partnered with data scientists to devise models to manage supply-chain disruptions, predict shortages of critical supplies, and enable quick changes to product mix and pricing policies. They also developed analytics to identify patients who

were at risk because they were putting off doctor visits. As the Covid crisis wore on, the value of AI became obvious to managers companywide.

Before this wave of AI adoption, Novartis's investments in technology consisted almost entirely of packaged enterprise applications, usually implemented by the IT department with the guidance of external consultants, vendors, or systems integrators. But to build companywide digital capability, under the leadership of then chief digital officer Bertrand Bodson, Novartis not only developed new capabilities in data science but also started to democratize access to data and technology well outside traditional tech silos. The company is now training employees at all levels and in all functions to identify and capitalize on opportunities for incorporating data and technology to improve their work. In 2021, the Novartis yearly AI summit was attended by thousands of employees.

The potential for employee-driven digital innovation is impossible to calculate, but according to the market research firm IDC's Worldwide IT Industry 2020 Predictions report, enterprises across the global economy will need to create some 500 million new digital solutions by 2023—more than the total number created over the past 40 years. This cannot be accomplished by small groups of technologists and data scientists walled off in organizational silos. It will require much larger and more-diverse groups of employees—executives, managers, and frontline workers—coming together to rethink how every aspect of the

business should operate. Our research sheds light on how to do that.

Idea in Brief

The Problem

Many companies struggle to reap the benefits of investments in digital transformation, while others see enormous gains. What do successful companies do differently?

The Journey

This article describes the five stages of digital transformation, from the traditional stage, where digital and technology are the province of the IT department, through to the platform stage, where a comprehensive software foundation enables the rapid deployment of AI-based applications.

The Ideal

The ideal is the native stage, whose hallmarks are an operating architecture designed to deploy AI at scale across a huge, distributed spectrum of applications; a core of experts; broadly accessible, easy-to-use tools; and investment in training and capability-building among large groups of businesspeople.

The Success Drivers

When we started our research, we wanted to understand why many companies struggle to reap the benefits of investments in digital transformation while others see enormous gains. What do successful companies do differently?

We looked at 150 companies in manufacturing, health care, consumer products, financial services, aerospace, and pharma/biotech, including a representative sample of the largest firms in each sector. Some were failing to move the needle, but many had made dramatic progress. Perhaps surprisingly, we found that outcomes did not depend on the relative size of IT budgets. Nor were the success stories confined to "born digital" organizations. Legacy giants such as Unilever, Fidelity, and Starbucks (where one of us, Satya, is on the board)—not to mention Novartis—had managed to create a digital innovation mindset and culture.

Our research shows that to enable transformation at scale, companies must create synergy in three areas:

Capabilities

Successful transformation efforts require that companies develop digital and data skills in employees outside traditional technology functions. These capabilities alone, however, are not sufficient to deliver the full benefits of transformation; organizations must also invest in developing process agility and, more broadly, a culture that encourages widespread, frequent experimentation.

Technology

Of course, investment in the right technologies is important, especially in the elements of an AI stack: data platform technology, data engineering, machine-learning algorithms, and algorithm-deployment technology. Companies must ensure that

the technology deployed is easy to use and accessible to the many nontechnical employees participating in innovation efforts.

Architecture

Investment in organizational and technical architecture is necessary to ensure that human capabilities and technology can work in synergy to drive innovation. That requires an architecture—for both technology and the organization—that supports the sharing, integration, and normalization of data (for example, making data definitions and characteristics consistent) across traditionally isolated silos. This is the only real, scalable way to assemble the necessary technological and data assets so that they are available to a distributed workforce.

Many large companies are making headway in each of these areas. But even leading companies tend to underestimate the importance of getting employees to pull transformation into their functions and their work rather than having central technology groups and consultants push the changes out to the business. As Eric von Hippel of MIT has advocated for many years, frontline users, who are closest to the use cases and best positioned to develop solutions that fit their needs, must take a central role, joining agile teams that dynamically coalesce and dissolve on the basis of business needs.

The Elements of Tech Intensity

TO ENABLE TRANSFORMATION, companies must create synergy in three key areas:

Capabilities

- Organizational culture
- Training and development
- Low-code/no-code tools
- Agile teams
- Organizational architecture
- Citizen developers
- Product management

Technology

- Machine learning
- Deep learning
- DevOps pipelines
- Data encryption
- Real-time analytics

Architecture

- Data platform
- Horizontal integration and normalization
- Data documentation
- API strategy
- Experimentation and risk

- Data governance

Building Tech Intensity

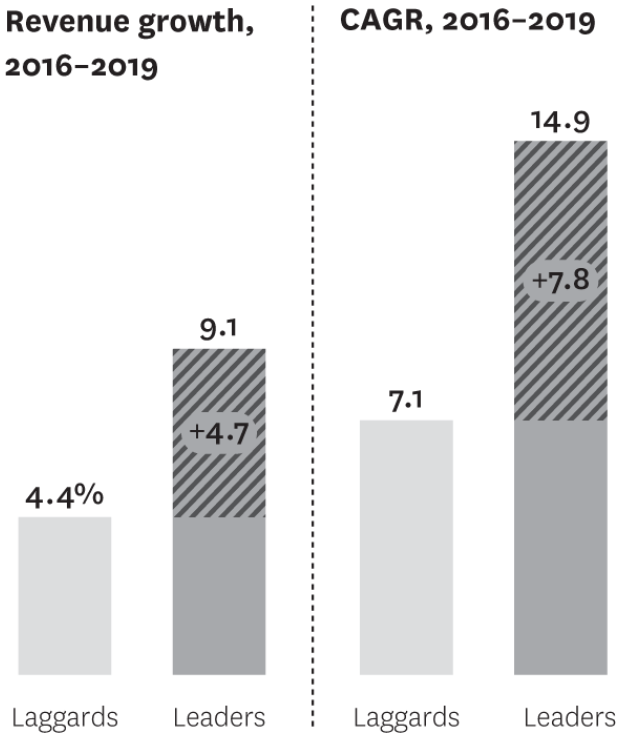
Our research unpacks how capabilities, technology, and architecture work together to build what we call *tech intensity*. Derived from the economics concept of intensive margin—how much a resource is utilized or applied—tech intensity refers to the extent to which employees put technology to use to drive digital innovation and achieve business outcomes. Our research found that companies that made good investments in technology and made tools accessible to a broad community of data- and tech-skilled employees achieved higher tech intensity—and superior performance. Companies that failed to develop tech- and data-related capabilities in their employees and offered only limited access to technology were left behind.

We ranked the tech intensity of the 150 firms in our study and found that the top quartile of the sample grew their revenues more than twice as fast as the bottom quartile. (See the exhibit "Digital transformation pays off.") We also found that technology, capability, and architecture indices correlated with other measures of performance, from productivity and profits to growth in enterprise value. Using an econometric technique known as *instrumental variables*, we also found evidence that the relationship between tech intensity and performance was causal:

That is, greater intensity (especially investments in technical and organizational architecture) powered higher revenue growth.

Digital transformation pays off

We studied 150 companies in a range of industries and found that revenue growth and compound annual growth rate among the leaders (the top quartile) in tech intensity were more than double that of the laggards (the bottom quartile).



Source: Keystone

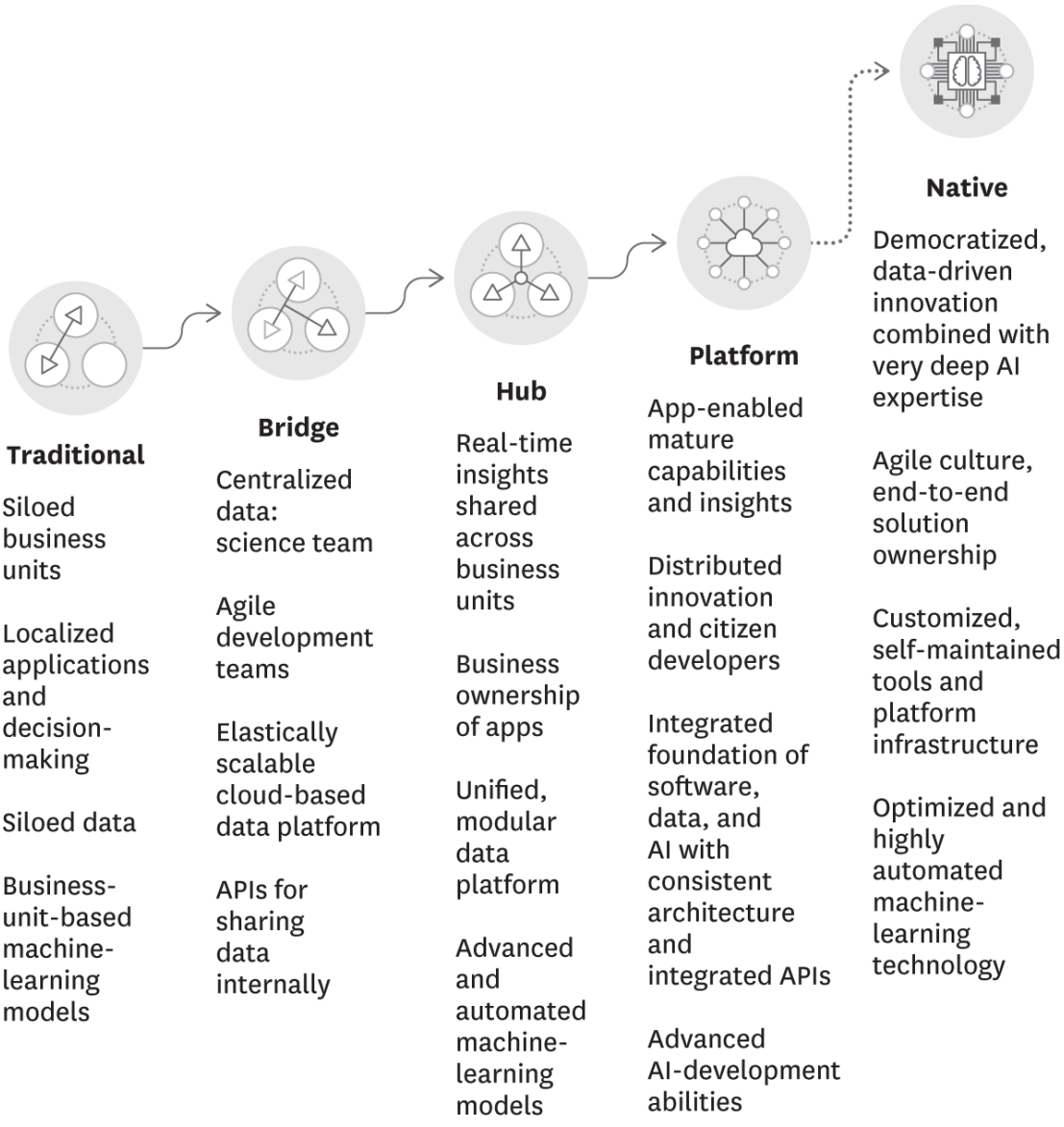
Staging the Transformation

Our analysis confirms that just spending money on technology does not result in more growth or better performance; in fact, in some cases it can actually damage the business if it accentuates

divisions and inconsistencies across groups. Instead, it is the architectural, managerial, and organizational approaches to transformation that best explain the substantial and enduring differences among firms. We found that companies typically progress through five stages on their transformation journey. (See the exhibit "The stages of digital maturity.")

The stages of digital maturity

Digital maturity is made up of these characteristics of organizational structure, process, tech architecture, and tech deployment. How does your company stack up?



Traditional model

Not surprisingly, many companies fit what we consider to be the traditional model of digital innovation, whereby digital and technology investments are the province of the IT department (or other technical specialist groups) and impact is scattered across groups, mostly in inconsistent ways. IT works with business units to fund projects and manage implementation—say, for the

deployment of an enterprise application or a data platform technology. The projects and their implementations are customized to the specific requirements of the individual silos, business units, or functions. The result is that over time, the technology and data infrastructure reflect the quirks of individual groups, without any consistency and connectivity. This sort of disjointed approach makes it virtually impossible to share, scale, or distribute innovation efforts across the organization.

Many businesses in the traditional model still spend a great deal of money on information technology. Consider a financial services firm we studied, whose tech and analytics budget is among the top in its industry, in both absolute and relative terms. The company has spent heavily on state-of-the-art data-platform technology and hired thousands of IT specialists and data scientists, who sit isolated in a separate IT group, while few (if any) employees on the business side are involved in the organization's digital innovation efforts. The company thus lacks the architecture and capabilities required to foster any intensity in tech adoption. Not surprisingly, the firm's IT and data sciences efforts have stalled, and business impact has been minimal.

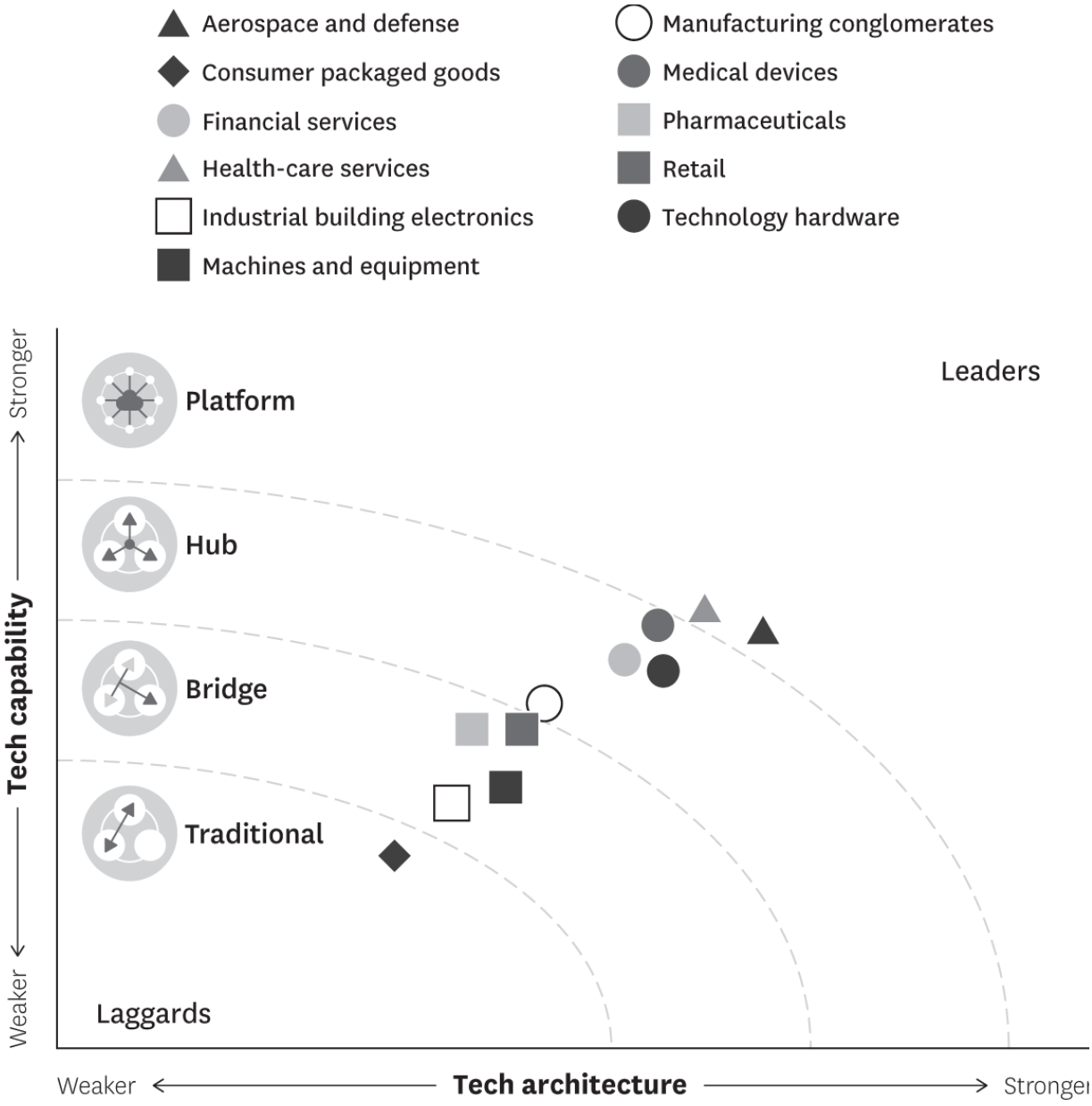
A telltale sign that a company is in the traditional stage is that perceptions of impact among technology and business employees are dramatically different. The former perceive impact to be high (as measured by the effort they put into their work), while the latter measure it as much lower (according to how their everyday activities have benefited).

Bridge model

To break free of the traditional constraints of silos—organizational and infrastructural—companies typically start by launching pilots that bridge previously separate groups and developing shareable data and technology assets to enable new innovations. They might first focus on specific functional opportunities such as optimizing advertising, manufacturing, or supply-chain capabilities. These companies are piloting not only technology but also a fundamentally different model of innovation in which executives, managers, and frontline workers from the business side work in collaboration with IT and data scientists. Victor Bulto, Novartis's head of U.S. pharmaceuticals, was instrumental in launching early pilots (focusing, for example, on identifying at-risk patients) and served as a champion for many initiatives as the organization moved through the bridge stage. Lori Beer, JPMorgan Chase's global CIO, likes to talk about the demonstrated impact of piloting AI to simplify expense reporting and approval—a process-improvement pilot that won over many employees.

Digital maturity by industry

We looked at 150 companies in a range of industries and plotted the average levels of technology capability and technology architecture for each industry. Companies in consumer packaged goods, for example, tended to be at the early stage of the transformation journey; aerospace and health-care firms were much more advanced.



Source: Keystone

Hubs

As more and more pilots demonstrate the success of the new approach, organizations form data and capability hubs and gradually develop the capacity to link and engage additional functions and business units in pursuit of opportunities for

transformation. As they progress down this path, leaders begin to realize that the bottleneck in innovation has shifted from investments in technology to investments in the workforce. The limiting factor at this stage is the number of business employees with the capability—the know-how and the access—to drive digital innovation. Companies thus need to invest in coaching and training a much larger community of employees.

Fidelity strives to develop what it calls *digital athletes*. It began to build hubs by creating centralized data assets (a companywide data lake, for example); now it is scaling up training for thousands of business employees, giving them the capacity to deploy digitally enabled solutions across the entire business. Digitally savvy investment specialists and tax experts, for example, are working closely with data scientists and technologists to create innovative solutions with a special focus on personalization and tailored customer impact. They've also created an app aimed at onboarding and engaging younger investors and another app for delivering AI-powered recommendations to Fidelity financial advisers, to name just a few examples.

Starbucks, too, is focused not only on technology and architecture but also on developing broad-based, agile innovation skills in its employees to power its hubs. CEO Kevin Johnson explains, "We've gone from large teams working in silos to smaller, cross-functional teams [everywhere], and from evaluating every idea as pass-fail to rapid iteration." Starbucks is now a digital innovation powerhouse, with sophisticated

customer apps enabling remote ordering, loyalty programs, and payment systems along with internal systems enabling AI-based labor allocation and inventory management.

Platform model

As companies enter the platform stage, data hubs merge into a comprehensive software foundation that enables the rapid deployment of AI-based applications. Firms focus on building sophisticated data-engineering capabilities and encouraging the reuse and integration of machine-learning models. Analytics-based prediction models are applied across the business, with an increasing focus on the automation of basic operational tasks. Organizations begin to function a bit more like software companies, developing comprehensive capabilities that enable product and program management and rapid experimentation.

Over the past five years, Microsoft has gone through almost every stage of this journey. Years ago, we were just as siloed as most companies, with each product-based organization segregating its own data, software, and capabilities. As we connected and normalized data from different functions and product groups, we were able to deploy integrated solutions in areas ranging from customer service to supply-chain management.

We integrated all our data in a companywide data lake, and we built what we call a *business process platform*, which provides software and analytics components that teams use to enable innovation in areas ranging from Xbox manufacturing to

managing advertising spend. We also invested in training programs for nontechnical employees, cultivating data-centric and machine-learning capabilities throughout the organization.

Native model

The most successful companies among the 150 in our study have deployed an entirely different type of operating architecture, centered on integrated data assets and software libraries and designed to deploy AI at scale across a huge, distributed spectrum of applications. Its hallmarks are a core of experts; broadly accessible, easy-to-use tools; and investment in training and capability-building among large groups of businesspeople. These companies are approaching the capacity of digital natives such as Airbnb and Uber, which were purpose-built to scale companywide analytics and software-based innovation. Airbnb and Uber are certainly not perfect, but they come close to the native ideal.

At Microsoft, we still have a lot to learn, but in some parts of the organization we are starting to approach the native model. As is common in any enterprise, the progress has not been uniform. Different groups have achieved different levels of capability, but the results overall are encouraging, as we see increasingly innovative solutions to internal and customer-facing problems. Most critically, our companywide approach to understanding, protecting, and working with data has progressed by light years.

The Imperative for Leaders

The mandate for digital transformation creates a leadership imperative: Embrace transformation, and work to sustain it. Articulate a clear strategy and communicate it relentlessly. Establish an organizational architecture to evolve into as you make the myriad daily decisions that define your technology strategy. Deploy a real governance process to track the many technology projects underway, and coordinate and integrate them whenever possible. Champion agility in all business initiatives you touch and influence. And finally, break free of tradition. Train and coach your employees to understand the potential of technology and data, and release the innovators within your workforce.

This mandate extends to technology providers. Despite much investment, technologies are still too complex and are often too hard to use and deploy. We need tools and technology that make driving transformation intuitive for frontline workers while keeping data secure. Let’s not forget that until recently many of us were relying on specialists in Fortran and Cobol to model business problems and even to perform basic mathematical operations. Spreadsheets brought about a revolution in mathematical modeling; we need technology providers to bring the same revolution to AI and make using a machine-learning application as easy as creating a pivot table.

Momentum is growing. But we must sustain the efforts to ensure that companies of all stripes make it across the digital

divide.

Originally published in May–June 2022. Reprint S22031

Why Companies Should Consolidate Tech Roles in the C-Suite

by Thomas H. Davenport, John Spens, and Saurabh Gupta

C-LEVEL TECHNOLOGY ROLES have proliferated over the past few decades. Chief information officers first appeared in the early 1980s. Next, chief technology officers became common in high-tech firms and then spread to other types of businesses. These roles were followed by chief information security officers, chief data officers, chief analytics officers, and chief digital officers. Most recently, chief artificial intelligence officers have appeared. Their numbers have greatly increased with the rise of generative AI during the past few years.

While the creation of these roles reflects the increasing importance of technology in business, such a dizzying array of technology leaders can also lead to confusion.

A recent study—sponsored by Thoughtworks (where one of us works) and the MIT Chief Data Officer/Information Quality Symposium—suggests that this is a serious issue. In a survey of 266 technology and data leaders, more than four out of five reported confusion in their organizations around the roles of different technology-oriented groups and where to turn for data- and technology-oriented services and issues. Only 12% reported no confusion.

As more organizations seek to integrate data and AI into their operations, this confusion may be having a negative impact. People in leadership roles are often unsure about their own responsibilities relative to other tech leaders: 30% of the surveyed leaders said they lacked clarity about how their role relates to other technology and data leadership roles. And although 78% of respondents reported that data and technologies are viewed as important or very important by their organizations, only 10% of respondents said they were very satisfied with their data foundations and capabilities for analytics and AI. In comparison, 30% said they were dissatisfied.

There are, of course, many challenges preventing success with data goals: people readiness, politics, diffusion of responsibility (heavily related to overlapping organizational structures in some organizations), budgets or financial constraints, technology readiness, and time—or just limited patience. Addressing all of these challenges will require clear, concerted leadership and a broad mandate for change.

In this study, however, we discovered that some organizations are recognizing the downside of C-level proliferation. To tackle the issue, they've begun to consolidate senior technology and data roles; we refer to them as "SuperTech" leaders, although it's important to note there is variation across organizations in the specifics of their roles and titles.

These individuals are most likely to be known as chief information officers, but they typically have broad responsibility for data and technology functions. Many also have operational responsibilities within the business. Leaders of specialty tech and data functions (data, analytics, AI, cybersecurity) usually report to them. Establishing SuperTech leadership roles will bring clarity and accountability to organizations' digital transformation initiatives. We interviewed several such leaders for the study and describe some of their backgrounds and perspectives below.

Rise of the SuperTech Leader

What skills should a combined-tech leader have? Survey participants named overall leadership and senior executive relationships (85%) and business strategy and vision (82%) as most important. Technology strategy and vision (72%) was somewhat less important. Skills involving managing the employees of the combined-technology function (38%) and technology execution skills (35%) were viewed as substantially

less important. In other words, SuperTech leaders should be heavily business-oriented, with a technical underpinning.

Idea in Brief

Recently, some organizations have begun to recognize the downside of C-level proliferation and decided to consolidate senior technology and data roles. These individuals—referred to as "SuperTech" leaders—are most likely to be known as chief information officers, but they also have broad responsibility for data and technology functions, and many also have operational responsibilities within the business. Leaders of specialty tech and data functions (data, analytics, AI, cybersecurity) report to them. Establishing SuperTech leadership roles will bring clarity and accountability to organizations' digital transformation initiatives.

The executives who hold multiple technology roles that we interviewed for this report reinforced this view. They talk and think as businesspeople first and tech people second in their day-to-day experience of these roles.

For example, Shamin Mohammad, who is both chief information and chief technology officer at CarMax, told us: "To be effective as a technology leader, I have to be a business leader. I am working side-by-side with the CMO, the COO—we are joined at the hip—and then the CFO, the CEO, and the board. Execution is table stakes. You have to deliver—it's not even up for discussion. It is more important for a technology leader to be a business leader, to understand what's happening in the industry, and to envision where the industry is going so you can enable that with technology and bring the rest of the

organization on board." Topics like "keeping the lights on," renewing technical infrastructure, or dealing with cybersecurity threats are important, but they are not his primary focus; his direct reports ensure they are taken care of.

Sean McCormack, a SuperTech leader at school busing company First Student, noted, "If you are not a value- and innovation-oriented IT executive, it won't work to oversee technology innovation, analytics and AI, and that sort of activity. If all you've done is package implementations, you will not succeed in a broader role." Sebastian Klapdor, until recently the head of technology, data, and analytics at marketing services company Vista, said that a strong business-value orientation is key to being asked to lead multiple technology- and data-related functions. When he was head of data and analytics for the company, he focused heavily on achieving measurable value through successful data products. He mentioned in an interview that he believes this was a key factor in his being asked to also assume the CTO role.

In the survey, respondents minimized "managing down" skills and technical execution skills, but the SuperTech leaders interviewed suggested they shouldn't be discounted. Several mentioned the importance of hiring strong specialists in each area reporting to them. Sastry Durvasula, whose title at TIAA is chief information and client services officer, leads all technology functions at TIAA and has direct reports dealing with data and AI, client services, and shared services, as well as chief

information and technology officers for TIAA's different business lines. In each area, he seeks out people who have the relevant technology skills but who also understand how to transform an organization. He reports to the company's CEO and said he still codes on occasion as well. Klapdor, who worked at Vista, said that he doesn't think of himself as highly technical, but he did create a small Arduino-based device for capturing sensor data from printing equipment. Understanding the technology helps him lead those who do it for a living. Both of these leaders have at least some degree of technology execution skills.

The broad survey of data and tech leaders suggests that many different groups within the organization are involved in overseeing or coordinating the introduction and use of generative AI. Chief data officers, chief information officers, chief technology officers, chief digital officers, and chief AI officers (in order of their perceived involvement in generative AI strategy and execution) all play a role. While this probably indicates that the potential value of the technology is viewed as high, it may be inefficient or confusing for so many groups to be involved and to have contention around the leadership roles for generative AI. This is another argument for consolidation of tech and data leadership.

Dissenters from This Model

Not all technology and data executives believe that an integrated organization and SuperTech leader is a good idea. In the interviews, we heard some objections that were based on individual attributes of leaders and some that were more principle-based.

Several data leaders told us that they would not be interested in reporting to a chief information officer who is primarily focused on infrastructure and "keeping the lights on," as opposed to bringing about substantial business change. Other leaders had different concerns with an integrated role. Scott Hallworth, the chief data and analytics officer at Hewlett-Packard, had been interim co-CIO at HP for much of 2023. His concerns about a fully integrated role—particularly the combination of chief digital and chief information officer roles—is that combining the roles would eliminate a check on important decisions within a company.

Of course, the strengths and weaknesses of organizational structures are specific to individual companies and the people who hold roles within them. It's possible that a particular structure and leader for technology and data would work better or worse depending on the organizational circumstances and the individuals involved.

Actions for the Future

The most obvious action that organizations can take to address these issues is to create an integrated technology and data leadership role with an effective, business-oriented executive occupying it. Judging from our interviews, many organizations have already taken that step and appear to have been successful with it.

If for some reason that isn't possible, some of the issues raised in this study can be addressed in other ways. One, for example, is to create greater role clarity for C-level tech and data leaders and make the role distinctions known to internal customers. Another is to foster more collaboration among tech and data leaders, which can lead to joint projects and higher levels of success in delivering them.

There are a variety of technologies and applications in use within organizations today, both familiar and emerging. There are times when specialists in particular technologies or management issues with them are needed. However, the trend toward specialized technology and data leaders has, we feel, gone too far. Greater coordination and integration are needed, and SuperTech leaders with a strong connection to the business can help make information technology a more effective business resource.

Adapted from hbr.org, September 13, 2024. Reprint H08DSC

Developing a Digital Mindset

by Tsedal Neeley and Paul Leonardi

WHEN THIERRY BRETON took over as CEO of the French IT services company Atos, in 2008, he knew that a massive and immediate digital transformation was necessary. Annual revenue had increased nearly 6% during the Great Recession, to $6.2 billion, but Atos wasn't growing as fast as its competitors were. The company suffered from siloed business and functional groups, had limited pooling of global resources, and needed more innovation throughout the company. Digital transformation was the only way forward.

But what would that look like for an IT giant? Breton began by scaling and globalizing the company, which provides online transactional services, systems integration, cybersecurity, and more. He doubled the size of the workforce to 100,000 people, hoping to fend off the competitors all around him, including digital-born start-ups from Silicon Valley, India, and China. Breton also laid out a plan to integrate AI and other data-driven technology into company processes and upskill the expanding workforce.

The three-year digital-transformation plan depended on creating a culture of continuous learning and required that employees develop what we call a *digital mindset*. Breton and his team debated options for how to approach those goals. Some believed a robust training program was the only way forward; others were convinced that people learn best on the job. They eventually created the Digital Transformation Factory upskilling certification program. The initial goal was to train 35,000 technical and nontechnical employees in digital technologies and artificial intelligence.

Notably, the upskilling program was voluntary. Breton's team launched an internal marketing campaign to encourage people to learn and get certified. It also instituted a peer and management nomination system to entice employees to join the program and offered rewards for achieving benchmarks in certification. The reasoning was that if employees got certified voluntarily, they would be more likely to internalize the new digital skills and modify their work behaviors accordingly. The learning programs accommodated everyone from data scientists and highly skilled engineers to people in traditionally nontechnical functions, such as sales and marketing.

The results exceeded expectations. Within three years, more than 70,000 people completed their digital certification, in large part because employees understood that growth at the company required digital fluency. Atos was clearly on the right track. Its revenue had reached close to $13 billion by the time Breton left

the company, in 2019, to become France's European Commissioner.

What Is a Digital Mindset?

Learning new technological skills is essential for digital transformation. But it is not enough. Employees must be motivated to use their skills to create new opportunities. They need a digital mindset. Psychologists describe mindset as a way of thinking and orienting to the world that shapes how we perceive, feel, and act. A digital mindset is a set of attitudes and behaviors that enable people and organizations to see how data, algorithms, and AI open up new possibilities and to chart a path for success in a business landscape increasingly dominated by data-intensive and intelligent technologies.

Developing a digital mindset takes work, but it's worth the effort. Our experience shows that employees who do so are more successful in their jobs and have higher satisfaction at work, they are more likely to get promoted, and they develop useful skills that are portable should they decide to change jobs. Leaders who have a digital mindset are better able to set their organizations up for success and to build a resilient workforce. And companies that have one react faster to shifts in the market and are well positioned to take advantage of new business opportunities.

Idea in Brief

The Problem

Learning technological skills is essential for digital transformation, but it is not enough. Employees must be motivated to use their new skills to create new opportunities.

The Solution

They need a digital mindset: a set of attitudes and behaviors that enable them to see how data, algorithms, and AI open up new possibilities and allow them to chart a path for success in an increasingly technology-intensive world.

Employees who do so are more successful in their jobs and more satisfied at work, and leaders who do so are better able to set their organizations up for success.

Maintaining Momentum

Digital transformation often encounters resistance, and missteps are inevitable. Companies do better when they focus on two areas: preparing people for a new digital organizational culture, and designing and aligning systems and processes.

Like any other change initiative, digital transformation often encounters resistance, and early missteps are inevitable. In our experience, companies do best when they focus on two critical areas: (1) preparing people for a new digital organizational culture and (2) designing and aligning systems and processes. In this article, we lay out the basic principles of this enormous undertaking, drawing lessons from Philips, Moderna, and Unilever. These companies offer a road map for developing digital mindsets in existing talent pools and aligning systems and processes to capitalize on digital proficiency.

Building a Continuous-Learning Culture

The health services company Philips recently transitioned its core competency from supplying health products to providing digital solutions. To bring employees along, it needed to create a continuous-learning environment. Philips partnered with Cornerstone OnDemand, a cloud-based learning and HR software provider, to build an AI-powered infrastructure that adapts to learners' specific needs and pace. Employees can share "playlists" of tailored lessons with colleagues, just as they share playlists on music-streaming services. The platform's social media function facilitates connection between new employees and more-experienced members who can serve as mentors, fostering more-organic peer-mentor relationships than formal matching programs do.

Philips's leaders, who serve as the continuous-learning program's teachers, have emphasized the need for not only new knowledge but a cultural shift. They assume responsibility for their team members' futures, not just for managing work tasks, and they share their expertise, knowledge, and passion during training sessions. The company collects data on how employees are using the platform, measures the correlation between continuous learning and performance, and examines how various tools help employees learn, in expected or unexpected ways.

The ability to develop a digital mindset depends on the extent to which employees internalize the undertaking. Thinking about how they will interact with and use new tools and how those

tools will help them attain superior performance is essential to a successful digital transformation.

Accelerating Adoption

Digital change is often radical, and it involves shifting shared values, norms, attitudes, and behaviors. That's a tall order, so it is helpful to kick things off with a bold stroke: an act that commands attention and prompts everyone in the company to understand that a new direction is required. (See "What Inexperienced Leaders Get Wrong [Hint: Management]" on hbr.org.) Examples include doing a major reorg, making an acquisition, reallocating resources, hiring a digital transformation czar who reports to the CEO, and announcing that a legacy system is being phased out.

While signaling the new order creates momentum, it isn't enough. A bold stroke must be followed by a long march, one that begins with assessing how employees feel about the plans for digital transformation. Some may be apprehensive about the unknown; others may worry about their own capacity to learn and apply the new technology and skills to their jobs. These anxieties will affect technical and nontechnical roles. Employees may also be dubious about whether the digital transformation matters—to the company and to their jobs.

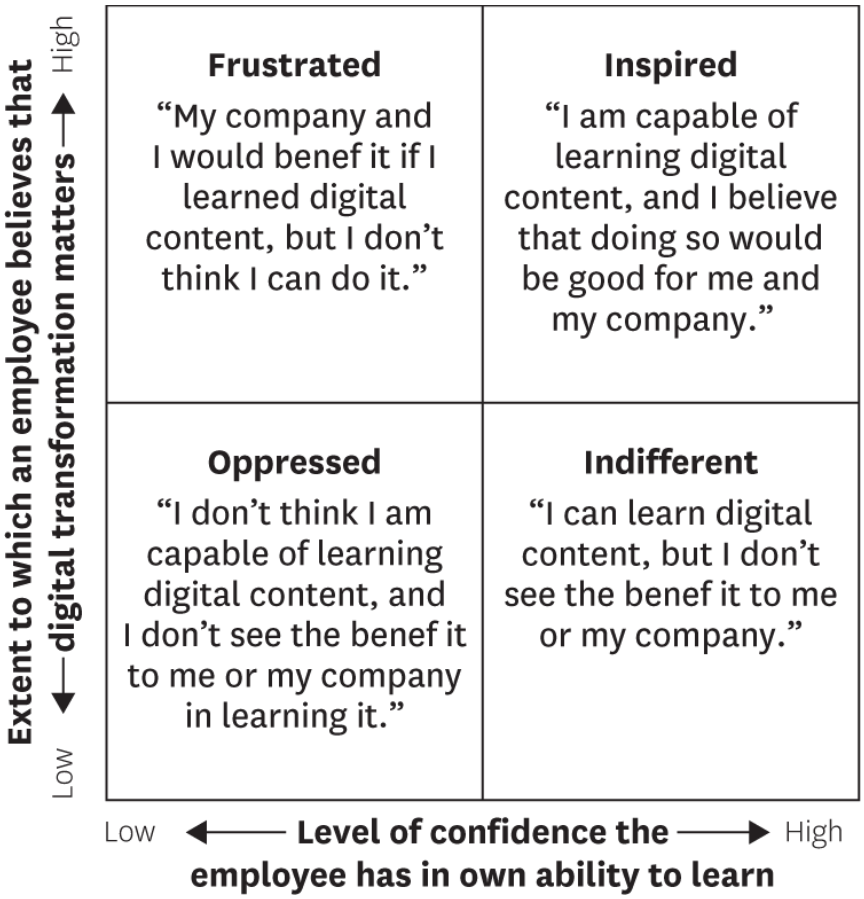
When implementing radical change, managers must carefully weigh these two key dimensions: buy-in (the degree to which people believe that the change will produce benefits for them and

the organization) and capacity to learn (the extent to which people are confident that they can gain sufficient literacy to pass muster). The highest levels of adoption occur when employees are motivated to develop competence because they fully buy into the transformation strategy and feel capable of helping make it a reality.

In a digital transformation, the two dimensions combine to produce the four quadrants of a matrix of responses: oppressed, frustrated, indifferent, and inspired. (See the exhibit "The adoption matrix.") In the best-case scenario, people will be in the top right quadrant, inspired by the change and believing that they have the capacity to learn digital content. Managers should assess which quadrant each of their team members falls into and then work to move individuals from one to another as needed.

The adoption matrix

Digital transformation sparks a range of responses in employees.



Promoting buy-in

To help engage people who don't see the value in gaining digital competencies (those in the bottom quadrants), leaders must increase messaging that stresses digital transformation as a critical frontier for the company. They should launch an internal marketing campaign to help employees imagine the potential of a company powered by digital technology. Managers should encourage their team members to view themselves as important contributors to the digital organization.

Promoting confidence

After establishing buy-in, managers should focus on boosting the confidence of team members in the two left quadrants. We have found that the more experience people have with digital technologies—whether through education or employment—the more confidence they gain. Sharing stories also helps: People can build confidence vicariously, by learning about the experiences of peers, managers, and others. With encouragement and reinforcement from company leaders and direct managers, employees can begin to believe in their own capabilities. (See the sidebar "The Elements of a Successful Employee-Training Program.")

The Elements of a Successful Employee-Training Program

CONTINUOUS LEARNING MARKS A NEW paradigm for education and career growth: The days when employees had one job and a fixed skill set for a whole career are gone. Companies that successfully upskill their workforce follow six practices.

1. Set a companywide goal for training.
2. Design learning opportunities that include all functional roles.
3. Prioritize virtual delivery, making learning scalable and accessible to everyone.
4. Motivate people to learn through campaigns, awards, and incentives.

5. Make sure managers understand the offerings so that they can effectively guide and inspire employees.
6. Encourage employees to participate in projects with digital components and hands-on learning opportunities.

It may seem that it would be more efficient to simply hire people who already have the technical skills needed to bring a workforce into the digital age. But as most companies know, the war for talent is fierce: Hiring enough digital talent to meet demand is nearly impossible in the current market. As a result, recruitment must be supplemented with an expansive effort to upskill existing talent.

Leaders should identify influencers in their ranks who have a digital mindset and recruit them to champion the transformation and become role models for those who are reluctant. Influencers can also be very helpful in identifying areas of concern among employees and ideas for improvements. They are likely to understand what kind of messaging will resonate with employees. Holding training sessions about the digital transformation and communicating new targets is also important.

Aligning Digital Systems

It is critical that organizational leaders understand how employees will deploy digital tools so that they can build

technology ecosystems and processes that foster a digital mindset and accelerate digital transformation.

Research by Harvard Business School professors Marco Iansiti and Karim Lakhani and colleagues identifies three of the main components of Moderna, the digital-born biotech and pharmaceutical company. The first, foundational layer is enormous access to data, which is the source of the company's value in developing vaccines and other therapeutics. The second is its reliance on cloud computing—a not only cheaper but faster and more agile solution than in-house servers. The third is its capacity for building AI algorithms to perform R&D processes with an accuracy and speed that is impossible to achieve manually. As Moderna's cofounder and CEO Stéphane Bancel told the scholars, Moderna is a "tech company that happens to do biology."

Historically, large pharmaceutical companies have operated in globally distributed, siloed units, but Moderna has a fully integrated structure in which data flows freely so that different teams can work together in real time. As Juan Andres, the company's chief technical operations and quality officer, has pointed out, "What's more important than having sophisticated digital tools or algorithms is integration at all levels. The way things come together is what matters about technology, not the technology itself."

In January 2020, when Moderna faced the urgent task of developing a vaccine for Covid-19, it was able to accelerate the

process because integration at all levels was already in place. Bancel had hired Marcello Damiani five years earlier to oversee digital and operational excellence, and Bancel was careful not to separate the two roles. “Enabling Marcello to design the processes was key,” he explains. “Digitization only makes sense once the processes are done. If you have crappy analog processes, you’ll get crappy digital processes.” Fully integrated systems and processes allowed Moderna employees to deploy existing digital solutions for the vaccine and build many others in-house, either designing algorithms from scratch or tweaking existing ones to perform deeper and more-specialized analyses. Only a few months after the Covid-19 outbreak, Moderna had developed some 20 algorithms for vaccine and therapeutic development and was working on many others.

Unilever, the consumer goods giant, has also adapted its sprawling global business for the digital age. For this manufacturer and retailer of household staples—more than 400 brands sold in 190 countries—success is a delicate balancing act between the specificities of local markets and the broad scale of global operations. The solution was agile teams, which could focus on customizing products to the “last mile” while simultaneously aligning their work across multiple countries using the company’s digital capacities. Rahul Welde, Unilever’s executive vice president for digital transformation and a 30-year veteran of the company, designed an agile-team structure that allowed members to remain globally distributed while making

strategic use of data for tailored initiatives within rapidly changing local markets.

Under Welde's leadership, Unilever formed 300 10-person agile teams that were remote and global and could operate at scale. According to Welde, the strategy had three parts. The first was using enabling technology and tools, which could reduce global-local divides. With digital platforms, brands could engage directly with customers in local markets on a vastly larger scale. The second was redesigning existing processes to adapt to new technology and tools. The third was about making sure that people had access to the technology and both the skills and the motivation to use it.

Who Selects Digital Tools?

Managers and business leaders must be heavily involved in selecting and implementing digital tools. To do that, they must understand what IT departments today can and cannot do. Historically technology groups have been well equipped to handle large, enterprisewide implementations of software and to make sure that the software undergirding a company's operations is maintained and works the way it should. That remains a key function of IT for implementations of bespoke tools or ERP systems. However, most of the technologies that companies adopt to enable digital transformation are cloud-based (SaaS). Teams can simply buy licenses, download the software, and get started without ever looping in IT.

Whereas IT is accustomed to managing support applications, business leaders are best suited to the task of defining new roles and routines—and effectively reshaping organizational culture and goals. They should begin by identifying which local activities will most effectively drive larger organizational goals, as this will inform the choice of digital tools and the direction of the transformation. As technology-driven process changes lead to new roles and responsibilities, new collaborative networks will open within the organization. These networks are the real positive drivers for the organization.

The company must continually gather data to monitor the transformation effort and assess whether employee behaviors are helping or hindering what we call the *work digitization process.* Leaders should study how information flows within the organization and remove institutional obstacles that might prevent employees from adopting the new process.

Change as a Constant

According to change management theory, organizations move from a current state to a transitional state and then on to a future state. The transitional state is typically considered to be a fixed period of time in which an organization shifts from familiar structures, processes, and cultural norms to new ones. People understandably experience strong emotions during the transition, because it requires them to make sense of new perspectives and ways of behaving. During this temporary state

of ambiguity, everyone's task is to negotiate between the organization's past and its future.

In a digitally driven world, however, there is no end point to the transitional phase: Digital tools change constantly and rapidly, as do the knowledge and skills necessary to use them. Organizational structures must be continually tuned to make use of new data insights, and leaders must keep working to bring employees along as the organization evolves.

Reconceiving of change as a constant process of transitioning rather than an activity that occurs between states helped Thierry Breton lead a successful digital transformation at Atos. It may be surprising that an IT company needed help with its own digital transformation, but that just underscores our point about how essential it is to cultivate a digital mindset. Just because employees have mastered one technology doesn't mean they are ready to adapt to the next one. Leaders need to view digital change as a state of constant transition that requires everyone to embrace the dynamism and uncertainty of permanent instability.

Digital technology and its impact on organizational structures, job roles, people's competencies, and customer needs is ever changing. A leader's task is not simply to adapt; it is to be adaptive. Digital transformation is not a goal that one achieves; it is the means to achieve one's unique goals. With a digital

mindset, employees across the organization are equipped to seize the opportunities our dynamic world presents.

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What Does It Actually Take to Build a Data-Driven Culture?

by Mai B. AlOwaish and Thomas C. Redman

MOST PEOPLE WHO WORK ON DATA SCIENCE, AI, and digital transformation are painfully aware that it is often culture, not technology, that stymies their efforts. Many even know the high-level steps they're supposed to take to fix this problem—invest attention and money into changing people's mindsets and how the company uses data. But once companies and leaders get into the nitty-gritty details of how to do this, it can be hard to know what implementing those steps actually looks like.

To understand what it takes to change a culture and encourage a digital mindset, it's helpful to see how another company is doing it. Which strategies worked and which were dead ends? What messaging landed with staff? Where should you actually start?

In this article, we begin to address this gap by summarizing the first two years of a new data program at Kuwait's Gulf Bank in which we worked to build a culture that embraced data. While

two years is far too short a time to claim the job complete, hundreds of people are doing their jobs differently and using data in new, exciting ways.

One of us, AlOwaish, was hired as Gulf Bank's first chief data officer in February 2021 as part of the strategic plan to launch a complete digital transformation of the bank's operations, with a mandate to provide data-driven customer experiences. Within that mandate her job was to sort out Gulf Bank's plans, build a small team, and execute. Though a successful technologist, she knew she would have to grow into the role. So she hired the other of us, Redman, to advise.

As she prepared to start her new job, she considered a common piece of advice: Score some quick wins, such as cleaning up a customer database, building a data lake to improve access, or improving regulatory reporting. But her boss, deputy CEO Raghu Menon, was an industry veteran and had seen too many data programs fail to launch when the low-hanging fruit turned out to be rotten. Instead, he counseled her to first "get the basics right."

We took two messages from Menon's insights. First, start with data quality. Many will view this as an odd choice, but in the data space, and especially for digital transformation and data-driven customer experience, nothing is more basic than quality. Bad data is the norm. And it is a vicious killer, adding enormous costs to day-in, day-out work and making monetization, analytics, and artificial intelligence far more difficult.

The second message was to think carefully about how we would get everyone involved, the culture we wished to create, and the organizational structures needed to be effective. Specifically, we wanted to drive home two things: that everyone needs data to do their job (they are data customers), and that they also create data used downstream (they are data creators). When people step into these roles, they work together to find and eliminate the sources of bad data and quality improves quickly. Along the way, attacking data quality in this manner leads people to engage and empower themselves with data.

Before charging off, we sought input from many long-term employees at all levels: Would employees find attacking data quality appealing? Would they find new roles as data customer and creator roles empowering? Their feedback told us that while some people would need convincing, plenty would find a lot to like, but we needed to provide some easy "getting started" assignments. It also encouraged us: Done well, employees said, these roles could transform the bank.

Idea in Brief

Building a data-driven culture is hard. To capture what it takes to succeed, the authors look at the first two years of a new data program at Kuwait's Gulf Bank in which they worked to build a culture that embraced data, and offer a few lessons. First, it is important to start building the new culture from day one, even as doing so is not the primary mandate. Second, to change a culture, you need to get everyone involved. Third, give data quality strong consideration as the

place to start. Finally, building this new culture takes courage and persistence.

Building the Extended Data Team

How could AlOwaish's small team get the entire bank of 1,800 people on board? To do so, we designed a "data ambassadors" program, essentially a network of people who would lead efforts to bring data quality to their teams. To build it, AlOwaish met the bank's management committee to explain Menon's charge, motivate the focus on quality, and describe the profile of people she sought. She also promised to provide training and support and that the entire bank would learn along the way. AlOwaish's "people, then technology" approach resonated with the committee, and its 13 members nominated 140 ambassadors-to-be.

Even though ambassadors-to-be had been nominated by senior leaders, as predicted, many were skeptical. They saw the role as nothing but added work. So AlOwaish and her team joined up with human resources to make the work interesting, rewarding, and fun. They did so in three ways:

1. **World-class training:** The ambassadors were told that they would learn and do things that would serve them well throughout their careers. Covid made delivery difficult, but the training—delivered face-to-face in five sessions—

explored their roles and responsibilities as data customers and creators, showed them how to make their first data-quality measurement, and provided a method to find and eliminate the root causes of error. The final session was a hands-on lab focused on self-serve analytics and data visualization. Each session featured an on-the-job assignment to help ambassadors get started.

2. **Media:** Ambassadors received lots of publicity, as internal newsletters, social channels, and local newspapers highlighted their work.
3. **Branding:** The data team reached out to marketing to create a logo for the data ambassadors program and build awareness by providing branded giveaways, such as a digital notebook.

Even the most skeptical ambassadors saw opportunities for personal empowerment by the end of the first session. They came to see that data and analytics weren't just for techies but something they could do on their own. And they carried these messages back to their teams.

Getting Everyone on Board

The next target was everyone else, with special focus on those working in branches, the call center, and on sales teams, on which so much bank customer experience depended. We

designed a “Data 101” program that explained their roles as data creators and customers, and highlighted the impact of data quality on the bank’s success at all levels. Interestingly, people in these roles create much of the bank’s most important data—but never knew why. Data was the furthest thing from their minds. Finally, AlOwaish worked to ensure that Data 101 is now included in all new-employee onboarding.

Understanding the broader reach of their work made it more exciting than the “just make the sale” approach in most banks. For example, Fahad AlRefaei, a direct sales representative, sought AlOwaish out after a training to explain how Data 101 changed his attitude. When opening a new account after closing a sale, he now pays extra attention to the data he doesn’t personally use, because he knows that data customers within the bank need it. Others provided similar feedback—once they learned how important quality data was, they took their responsibilities as data creators seriously. They felt empowered and better connected to the bank’s overall success. Thousands of such small steps make it easier for everyone to bring more, and more-trusted, data to customer engagements.

Innovation to the Fore

Empowerment is a beautiful thing! As we expected, ambassadors and others across the bank began working together, making measurements, targeting data cleanups, and

eliminating root causes of error. Then, somewhat organically, ambassadors and regular employees began using methods and tools provided in the training in new ways, to innovate on their own. For example, two ambassadors joined forces to improve anti-money-laundering models, enhancing the customer experience in the branch, while simultaneously reducing risk and operational expense. Earlier this year, AlOwaish and her team organized the inaugural "innovation tournament" at Gulf Bank. Hundreds competed—a sure sign that engagement and empowerment are taking root.

As noted above, two years is too soon to claim that a data culture has become fully embedded at Gulf Bank. Much can still go wrong. Moreover, AlOwaish and Gulf Bank have larger ambitions, including artificial intelligence, shared language, data-driven innovation, data-supply-chain management, and monetization. Many of these efforts will require big data, advanced technologies, professionals with advanced degrees, and support from ambassadors and others.

Lessons Learned

We are quite certain that there are many paths to building a great data culture. The U.S. Department of State has adopted a "surge philosophy," focusing on one department at a time, and others may do so by taking advantage of the excitement around

artificial intelligence.[1] Still, we think Gulf Bank's experiences illustrate some important points.

It is hard to change an existing culture and harder still if you're fighting it every step of the way. So instead, look for things that the existing culture will embrace and that will move the data culture you desire forward. For example, people working in health care may buy into "helping people lead longer, healthier lives." Explaining how a data program will advance that mission increases your chances.

It is important to start building the new culture from day one, even as doing so is not the primary mandate. This runs counter to conventional wisdom, which advises that you need quick wins to help build support. But quick-win efforts often take shortcuts, running roughshod over people and culture and increasing the likelihood that these projects fail. Further, successful quick wins may lead companies to falsely conclude they need not worry about people and culture, setting themselves up for future failures. Instead, aim for "significant wins" that fully embrace business results, structure, people, and culture.

Second, to change a culture, you need to get everyone involved. At Gulf Bank, we sought out the management committee, human resources, marketing, and corporate communications and received timely contributions from all. We emphasized data's importance by delivering the training face-to-face and tailoring it to each group. Indeed, there were more than

20 versions of Data 101. Further, cultures change through deeds, not words. So we were explicit about what we wanted people to do, not just how we wanted them to think or feel. The assignments provided in our training helped people jump-start the efforts.

Third, give data quality strong consideration as the place to start, as we did. While many view data quality as the least sexy topic in all of data, it is a great way to get everyone involved and it is foundational. You cannot build a great data program atop bad data.

Finally, building a culture takes persistence and courage. Expect to have some bad days, but keep the larger prize fully in mind.

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Note

1. Matthew Graviss and Garrett Berntsen, "How Do You Make the State Department Data-Driven? One Campaign at a Time," FedScoop, September 28, 2022, https://fedscoop.com/opinion-how-do-you-make-the-state-department-data-driven-one-campaign-at-a-time/.

When Data Creates Competitive Advantage

. . . and When It Doesn't *by Andrei Hagiu and Julian Wright*

MANY EXECUTIVES AND INVESTORS ASSUME that it's possible to use customer-data capabilities to gain an unbeatable competitive edge. The more customers you have, the more data you can gather, and that data, when analyzed with machine-learning tools, allows you to offer a better product that attracts more customers. You can then collect even more data and eventually marginalize your competitors in the same way that businesses with sizable network effects do. Or so the thinking goes. More often than not, this assumption is wrong. In most instances people grossly overestimate the advantage that data confers.

The virtuous cycles generated by data-enabled learning may look similar to those of regular network effects, wherein an offering—like a social media platform—becomes more valuable as more people use it and ultimately garners a critical mass of users that shuts out competitors. But in practice regular network

effects last longer and tend to be more powerful. To establish the strongest competitive position, you need them *and* data-enabled learning. However, few companies are able to develop both. Nevertheless under the right conditions customer-generated data can help you build competitive defenses, even if network effects aren't present. In this article we'll walk you through what those conditions are and explain how to evaluate whether they apply to your business.

What Has Changed?

Companies built on data have been around for a long time. Take credit bureaus and the information aggregators LexisNexis, Thomson Reuters, and Bloomberg, just to name a few. Those companies are protected by significant barriers to entry because of the economies of scale involved in acquiring and structuring huge amounts of data, but their business models don't involve gleaning data from customers and mining it to understand how to improve offerings.

Gathering customer information and using it to make better products and services is an age-old strategy, but the process used to be slow, limited in scope, and difficult to scale up. For automakers, consumer-packaged-goods companies, and many other traditional manufacturers, it required crunching sales data, conducting customer surveys, and holding focus groups. But the sales data often wasn't linked to individual customers,

and since surveys and focus groups were expensive and time-consuming, only data from a relatively small number of customers was collected.

That changed dramatically with the advent of the cloud and new technologies that allow firms to quickly process and make sense of vast amounts of data. Internet-connected products and services can now directly collect information on customers, including their personal details, search behavior, choices of content, communications, social media posts, GPS location, and usage patterns. After machine-learning algorithms analyze this "digital exhaust," a company's offerings can be automatically adjusted to reflect the findings and even tailored to individuals.

These developments make data-enabled learning much more powerful than the customer insights companies produced in the past. They do not, however, guarantee defensible barriers.

Idea in Brief

The Assumption

Companies can build winner-take-all positions by collecting and analyzing customer data. The more customers a firm has, the more data it can gather and mine; the resulting insights allow it to offer a better product that attracts even more customers, from which it can collect still more data.

The Reality

Even when customer data does confer a competitive advantage, it gives rise to network effects only infrequently. And that advantage may not last.

The Solution

To understand the edge that data-enabled learning can provide, companies should answer seven questions, which examine the value of the data; whether its marginal value drops quickly; how fast it becomes obsolete; whether it's proprietary; whether the improvements from it can be easily imitated; whether they enhance the product for current users, other users, or both; and how fast insights can be incorporated into products.

Building Moats with Data-Enabled Learning

To determine to what degree a competitive advantage provided by data-enabled learning is sustainable, companies should answer seven questions:

1. How much value is added by customer data relative to the stand-alone value of the offering?

The higher the value added, the greater the chance that it will create a lasting edge. Let's look at a business where the value of customer data is very high: Mobileye, the leading provider of advanced driver-assistance systems (ADAS), which include collision-prevention and lane-departure warnings for vehicles. Mobileye sells its systems mainly to car manufacturers, which test them extensively before incorporating them into their products. It's crucial for the systems to be fail-safe, and the testing data is essential to improving their accuracy. By gathering it from dozens of its customers, Mobileye has been able to raise the accuracy of its ADAS to 99.99%.

Conversely, the value of learning from customers is relatively low for makers of smart televisions. Some now include software that can provide personalized recommendations for shows or movies based on an individual's viewing habits as well as what's popular with other users. So far, consumers don't care much about this feature (which is also offered by streaming service providers such as Amazon and Netflix). They largely consider TV size, picture quality, ease of use, and durability when making purchasing decisions. If learning from customers was a bigger factor, perhaps the smart TV business would be less competitive.

2. How quickly does the marginal value of data-enabled learning drop off?

In other words, how soon does the company reach a point where additional customer data no longer enhances the value of an offering? The more slowly the marginal value decreases, the stronger the barrier is. Note that when answering this question, you should judge the value of the learning by customers' willingness to pay and not by some other application-specific measure, such as the percentage of chatbot queries that could be answered correctly or the fraction of times a movie recommendation was clicked on.

Let's say you graphed the accuracy of Mobileye's ADAS as a function of customer usage (total miles driven by car manufacturers testing it) and found that a few manufacturers

and a moderate level of testing would be sufficient to achieve, say, 90% accuracy—but that a lot more testing with a bigger set of car manufacturers would be needed to get to 99%, let alone 99.99%. Interpreting that to mean that the customer data's marginal value was rapidly decreasing would, of course, be incorrect: The value of the additional 9-percentage-point (or even a 0.99-point) improvement in accuracy remains extremely high, given the life-or-death implications. It would be difficult for any individual car manufacturer—even the largest one—to generate the necessary amount of data on its own or for any potential Mobileye competitors to replicate the data. That's why Mobileye was able to carve out a dominant position in the ADAS market, making it a highly attractive acquisition for Intel, which bought it for $15 billion in 2017.

When the marginal value of learning from customer data remains high even after a very large customer base has been acquired, products and services tend to have significant competitive advantages. You can see this with systems designed to predict rare diseases (such as those offered by RDMD) and online search engines such as Baidu and Google. Although Microsoft has invested many years and billions of dollars in Bing, it has been unable to shake Google's dominance in search. Search engines and disease-prediction systems all need huge amounts of user data to provide consistently reliable results.

A counterexample of a business where the marginal value of user data drops off quickly is smart thermostats. These products

need only a few days to learn users' temperature preferences throughout the day. In this context, data-enabled learning can't provide much competitive advantage. Although it launched the first smart thermostats that learn from customer behavior in 2011, Nest (acquired by Google in 2014) now faces significant competition from players such as Ecobee and Honeywell.

3. How fast does the relevance of the user data depreciate?
If the data becomes obsolete quickly, then all other things being equal, it will be easier for a rival to enter the market, because it doesn't need to match the incumbent's years of learning from data.

All the data Mobileye has accumulated over the years from car manufacturers remains valuable in the current versions of its products. So does the data on search-engine users that Google has collected over decades. Although searches for some terms may become rare over time while searches for new ones might start appearing more frequently, having years of historical search data is of undeniable value in serving today's users. Their data's low depreciation rate helps explain why both Mobileye and Google Search have proved to be very resilient businesses.

With casual social games for computers and mobile devices, however, the value of learning from user data tends to decrease quickly. In 2009 this market took off when Zynga introduced its highly successful FarmVille game. While the company was famous for relying heavily on user-data analytics to make design

decisions, it turned out that the insights learned from one game did not transfer very well to the next: Casual social games are subject to fads, and user preferences shift quickly over time, making it difficult to build sustainable data-driven competitive advantages. After a few more successes, including FarmVille 2 and CityVille, Zynga stopped producing new hits, and in 2013 it lost nearly half its user base. It was superseded by game makers like Supercell (Clash of Clans) and Epic Games (Fortnite). After reaching a peak of $10.4 billion in 2012, Zynga's market value languished below $4 billion for most of the next six years.

4. Is the data proprietary—meaning it can't be purchased from other sources, easily copied, or reverse-engineered?

Having unique customer data with few or no substitutes is critical to creating a defensible barrier. Consider Adaviv, a Boston-area startup we've invested in, which offers a crop-management system that allows growers (now primarily of cannabis) to continuously monitor individual plants. The system relies on AI, computer-vision software, and a proprietary data-annotation technique to track plant biometrics not visible to the human eye, such as early signs of disease or lack of adequate nutrients. It then translates the data into insights that growers can use to prevent disease outbreaks and improve yields. The more growers Adaviv serves, the broader the range of variants, agricultural conditions, and other factors it can learn about, and the greater the accuracy of its predictions for new

and existing customers. Contrast its situation with that of spam-filter providers, which can acquire user data relatively cheaply. That helps explain the existence of dozens of such providers.

It's important to keep in mind that technological progress can undermine a position based on unique or proprietary data. A case in point is speech-recognition software. Historically, users needed to train the software to understand their individual voices and speech patterns, and the more a person used it, the more accurate it became. This market was dominated by Nuance's Dragon solutions for many years. However, the past decade has seen rapid improvements in speaker-independent speech-recognition systems, which can be trained on publicly available sets of speech data and take minimal or no time to learn to understand a new speaker's voice. These advances have allowed many companies to provide new speech-recognition applications (automated customer service over the phone, automated meeting transcript services, virtual assistants), and they're putting increasing pressure on Nuance in its core markets.

5. How hard is it to imitate product improvements that are based on customer data?

Even when the data is unique or proprietary and produces valuable insights, it's difficult to build a durable competitive advantage if the resulting enhancements can be copied by competitors without similar data.

A couple of factors affect companies' ability to overcome this challenge. One is whether the improvements are hidden or deeply embedded in a complex production process, making them hard to replicate. Pandora, the music-streaming service, benefits from this barrier. Its offering leveraged the firm's proprietary Music Genome Project, which categorized millions of songs on the basis of some 450 attributes, allowing Pandora to customize radio stations to individual users' preferences. The more a user listens to his or her stations and rates songs up or down, the better Pandora can tailor musical selections to that user. Such customization cannot be easily imitated by any rival because it is deeply tied to the Music Genome Project. In contrast, the design improvements based on learning from the customer use of many office-productivity software products—such as Calendly for coordinating calendars and Doodle for polling people about meeting times—can be easily observed and copied. That's why dozens of companies offer similar software.

The second factor is how quickly the insights from customer data change. The more rapidly they do so, the harder they are for others to imitate. For example, many design features of the Google Maps interface can be easily copied (and they have been, by Apple Maps, among others). But a key part of Google Maps' value is its ability to predict traffic and recommend optimal routes, which is much harder to copy because it leverages real-time user data that becomes obsolete within minutes. Only companies with similarly large user bases (such as Apple in the

United States) can hope to replicate that feature. Apple Maps is closing the gap with Google Maps in the United States, but not in countries where Apple has a relatively small user base.

6. Does the data from one user help improve the product for the same user or for others?

Ideally, it will do both, but the difference between the two is important. When data from one user improves the product for that person, the firm can individually customize it, creating switching costs. When data from one user improves the product for other users, this can—but may not—create network effects. Both kinds of enhancements help provide a barrier to entry, but the former makes *existing* customers very sticky, whereas the latter provides a key advantage in competing for *new* customers.

For example, Pandora was the first big player in digital music streaming but then fell behind Spotify and Apple Music, which are still growing. As we noted, Pandora's main selling point is that it can tailor stations to each user's tastes. But learning across users is very limited: An individual user's up-or-down votes allow Pandora to identify music attributes that the user likes and then serve that person songs sharing those attributes. In contrast, Spotify focused a lot more on providing users with sharing and discovery features, such as the ability to search and listen to other people's stations, thereby creating direct network effects and luring additional customers. Pandora's service remains available only in the United States (where it has a base

of loyal users), while Spotify and Apple Music have become global players. And though Pandora was acquired by Sirius XM for $3.5 billion in February 2019, Spotify became a public company in April 2018 and as of early November 2019 was worth $26 billion. Clearly, customization based on learning from an individual user's data helps keep existing customers locked in, but it doesn't lead to the type of exponential growth that network effects produce.

7. How fast can the insights from user data be incorporated into products?

Rapid learning cycles make it hard for competitors to catch up, especially if multiple product-improvement cycles occur during the average customer's contract. But when it takes years or successive product generations to make enhancements based on the data, competitors have more of a chance to innovate in the interim and start collecting their own user data. So the competitive advantage from customer data is stronger when the learning from *today's* customers translates into more-frequent improvements of the product for those same customers rather than just for *future* customers of the product or service. Several of the product examples we've discussed already—maps, search engines, and AI-based crop-management systems—can be quickly updated to incorporate the learning from current customers.

A counterexample is offered by direct online lenders, such as LendUp and LendingPoint, which learn how to make better loan decisions by examining users' repayment history and how it correlates with various aspects of users' profiles and behavior. Here, the only learning that is relevant to *current* borrowers is that from *previous* borrowers, which is already reflected in the contracts and rates that current borrowers are offered. There's no reason for borrowers to care about any future learning that the lender may benefit from, since their existing contracts won't be affected. For that reason, customers don't worry about how many other borrowers will sign up when deciding whether to take a loan from a particular lender. Existing borrowers might prefer to stick with their current lenders, which know them better than other lenders do, but the market for new borrowers remains very competitive.

Does Data Confer Network Effects?

The answers to questions 6 and 7 will tell you whether data-enabled learning will create true network effects. When learning from one customer translates into a better experience for other customers *and* when that learning can be incorporated into a product fast enough to benefit its current users, customers will care about how many other people are adopting the product. The mechanism at work here is very similar to the one underlying network effects with online platforms. The

difference is that platform users prefer to join bigger networks because they want more people to interact with, not because more users generate more insights that improve products.

Let's look at Google Maps again. Drivers use it in part because they expect many others to employ it too, and the more traffic data the software gathers from them, the better its predictions on road conditions and travel times. Google Search and Adaviv's AI-based crop-management system also enjoy data-enabled network effects.

Like regular network effects, data-enabled ones can create barriers to entry. Both types of effects present a huge cold-start, or chicken-or-egg, challenge: Businesses aiming to build regular network effects need to attract some minimum number of users to get the effects started, and those aiming to achieve data-enabled network effects need some initial amount of data to start the virtuous cycle of learning.

Despite these similarities, regular network effects and data-enabled network effects have key differences, and they tend to make advantages based on the regular ones stronger. First, the cold-start problem is usually less severe with data-enabled network effects, because buying data is easier than buying customers. Often, alternative sources of data, even if not perfect, can significantly level the playing field by removing the need for a big customer base.

Second, to produce lasting data-enabled network effects, the firm has to work constantly to learn from customer data. In

contrast, as Intuit cofounder Scott Cook has often said, "products that benefit from [regular] network effects get better while I sleep." With regular network effects, interactions between customers (and possibly with third-party providers of complementary offerings) create value even if the platform stops innovating. Even if a new social network offered users objectively better features than Facebook does (for instance, better privacy protection), it would still have to contend with Facebook's powerful network effects—users want to be on the same social platform as most other users.

Third, in many cases nearly all the benefits of learning from customer data can be achieved with relatively low numbers of customers. And in some applications (like speech recognition), dramatic improvements in AI will reduce the need for customer data to the point where the value of data-enabled learning might disappear completely. Regular network effects, on the other hand, extend further and are more resilient: An additional customer still typically enhances value for existing customers (who can interact or transact with him or her), even when the number of existing customers is already very large.

As even mundane consumer products become smart and connected—new kinds of clothing, for instance, can now react to weather conditions and track mileage and vital signs—data-enabled learning will be used to enhance and personalize more

and more offerings. However, their providers won't build strong competitive positions unless the value added by customer data is high and lasting, the data is proprietary and leads to product improvements that are hard to copy, or the data-enabled learning creates network effects.

In the decades ahead, improving offerings with customer data will be a prerequisite for staying in the game, and it may give incumbents an edge over new entrants. But in most cases it will not generate winner-take-all dynamics. Instead, the most valuable and powerful businesses for the foreseeable future will be those that are both built on regular network effects and enhanced by data-enabled learning, like Alibaba's and Amazon's marketplaces, Apple's App Store, and Facebook's social networks.

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Building an Insights Engine

by Frank van den Driest, Stan Sthanunathan, and Keith Weed

OPERATIONAL SKILL USED TO CONFER long-term advantage. If you had leaner manufacturing, made higher-quality products, or had superior distribution, you could outrun competitors. But today those capabilities are table stakes. The new source of competitive advantage is customer centricity: deeply understanding your customers' needs and fulfilling them better than anyone else.

You need data to accomplish this. Yet having troves of data is of little value in and of itself. What increasingly separates the winners from the losers is the ability to transform data into insights about consumers' motivations and to turn those insights into strategy. This alchemy requires innovative organizational capabilities that, collectively, we call the "insights engine."

The vital role of the insights engine was revealed in a global market-research study led last year by the strategy consultancy

Kantar Vermeer. The study, called Insights2020 (i2020), involved interviews and surveys of more than 10,000 business practitioners worldwide (see the sidebar "About the Insights2020 Research"). Of the factors that were found to drive customer-centric growth, none mattered more than a firm's insights engine, embodied in its insights and analytics function. (While these go by many names—including "I&A," "consumer and market insights," and "customer intelligence"—for simplicity we refer to them as insights functions here.)

About the Insights2020 Research

TO UNDERSTAND THE ORGANIZATIONAL strategies, structures, and capabilities required to drive customer-centric growth, a global team launched the Insights2020 initiative in 2015.

Researchers conducted in-depth interviews with more than 350 business, marketing, and insights and analytics (I&A) leaders, along with online surveys of more than 10,000 practitioners in 60 countries. Kantar Vermeer led the project in partnership with Esomar, the Advertising Research Foundation, LinkedIn, and Korn Ferry. The Wharton School engaged an expert panel to validate and extend the research, and an analysis of LinkedIn's 400-million-member database discerned the interaction behaviors of executives at the most customer-centric firms.

Respondents were divided into two groups—overperformers and underperformers—on the basis of their companies' three-year revenue growth relative to their competitors'. The firms were then compared across an array of marketing and I&A dimensions. The results confirmed a strong correlation between customer centricity and revenue growth and identified the essential features of high-

performing, customer-centric organizations. These fell into three broad categories: a superior ability to provide a consistent, personalized, meaningful experience across all touchpoints; a singular commitment from every department to meeting customer needs; and the presence of an insights engine—typically an independent I&A function that participates fully in business planning and organizational strategy. Of these three categories, the insights engine—the focus of this article—had the largest impact on revenue growth.

In this article we describe the elements of the insights engine and show how it works at consumer goods giant Unilever. The firm's 400-plus brands, which include Dove, Knorr, and Axe, generated $60 billion in revenue in 2015, propelling underlying sales growth of 4.1% for the year. Performance at that level requires the full engagement of the company's 169,000 employees, who span functions from supply chain and R&D to marketing and finance. But as we'll show, it's the insights engine, manifested in the firm's Consumer and Market Insights (CMI) group, that underpins Unilever's customer-centric strategy.

A New Strategy

When Unilever released its first-quarter results in April 2016, CFO Graeme Pitkethly, addressing analysts, announced a major new initiative to shift resources to local markets around the world. He noted that consumers are increasingly seeking brands and products that align with their cultural identity and lifestyle.

The result is that local firms, particularly in emerging markets, are growing fast and strengthening their competitive positions. The new program, he explained, would clarify accountability and make Unilever's marketing teams more agile both globally and locally.

Country business heads had recognized the rising popularity of local brands, and the implications were being discussed separately at many levels across the firm. A presentation to the operating board by CMI's head, coauthor Stan Sthanunathan, drew on this intelligence and on CMI's own review of what was happening. Sthanunathan walked the board members through an analysis of why local brands were growing, what threat this posed, and how Unilever could compete. The presentation focused attention, catalyzed the conversation about strategy, and ultimately led to changes in both organization and mindset.

Unilever's new initiative showcases the type of high-level advisory role that leading insights functions are increasingly taking. A decade ago, this sort of strategic involvement by a customer intelligence operation was almost unheard of. The market research department typically was a reactive service unit reporting to the marketing function, fielding marketing requests, and producing performance management reports. Over time, however, market research departments have been shifting from merely supplying data to interpreting it—distilling insights about consumers' motivations and needs on the basis of their behavior.

Idea in Brief

The Imperative

Operational skill once conferred competitive advantage; now it's table stakes. The new source of advantage is customer centricity: deeply understanding your customers' needs and fulfilling them better than anyone else.

The Research

A study involving more than 10,000 practitioners examined the strategies, structures, and capabilities that distinguish high-performing, customer-centric companies. Having an independent insights and analytics function that participates fully in business planning and strategy is key.

The Case

Unilever's CMI group embodies the so-called insights engine through its expertise in synthesizing data, close collaboration with other functions, innovative use of new technologies and programs, and whole-brain mindset that balances creative and analytical thinking.

Driven by the imperative to become customer-centric, leading firms are now completing the transformation of market research groups into true insights engines with a fundamentally strategic role. At Unilever, CMI's prominently communicated mission is "to inspire and provoke to enable transformational action." Note that the word "insight" is missing—intentionally. That's because insights merely provide a means to the desired end: action that drives business growth.

In the pages that follow, we describe 10 characteristics of superior insights engines, gleaned from the i2020 research and our experience at Unilever. We divide these into two broad groups: *operational characteristics*, such as functional independence and experimental orientation, and *people characteristics*, such as business acumen and well-balanced analytic and creative thinking styles.

Operational Characteristics

Seven of the key characteristics relate to the way insights engines operate.

Data synthesis

Until recently, large firms had an advantage over smaller rivals simply because of the scale of their market research capability. Today research that once took months and cost millions can be done for a fraction of that price and in mere days. What matters now is not so much the quantity of data a firm can amass but its ability to connect the dots and extract value from the information. This capability differentiates successful organizations from less successful ones: According to the i2020 research, 67% of the executives at overperforming firms (those that outpaced competitors in revenue growth) said that their company was skilled at linking disparate data sources, whereas

only 34% of the executives at underperformers made the same claim.

This proficiency in using data is evident in high-performing firms across industries, including pharmaceuticals, financial services, hospitality, and consumer packaged goods. And to improve, many firms are creating dedicated data groups, under senior executive leadership, to consolidate, manage, and analyze data and distribute it throughout the organization. At Unilever, CMI has taken on this role.

For any insights group that serves as a data aggregator, interpreter, and disseminator, the first challenge is to integrate massive and disparate sets of both structured and unstructured data from such sources as product sales figures, spending on media, call-center records, and social media monitoring. This may amount to tens of millions of pieces of data. The data sets are customarily owned by different teams—sales data by sales, media spending by marketing, customer interactions by customer service, and so on.

Working closely with IT, CMI implemented a global marketing-information system, accessible to all marketers throughout the company, that integrates data and presents it in consistent formats. This ensures that all users, wherever they reside in the firm, see the same information in the same way—what CMI calls “one version of the truth.” Thus if marketing and finance are both looking at first-quarter shares of Dove soaps in any market segment, they’re viewing the same numbers and

units, derived using the same methodology and displayed in the same manner. Likewise, they see precisely the same picture when they look at data across brands, retailers, or regions.

Unilever's global marketing-information system has dramatically reduced the debates about data definitions, methodology, and interpretation that led to competing (and sometimes wrong) conclusions. It has also freed CMI from much of the resource-intensive reporting work that mires many firms' insights groups, allowing it to shift its focus from simply providing data to delivering insights and recommendations for action.

Consider CMI's role in Unilever's campaign to improve consumers' heart health. The firm was selling cholesterol-lowering spreads and drinks, but the hurdle was getting consumers to consistently use them. CMI's research generated quantities of data about consumption patterns. The initial insight was that for behavioral change to stick, people had to use the products for at least three weeks. The further insight was that the best way to get that long-term commitment was through peer pressure—engaging a group to work together. That insight then powered the marketing team to create a program called It Takes a Village, which challenges the people of an entire town to lower their cholesterol. The program, now in communities in more than 10 countries, includes cholesterol testing, nutrition advice, cooking guidance (involving the firm's

products), and group breakfasts and exercise. To date, 85% of people taking the challenge have lowered their cholesterol.

CMI's approach to data gathering and analysis is often technology-intensive. For example, while monitoring Twitter chatter in response to a Ben & Jerry's "free cone" promotion, a CMI team noticed a strong relationship between chatter and sales increases in most regions—but not all. A real-time analysis of the slow spots revealed that stockouts there were inhibiting sales, allowing Unilever to head off similar problems with future promotions.

A full accounting of how CMI marshals technology to synthesize data is beyond the scope of this article, but two major programs are illustrative. The first, CMI's People Data Centre, combines social media and business analytics with data mining of Unilever's customer-care lines and digital marketing channels, which capture millions of conversations a day in 40 languages. CMI can rapidly turn raw data from those sources into business impact. When the firm's Knorr brand launched its "Love at First Taste" campaign, for example, it was inspired by research showing that most people are attracted to others who like the same flavors they do. So Knorr found singles with shared tastes, set them up on food-based blind dates, and filmed the results. Then it released the video on social media and engaged with people who'd been identified as "food influencers." In the first three weeks, the video received 100 million views.

Another CMI program, PeopleWorld, addresses the problem "If only Unilever knew what Unilever knows." Often the answer to a marketing question already exists in the firm's historical research; finding it is the challenge. But using an artificial intelligence platform, anyone within Unilever can mine PeopleWorld's 70,000 consumer research documents and quantities of social media data for answers to specific natural-language questions. For example, a brand manager might ask, "What hair-care problems concern middle-aged men in India?" PeopleWorld computers would intuit what's needed, search the vast repository of information on hair loss, dandruff, and similar topics, and instantly deliver a high-level overview. Through a set of related queries, the manager could get a clear picture of the distinct and overlapping hair-care concerns of younger or older men and those in different countries—information that might yield insights about consumer needs in various markets and how to meet them.

Independence

Superior insights groups sit decisively outside marketing and other functions and often report to someone in the C-suite—the CEO, the chief strategy officer, or the chief experience officer. The i2020 research shows that insights leaders in overperforming organizations report to these senior executives more than twice as often as their counterparts in underperforming organizations do (29% versus 12%). Kantar

Vermeer's work with dozens of firms across industries indicates that this number is increasing, and we expect that in time this will be the typical arrangement.

At Unilever, Stan Sthanunathan reports to a member of the executive board—coauthor Keith Weed, who leads marketing, communications, and sustainable business functions. This reporting structure makes CMI a fully independent function with direct lines to the CEO. In this position, CMI can be objective, collaborate on an equal footing with other functions, and challenge or even set the direction of functional and organizational projects and strategy.

Take CMI's push to make advertising pretesting a standard procedure. Because Unilever is the world's second-largest media spender, improving advertising performance by even a few percentage points can translate into hundreds of millions of dollars in reduced costs and new revenue. And yet in the past, ads were often launched without hard data about their effectiveness. To change that, CMI implemented a disciplined testing program; using consumer surveys and software that reads facial expressions, the CMI team can now see if people find the ads authentic, relevant, and conversation-worthy—before they're aired. Poor ads are killed while powerful ones are given the go-ahead, and CMI collaborates with marketing to boost their performance. Ad creators originally saw the testing program as a threat to creativity and resisted it. But it proved so effective that marketers now embrace it, knowing that it helps

them do their best work and that successful ads figure into their bonus computations.

CMI's independence is enabled by having autonomy over its own budget, a mandate to drive business performance, and accountability for helping other functions achieve business targets. Thus when CMI recommends, for example, extending a brand into new local markets, it works in close partnership with marketing on the strategy and execution, because falling short would be as much CMI's responsibility as marketing's.

Integrated planning

For most companies, the business- and brand-planning cycle is the driving force behind strategy development and execution. This is where decisions are made about where to play and how to win. And it's here that resource allocation and budgeting are formalized and performance is monitored against goals. If insights groups are to help drive strategy, their activities must be aligned during the planning cycle with those of strategic planning, marketing, finance, sales, and other functions. That's why substantially more overperforming firms than underperforming ones (61% versus 46%) include insights leaders at all key stages of the planning cycle. We find that insights-function involvement in the cycle varies by industry; it's especially strong in retail.

Here's how CMI participates in the planning cycle: "Where to play?" is fundamentally a question of where to direct growth

investments—in existing, adjacent, or new markets. To help determine this, CMI uses a bespoke software tool called Growth Scout, which mines millions of data points on consumer demand across demographics, regions, and countries to quantify the potential value of deeper category or brand penetration. A typical application might be to gauge the impact of, say, increasing the penetration of shower gels by 10% in Thai markets. The results could help Unilever prioritize growth opportunities and decide where it could most profitably invest additional marketing or product-development resources. Recently, the CMI home-care team used Growth Scout to uncover potentially lucrative new markets for Unilever detergent brands by identifying demographic segments with weak penetration.

Once decisions have been made about where to play, another custom-built software tool, called Growth Cockpit, helps guide "How to win?" strategies. The tool provides a one-screen overview of a brand's performance in a market relative to the category. By rapidly building a visual picture of how the brand compares on a host of metrics—market share, penetration, pricing, media spending, and more—it points managers to growth opportunities.

Additionally, CMI employs other tools to help answer questions about which product benefits marketing should emphasize, which ads are most effective, what marketing budget allocations will yield the highest return on investment,

and what pricing is optimal. CMI then plays a central role in tracking the performance of marketing initiatives against targets and advising on tactical adjustments that may improve performance.

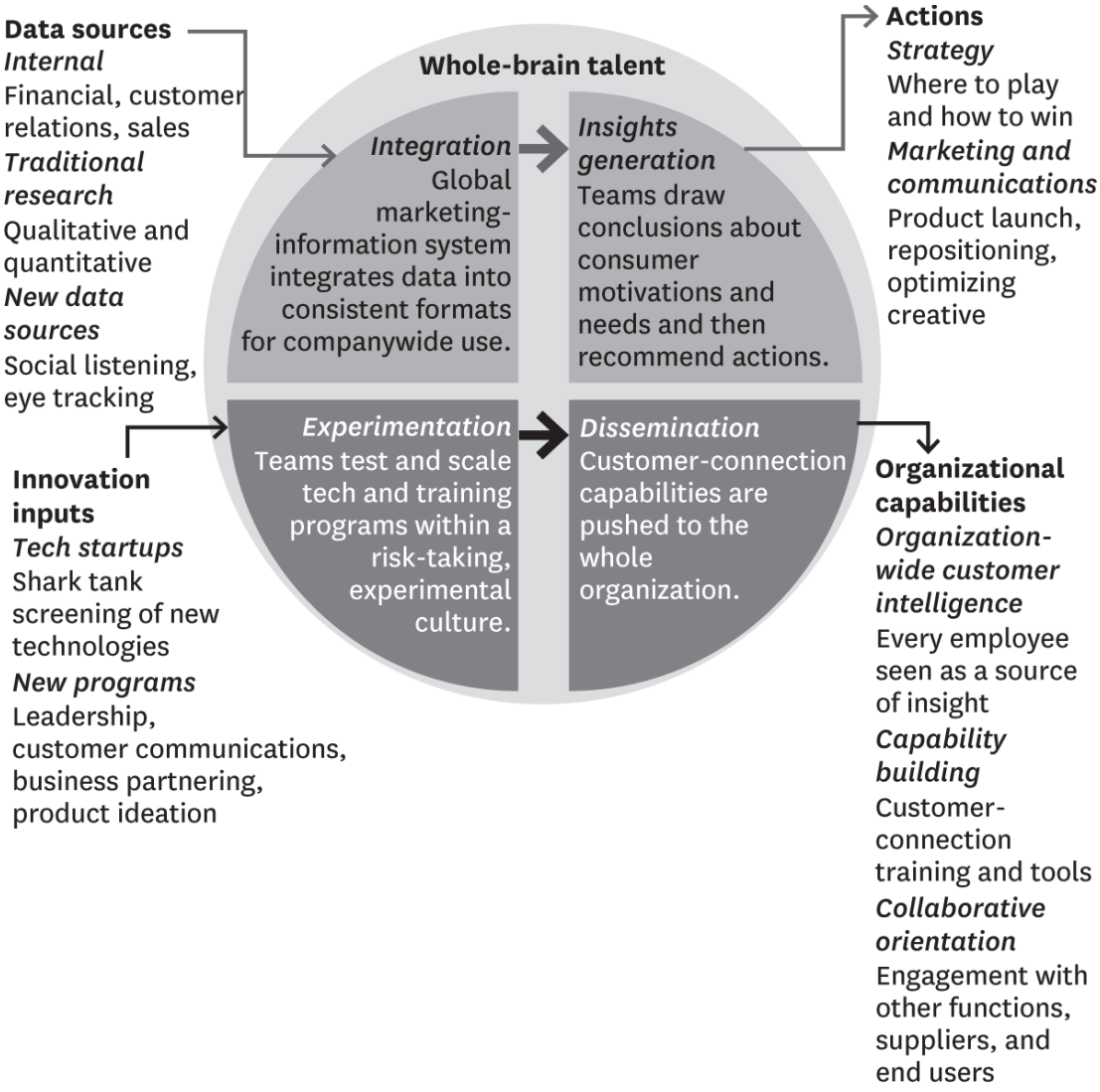
Collaboration

The i2020 study found that on average, 69% of respondents from overperforming firms said they work closely with other functions and customers, compared with just 52% of those in underperforming companies. This emphasis on collaboration is evident particularly among tech startups, but we're also seeing it among giants such as Alibaba and Google, and it's certainly the norm at Unilever and other large CPG firms.

The Workings of the Insights Engine

WHOLE-BRAIN THINKING IS at the core of the insights engine, powering two parallel processes that allow companies to understand and fulfill customers' needs.

The first involves gathering data from disparate sources, integrating it into a universally accessible database, mining it for insights, and then recommending concrete actions related to strategy and marketing. The second process starts with the use of new technologies and other inputs to foster a culture of experimentation. The insights engine team then strives to spread a customer-centric mindset throughout the organization, expanding employees' capabilities with training, tools, and an emphasis on collaboration.



In traditional market-research functions, the emphasis isn't so much on collaboration as on being an effective service provider. Insights functions like CMI have a distinctly different role that emphasizes shared goals and partnerships. We saw this in CMI's work with IT to create "smart" information-sharing platforms, like PeopleWorld, that anyone at Unilever can use. Similarly, CMI consciously collaborated with marketing, shedding its image as a "policeman" monitoring performance and instead

coming to be seen as a helpful partner in creating effective communications.

More broadly, CMI's structural alignment with the rest of the organization and its integration into the planning cycle create natural channels for often-daily collaboration. For example, CMI's organizational structure includes teams that focus on personal care, home care, foods, and refreshments, and the team leaders are colocated with the presidents of the same product categories in the broader organization. This helps ensure that when the strategy discussion turns to, say, expanding a personal-care brand into a new market, CMI and other functions are participating in conversations together and working as partners. Being held accountable for business results also provides an incentive for CMI to collaborate with all commercially oriented teams, since that is the best way to influence the key performance indicators for each team's operations.

It's understood across the firm that insights can come from anyone at any time. Therefore, CMI encourages every employee to engage with customers to gain insights about their needs and the role of Unilever products in their lives, and it provides tools to help. Through a program called People Voice, for example, all employees, from factory workers in Asia to members of global brand teams and up to the CEO, can connect directly with customers at events with themes such as "sustainability" and "shopper experience." Another option is for employees to use an

“always-on” platform, provided through a startup called Discuss.io, to arrange virtual meetings with consumers anywhere. A typical request might be: “I want to meet a South African soup lover next week at 4 PM.” The employee then gets an automated calendar invite to a live video chat. Some category presidents use the platform to engage with people in a country they plan to visit, asking about their needs and exploring opportunities for Unilever. This helps the presidents focus their conversations with local managers when they arrive.

To record their insights, employees use an in-house app that captures their observations from live chats or other consumer interactions. For example, an employee might note that people she talked with in Algeria equated “sustainability” with water conservation. Such notes, stories, pictures, and videos of employees’ communications are stored centrally and analyzed by CMI, which uses video mining and other technologies to identify behavioral patterns across regions and groups and to generate insights about consumer needs. For instance, reports from employee visits to customers’ kitchens in China revealed that because of high heat and tight space, grease buildup on surfaces is a common problem. Brand teams are now trying to determine what product innovations and messaging can help provide a solution.

About 30,000 people participate in People Voice programs annually. In addition to helping Unilever understand consumers’ needs, the programs reinforce the idea that it’s

everyone's job to uncover insights—a challenge that motivates and engages employees at every level.

Experimentation

Overperforming companies are three times as likely as underperformers to embrace a culture of experimentation, the i2020 research shows (40% versus 13%), and B2B firms in general are more experimental than B2C companies. Unilever is an exception in the B2C world, having formalized experimentation in a variety of ways, most visibly in its 2014 launch of the Foundry. Originally a marketing-technology startup incubator, the Foundry has since expanded to include hackathons, a collaboration platform for addressing sustainability issues, another platform that sources and gives prizes for creative marketing concepts, and a mentoring program that connects startups with Unilever experts who advise on product and brand development and marketing strategy. Much of the Foundry's work revolves around the "challenges" it posts on its site—requests for proposals to address a specific problem, such as consumers' quandaries over what to cook for dinner or how to live a more sustainable lifestyle.

Under the Foundry's aegis, CMI's Shark Tank initiative applies a technique borrowed from the CNBC show of the same name. A dozen or so startups pitch new technologies to a CMI executive team. Each has five minutes to tell its story, followed by five

minutes of Q&A. After the presentations, the team votes on which ideas to pilot and which to reject. Since its inception two years ago, Shark Tank has screened more than 650 technologies, piloted more than 175, and scaled up 37.

One of the startups brought in was Discuss.io, the online consumer-connection video platform. Another was weseethrough, which uses wearable technology to observe what consumers actually do—which is often not what they claim to do. Test subjects for weseethrough wear Google Glass while engaging in routine tasks, such as cleaning, cooking, or shopping. The company then analyzes the video captured by the headsets to discern behaviors that consumers themselves may be unaware of. For example, people may think it takes longer to clean the living room than the bathroom, but in fact the reverse is true. Insights like that have helped Unilever adapt its portfolio of products to address consumers' unarticulated cleaning needs.

Forward-looking orientation

To get a handle on the future, market researchers traditionally focused on the past. They might have reviewed a project launch months after the fact, for instance. Most firms today have shifted substantial attention to studying the present, monitoring consumers in real time to anticipate what they'll do next. The most sophisticated practitioners—those with insights engines like CMI—take the next step, using predictive analytics and other technologies, along with new organizational structures, to

both anticipate and influence behavior. Though overperformers currently aren't far ahead of underperformers in this regard (32% versus 28%), the i2020 research suggests that the gap is widening, and we expect the trend to continue.

Consider how CMI worked with Google and Razorfish to develop a program that leveraged real-time media monitoring to anticipate hairstyle trends and shape demand for related products. Unilever is one of the largest players in the global hair-care market, with brands including Suave and TRESemmé, but like its competitors, it had struggled to differentiate itself. Using a custom tool to analyze hair-related Google searches (there are about a billion a month), the program identifies styling trends and rapidly creates how-to videos featuring (but not directly promoting) Unilever products on a YouTube channel called All Things Hair. There visitors can browse by hair type and buy relevant Unilever products. Now live in 10 markets, All Things Hair has had more than 125 million views since its launch in 2013, and the research shows that it's three times as likely to drive purchases as conventional advertising is.

At a broader level, CMI created a team called Human and Cultural Futures (HCF), dedicated to imagining the future, examining developments in key regions, and exploring the implications for strategy. The team has identified certain societal, technological, environmental, political, and economic pressures, or "macro forces," that are shaping the world—including a shift of economic and technological growth to the

East (India and China) and South (Africa and South America), and growing environmental stress. Among its programs, HCF runs cultural awareness workshops and prompts brand and category teams to discuss how various macro forces might affect both consumers and Unilever. In one conversation about increased mortality among children under five, the Lifebuoy soap brand team zeroed in on data showing that over 40% of the deaths occur among infants less than a month old, and many could be prevented with handwashing. This has led to a sweeping handwashing education program that has changed the behavior of 337 million people in 28 countries. In villages in India, mothers reported that the incidence of diarrhea in family members dropped from 36% in 2013 to 5% in 2014.

Affinity for action

The most influential insights functions focus as much on strategy as on data. Indeed, i2020 found that 79% of insights functions at overperforming companies participated in strategic decision-making at all levels of the organization, compared with just 47% at underperforming companies.

CMI's action orientation manifests itself in two broad ways: in its specific recommendations to other functions and in the recruitment and training of "action-oriented" employees.

Look, for example, at CMI's engagement with marketing regarding market development. CMI pointed out the large "size of the prize" that Unilever stood to gain by expanding the

markets it operated in. Company leaders acknowledged this as the firm's biggest growth opportunity. CMI helped break the challenge into three parts—generating more product users, more usage, and more benefits for users—and then helped identify ways to attack those challenges. For instance, in the area of more usage, CMI suggested that promoting nighttime use of toothbrushes and toothpaste could boost business growth and tie in with Unilever's social mission of improving oral hygiene. CMI facilitated a workshop that highlighted the importance of dads in teaching their children to brush. That resulted in a marketing campaign with a song encouraging kids to brush at night as a way to have fun and bond with their fathers.

On the staffing side, from top to bottom, CMI invests in development programs designed to expand people's capabilities beyond the expected functional skills (research and analysis) to "action" skills—communicating, persuading, facilitating, leading. The idea is to help employees become better at turning insights into business results, whether by conceiving of a new business opportunity or by selling it within the organization.

People Characteristics

The operational characteristics that distinguish superior insights engines are complemented by three traits characterizing the people who are part of them.

Whole-brain mindset

For an insights engine to be collaborative, experimental, and so on, it needs a culture that breaks from the past. Historically, the members of insights organizations focused on analytics. That left-brain orientation served them well, but today's insights teams must think holistically, exercising creative, right-brain skills as well.

High-performing organizations are particularly adept at integrating the two types of approaches; far more respondents from overperformers than from underperformers agreed that their insights functions were skilled at whole-brain thinking (71% versus 42%). Achieving balance between right- and left-brain thinking requires a two-pronged effort: recruiting whole-brain talent and encouraging the mindset across the existing organization. Few people are purely right- or left-brained. But organizational work often favors analytical thinking, so conscious efforts to unleash people's creative side are particularly vital.

One approach that CMI uses is Upping Your Elvis workshops, run by a company of the same name. The energetic and interactive training pushes people out of their default thinking styles and gets them to engage in creative problem solving with colleagues they might not normally connect with. A recent workshop, for example, brought together people from marketing, R&D, CMI, and other areas and asked them to brainstorm ways to boost hair-conditioner sales in Southeast

Asia. Their insight was that consumers were reluctant to risk buying a product when they weren't sure of its benefits. This led to the idea of launching an inexpensive trial-size packet.

In other CMI workshops, the focus is on linking data about markets and brand performance to the actual consumer experience. Marketers, R&D staff, and others in the organization will go to people's homes to wash clothes or cook a meal, seeing first-hand how users engage with Unilever products. Workshop participants also connect directly with both loyal and lapsed customers and hear outside speakers present case studies on customer engagement. And they join in ideation sessions with colleagues across functions to imagine new growth programs and develop detailed action plans.

In all cases, employees leave these workshops with new collaboration tools, and they become role models and evangelists for whole-brain thinking.

Business focus

Historically, organizations' right-brain thinkers—marketing creative teams, for example—have not naturally focused on the business side. But i2020 found that respondents from high-performing firms were much more likely than those from low-performing firms to believe that their insights functions were business-focused (75% versus 50%).

At Unilever, CMI has implemented an array of programs to build business acumen. Recall that the vision of the CMI team is

“to inspire and provoke to enable transformational action.” CMI sees developing insights as a means to an end—customer-centric business growth. To reinforce the connection between insights and growth, staff bonuses are linked to the wider business unit performance. This creates shared accountability with other functions, encourages CMI teams to take responsibility for growth, and motivates them to go the extra mile. Teams are trained to think outside their traditional areas through “CMI Academy” courses on topics such as finance for nonfinance managers and effective business partnering. As a result of these and other programs, teams now instinctively consider the business impact of their work and of every recommendation they make.

Storytelling

The i2020 research imparts a final lesson about what makes for a strong insights engine: good storytelling. At overperforming firms, 61% of surveyed executives agreed that people in their insights functions were skilled at conveying their messages through engaging narratives; at underperforming firms, only 37% agreed.

At Unilever, CMI has embraced storytelling. Traditionally its presentations were data-intensive, built on the assumption that a fact-filled talk would be more persuasive than a fact-based one with less data and more narrative. Although data has its place, CMI has moved away from charts and tables and toward

provocative storytelling, embracing an ethos of “Show, don’t tell.” Increasingly, CMI is making its points with memorable TED-style talks and other experiential approaches.

For example, early in the business-planning cycle, CMI does market-by-market presentations to leadership and staff, including the heads of Unilever’s personal care, foods, and other categories. These describe global demographic, consumption, and other trends that are relevant to each category. Rather than bludgeon audiences with data, the presentations include compelling imagery and vignettes to advance a story line that has implications for strategy.

For an initiative targeting senior citizens, the CMI team found a novel way to bring the experience of older consumers to life. Instead of simply reporting how seniors struggle with products, CMI had marketing executives don old-age simulation equipment and then try to read labels and handle Unilever products such as shampoo. Encumbered by gear that reduced their mobility and vision, the marketers gained a real appreciation for the obstacles the elderly face. One outcome of the event is newly designed ice cream packaging that’s easier to read.

Much of what insights engines at any firm do is gather and analyze data. But today that is the minimum needed for success. Being able to translate this capability into customer-centric

growth is what distinguishes winners from losers. The insights engine is critical to this process—in fact, it's the most important driver identified by the i2020 research. But by itself, even the most advanced insights engine can't make a firm customer-centric. That requires leadership from the top to ensure that every function, from R&D to marketing to CMI itself, maintains a singular focus on understanding and meeting consumers' fundamental needs.

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Personalization Done Right

by Mark Abraham and David C. Edelman

THE SPOTIFY APP KNOWS what you want to hear. It uses AI to process a vast array of your engagement data, including the songs, podcasts, and audiobooks you've listened to, when you listened to them, and what led you to them. Its library is tagged by genre, era, tempo, mood, and a long list of other characteristics. The tags allow Spotify to collate playlists based on your listening habits. The app is always learning, constantly running micro tests with user groups. Spotify attributes a large part of its success to its personalized recommendations—its user base and revenues have both increased by 1,000% in the past decade to more than 600 million users and $14 billion, respectively.

Despite investing heavily in technology that could make personalization easier, faster, and smarter, most companies struggle to achieve Spotify's degree of personalization. In a BCG survey of 5,000 global consumers designed to gauge the effectiveness of personalized recommendations, two-thirds of respondents say they have recently experienced a recommendation that was inappropriate, inaccurate, or invasive.

Despite such negative experiences, more than 80% of respondents say they want and expect personalized experiences.

Business leaders aren't responding quickly enough. A BCG survey of more than 1,400 global C-suite executives found that 85% of business leaders plan to increase spending on AI in 2024. But most of that is earmarked for cost-saving initiatives rather than personalization. That is a mistake. We believe that personalization will be the most exciting and most profitable outcome of the emerging AI boom—but only for companies that make it a strategic focus.

Unfortunately, most companies lack a clear understanding of what great personalization should look like. We attempt to remedy that problem in this article. Drawing on decades of work consulting on the personalization efforts of hundreds of large companies, we have built the Personalization Index. This metric, a single score from 0 to 100, measures how well companies deliver on the promises they implicitly make to customers when they personalize an interaction.

To be clear, personalization isn't as simple as automatically plugging customer names into emails. True personalization requires creating experiences at scale, which get fine-tuned with each successive interaction and empower customers to get what they want—faster, cheaper, or more easily. Doing so requires delivering on five implicit promises that shape customers' expectations: *empower me, know me, reach me, show me,* and *delight me*. We'll describe how the components of the index are

based on these promises and show businesses how to assess their own Personalization Index score.

Building the Personalization Index

To develop the index, we surveyed 200 marketers, acted as mystery shoppers to observe how companies delivered experiences across channels, and performed a financial analysis of the impact of personalization on companies' growth and shareholder returns. We developed, tested, and refined the index's rubric internally at BCG and in our work with hundreds of clients. It uses objective questions, which allows scoring to remain consistent, whether the assessment was conducted by someone at BCG or by clients themselves.

Our research uncovered a set of performance drivers for each component of the Personalization Index. Let's look at each one in turn and examine how personalization leaders—companies that score in the top quartile of the index overall and excel in at least one area—are getting it right.

Idea in Brief

The Problem

In a survey of more than 5,000 consumers, more than 80% say they want and expect personalized experiences. But two-thirds say they've experienced personalization that is inappropriate, inaccurate, or invasive.

The Opportunity

Companies can use AI to create and continually refine personalized experiences at scale—empowering customers to get what they want faster, cheaper, or more easily.

The Assessment

The Personalization Index allows companies to assess how well they deliver on the promises they implicitly make to customers—and discover where they need to improve.

Empower me

Personalization leaders start by asking: How can I make the customer's experience better by personalizing it? They understand customers' unique needs at every step of their journey and decide how personalization can best help them. This is the most important component of the Personalization Index. To determine this score, we use mystery shopping activities and customer surveys to assess the level and quality of personalization that companies are delivering.

A leader in this area is SonderMind, a mental wellness company. According to the National Institute of Mental Health, almost a quarter of adults in the United States today struggle with mental health. It can be confusing to navigate the various treatment options, so SonderMind set out to empower its users to find help more easily. SonderMind's app personalizes patient journeys on the basis of interactive experiences that assess 12 brain functions (such as mental-processing speed and a predisposition to anxiety and depression), and patients log their progress against goals they set. While protecting individual

patient privacy, the app learns from the experiences of its many users and can suggest simple actions for each user to take (in addition to connecting with a therapist)—for example, practicing a meditation exercise or keeping a diary of one's feelings. It also arms therapists with tools to develop the optimal treatment plan, based on anonymized data about what has worked for similar patients. The result has been a win-win: lower costs for insurers and better outcomes for patients.

Know me

Personalization leaders win their customers' trust; they are granted permission to securely collect their data and use insights culled from it to improve the customer experience. Here, we look at the number and depth of digital relationships a company has with customers and its ability to retain them. We also look at the maturity of its customer data management and identity matching.

Spotify is one of a dozen or so large brands that have an average of upward of 600 million monthly users. But personalization leaders don't have to be the largest in their categories to deliver on this promise. Sweetgreen, a newcomer to the restaurant business relative to the largest chains, illustrates this point well. Right from its start, in 2007, it invested in building digital relationships with customers. It launched a mobile app in 2013, ahead of many large restaurant chains, and progressively added features such as mobile ordering, delivery, personalized offers and challenges, and a loyalty program to drive digital

engagement. Sweetgreen stands out not only in terms of its share of sales through digital channels (around 60% in 2023) but also in terms of engagement: Customers who use its digital channels order more frequently, and spend more on average, than those who come in to order salads at one of its brick-and-mortar locations.

Reach me

Having the data to know the customer is not enough. Leaders use AI to identify triggers to reach out, such as when a customer browses online or makes an inquiry. Moreover, they invest in orchestrating touches across channels and use smart frequency management to ensure that their touches are coordinated, not overwhelming. To assess an organization's maturity in this area, we examine the sophistication of its targeting intelligence, how advanced its experimentation capabilities are (as measured by the volume, speed, and scale of its experiments), and how well it orchestrates actions across channels.

Although Cisco focuses primarily on B2B customer experiences, it turned to personalization to enable its account teams and customers to navigate its ever-growing range of products and services. It recognizes that every client has a different context and that clients' situations can change rapidly. So it combined a wide range of data sources across the company, including product use, service requests, marketing interactions, and even external triggers such as financial news. Coordinating all this data has made Cisco a personalization leader. Its sales team knows whom

to contact, when, and about what, and comes armed with relevant content and demos. Because Cisco's sales and marketing teams are closely linked, customers get coordinated exposure to content that supports their needs and that opens up sales dialogues. Leveraging data on interactions around the world, Cisco uses AI models to understand the appropriate level of engagement for each account and to suggest the right timing to approach contacts. The models also make recommendations that marketers and salespeople can act on. Adoption of these tools quickly accelerated as salespeople found they could grow their account portfolios without losing momentum with their existing key accounts.

Show me

Personalization leaders build and manage robust content libraries that they tailor to the unique needs of each customer—while staying true to their brand voice. To do this, many are leveraging generative AI tools to create content. To evaluate companies on this promise, we looked at how sophisticated their content creation and management capabilities were and how well they personalized text, images, and videos for individual customers.

The global jewelry brand Pandora thrives by sparking customer interest with inspirational content. As part of its strategy, it uses AI-generated content to tailor its messaging to each customer and cut cycle times for certain types of content creation from 12–14 months to a mere 10 days. The company learned that personalizing the background and model image for each

individual—and coordinating how the customer sees those images across emails, websites, and other ads—substantially improved conversion rates.

Delight me

Personalization leaders adopt agile ways of working to accelerate the testing and learning that improve the intelligence behind each customer interaction. As noted earlier, true personalization requires that the quality and value of customer experiences be continually fine-tuned. Our research shows that the *delight me* promise is the hardest to deliver on. It requires relentless experimentation. The more customers you test with, in ever-faster cycles, the more learning you capture and feed into your AI engines, optimizing what you can deliver at increasingly granular levels. This level of personalization requires technology, but it also requires new operating processes that marry AI with organizational intelligence. Therefore, we measure the speed and scale of the company's test-and-learn process, the sophistication and automation of its measurement, and how the organization is set up for personalization—for instance, the prevalence of cross-functional agile teams, a clear owner of the initiative, and committed funding.

Many of the personalization leaders that set the bar in this area are digital natives. DoorDash runs hundreds of micro-experiments in its app, testing things such as improvements in add-to-cart and product-substitution recommendations. It has designed its app and its tech infrastructure to quickly run and

measure such experiments, and it operates agile teams that act on the learnings to improve each personalized digital experience.

More-traditional companies often fail a few times before they manage to emulate these approaches. We recently worked with a large U.S. bank that had created a team charged with breaking down silos to deliver greater agility throughout the company. But all it did was create yet another silo. We disbanded it and replaced it with a cross-functional "agile" pod, made up of in-house representatives from marketing, analytics, IT, creative, data science, and engineering. The participants had the authority and the budget to take meaningful action. They were tasked with a common goal of increasing the speed of testing. Within two months they cut the bank's traditional 12-week campaign-development process down to three days. Capitalizing on that drastically shorter cycle time, they launched hundreds of new tests, exponentially increasing their rate of learning. Those actions cut the bank's credit card churn rates in half and doubled the rate of converting leads into new active card members.

As those examples illustrate, getting personalization right has many components, but a marketer who is reasonably familiar with the company's personalization capabilities can complete a reliable self-assessment with a 15-minute survey. A full assessment can be done in a couple of weeks. We believe that it is worth your time and effort.

The Power of Personalization

The nature of competition is changing. Instead of competing on manufacturing scale, companies are competing on the scale of customer interactions, by building up their loyalty programs, app registrations, authenticated website users, and reasons to interact across the customer journey, to name a few examples. They compete on the speed at which they learn from interactions by testing variations in every interaction. They also compete on their ability to use their learning to tailor the customer experience by having cross-functional teams that constantly experiment with new experience designs, content, timing, channels, and the like. Here are several high-level takeaways from our research into personalization that can help businesses move forward.

Personalization leaders can be found in every industry

Across the 12 industry segments we reviewed, companies today score, on average, only 49 on the Personalization Index. The average score of companies in the top decile is 72. Unsurprisingly, the index leaders tend to be digital natives, such as Netflix, Uber, Alibaba, and Amazon, along with other early movers in the personalization space, such as Starbucks and Sephora. But we found leaders in almost all sectors.

We have already highlighted some in media (Spotify), healthcare (SonderMind), restaurants (Sweetgreen), B2B technology (Cisco), fashion (Pandora), and delivery services (DoorDash, a digital native). The differences within sectors are more pronounced than those across sectors. In addition to digital natives, the food, drug, and mass retail sectors also rate highly as

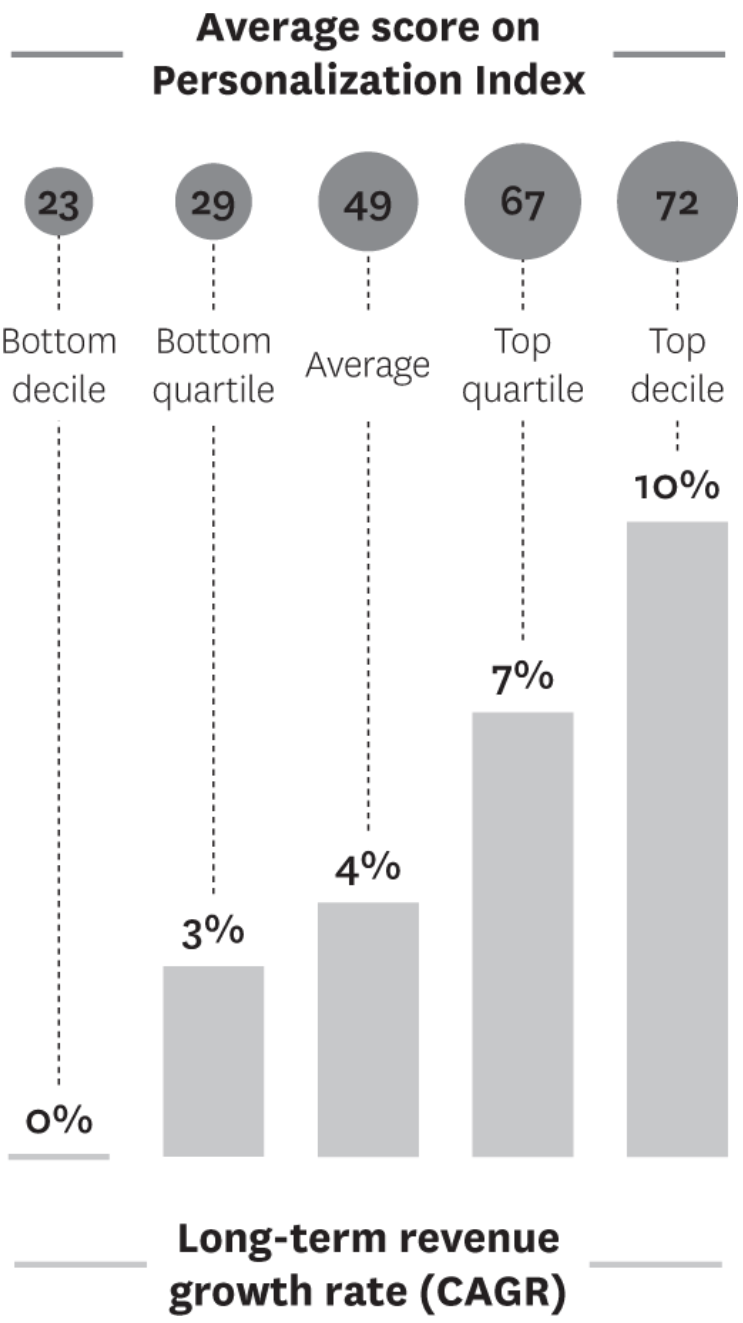
a group—not surprising, given the data-rich loyalty programs, high purchase frequencies, volume of touchpoints, and number of products in those sectors, all of which enable companies to collect more data in less time. Although companies in more highly regulated industries, such as insurance, financial services, and health care, tend to score lower on the index, some firms in those sectors have leapfrogged their competitors.

Advanced personalization generates lots of revenue

Companies that put personalization and AI at the center of their customer strategy are growing 10 percentage points faster than personalization laggards and six points faster than companies on average (see the exhibit "Better personalization leads to faster growth"). The data shows that personalization leaders have more digital customer relationships, and their customers spend 30% more than customers as a whole in their category. The root cause of this difference is that their customers engage three times as often (not just by transacting but also pre- and post-purchase) as do the customers of their competitors. As a result, the leading companies generate more data and insights on which to base future personalized interactions.

Better personalization leads to faster growth

Companies that excel at personalization grow faster than those that don't. When we compared companies that scored in the top decile on the Personalization Index with those that had an average score, we found a difference of six percentage points in their compound annual growth rate.



Note: We looked at the CAGR of publicly listed companies from 2018 to 2023, where data was available and excluding companies that had made major acquisitions.

Source: BCG Personalization Index research; *Personalized: Customer Strategy in the Age of AI* (Harvard Business Review Press, 2024)

Scaling personalization across customer interactions is crucial

Achieving the highest levels of customer satisfaction is possible only when a company delivers on personalization at scale—that is, in more interactions, with as many customers as possible, and while leveraging the benefits of detailed data in every interaction. When we examined the link between customer satisfaction (measured by a company's Net Promoter Score) and the Personalization Index, we observed a positive correlation. For every 10 points higher that a company scored on the index, its Net Promoter Score tended to increase by an average of seven points. The levels of customer satisfaction at companies that score low on the index vary widely. Indeed, some companies with exceptional service and value don't rely on personalization—think of a low-cost retailer known for its private-label quality or a no-frills airline known for on-time arrivals. However, the relationship between index ranking and customer satisfaction is clearest among personalization leaders. All top-scoring companies in the Personalization Index had NPS scores above 30, and only companies with high index scores had NPS scores above 40. By building personalization programs that scale up, companies can truly stand out among the world's best and achieve the highest levels of customer satisfaction.

Personalization leaders deliver superior value creation

We examined the value creation track record of personalization leaders over time. The top quartile of companies in the Personalization Index outperformed not only the laggards but also the market index over three, five, and 10 years. The

performances of the leaders and the laggards were already markedly different before 2020, but the Covid-19 pandemic accentuated the gap. One dollar invested in a personalization leader would have yielded three dollars after five years; the same investment in a personalization laggard would be worth only 50 cents.

These results show that companies that successfully execute personalization at scale are growing faster, delighting their customers, and creating superior value as they capture share from competitors. After crunching the numbers on the relative growth rates of personalization leaders and laggards by sector and taking into account the size of each sector, we estimate that a $2 trillion prize is waiting to be captured across industries by personalization leaders.

Voya Financial Keeps the Promises of Personalization

One company that's realizing the potential of personalization is Voya Financial. Recognizing that employee benefits were becoming a no-win commodity game, Voya's leaders set out to differentiate the firm. They positioned it as a partner that would improve how its clients' employees managed their finances while delivering superior economics for employers. To do that, they focused on pulling together a complete financial picture for registered users—including their retirement plans, personal financial activity, and health-care benefits. Voya examined how people make financial decisions and discovered numerous

opportunities to be a better partner. If it could get a more complete picture of customer financials, executives realized, it could dramatically improve its ability to inform customers about their situations and recommend options.

Voya's leaders started by recognizing that most people don't really understand how their disparate financial relationships fit together. Their research showed that few people had thought through how their health care costs were likely to affect their financial plans, or why their retirement and personal savings should have different goals and get different tax treatment. So they first focused on bringing together all the data, educating their members, and providing a single app, My Voyage, where members could see and add to their financial profiles.

Then Voya experimented with new customer experiences that were driven by personalization. When customers gave Voya a full picture of their financial portfolios, the company quickly produced recommendations for rebalancing the portfolios for optimal health, even if it meant money would be moved out of Voya accounts. By being objective, Voya built trust, and members felt encouraged to add more of their accounts outside Voya into the app.

Next Voya built a "test engine" to relentlessly experiment in order to learn how best to encourage members to take a useful action, such as increasing their retirement contributions, setting up a health savings account, or moving personal savings into an account with better interest rates. The company started with its

own employees, working through the execution issues and building its base of data. Then it scaled up those efforts and continually learned from its experiments. AI capabilities helped spot which members would benefit, helped set up the cells for testing multiple variables, and generated creative variants for the program team to consider.

Taken together, these personalization efforts contributed to not only a growth in accounts and double-digit revenue growth but also widespread accolades. Voya was named one of the "Most Trusted Financial Services" brands by *Newsweek* in 2023 and one of the "World's Most Ethical Companies" by Ethisphere.

Calculating Your Index Score

Now that we've proved the power of personalization, we'll help you determine your company's index score.

Sample Personalization Self-Assessment

TO COMPLETE A SELF-ASSESSMENT, companies must answer dozens of questions related to their personalization efforts. The following is a small sample of questions marketers should ask to determine the maturity of their personalization efforts. You can take the Personalization Index survey at on.bcg.com/personalization.

Empower Me

- Do we personalize experiences at each step of engagement throughout the customer journey?
- Do we provide cross-channel personalization capabilities?
- Is every communication channel used for personalization?

Know Me

- Does customer data live in a single repository?
- Is customer data integrated throughout our other marketing systems?
- Do we have high-quality identity resolution in place?

Reach Me

- Do we conduct A/B testing on our personalization efforts?
- Have we deployed a next-best-action decisioning or product recommendation engine?
- Do we automate customer segmentation?

Show Me

- Do we offer tailored experiences on a 1:1 basis for every customer? 10:1? 100:1?
- Are we able to conduct personalized experiences or campaigns quickly from ideation to launch?
- Do we use gen AI to automate the production of marketing collateral?

Delight Me

- Do organizational or technological impediments keep us from measuring progress on personalization?

- Do we run weekly tests related to personalization?
- Do we implement improvements on a weekly or more frequent basis?

Empower me (50 points). Begin by assessing the actual level of personalization you deliver to your customers in each step of their journey across all your channels. Because delivering on the first promise of personalization, *empower me*, is the most important, this score makes up half the Personalization Index score. To determine your score, identify the steps of your customer's journey and the channels where personalization truly matters and can make the experience better, faster, or more convenient. (That will depend on the categories you play in, your customers' expectations, and how well your competitors are personalizing.) Then look at what real customers recently experienced and evaluate whether you are personalizing "fully 1:1," "extensively," "moderately," or "not at all," and assign yourself points accordingly.

Next take an honest look at your organization's capabilities in delivering personalization at scale.

Know me (10 points). Measure the number and quality of the digital customer relationships you have relative to the total customer base in the markets you serve. Ask yourself how easily you can contact these customers across channels and how advanced your data management practices are. For example, how

do you handle data governance? And have you built a customer-360 database leveraging universal customer-identity matching?

Reach me (10 points). Measure how finely tuned your targeting models are. Are you using automated experimentation and machine learning to optimize the next best action for your customers and orchestrating across channels? Leaders leverage machine-learning AI solutions while laggards pull customer-targeting lists manually, with limited segmentation.

Show me (10 points). The length of time it takes you to launch a personalized campaign is a good barometer of your company's maturity on this dimension. (Laggard organizations take 12 or more weeks, but leaders take a few days.) It is also worth assessing how advanced your automated content creation and management capabilities are. Leaders leverage well-tagged content libraries and the latest gen AI solutions, paired with automated content creation processes, whereas laggards manually create each campaign, with little reuse of assets.

Delight me (10 points). Measure how well and how quickly you can test, learn from, and improve the customer experience. Diagnose whether you have organizational or technological impediments to measuring progress on personalization and implementing improvements on a weekly or more frequent basis. Leaders learn 10 times as fast as laggards do; they run hundreds

of experiments and turn around improvements in days, while laggards take many months to achieve the same feat.

And finally, be accountable (10 points). After you've assessed your performance on the five implicit promises, ask yourself whether your organization has a senior leader who is accountable for personalization, with P&L responsibility and a cross-functional mandate. In the large companies we studied, we have often found this to be what makes the difference between success and failure in personalization. That's why having such a senior leader in place is worth the final 10 points on the index.

Lessons from the Leaders

Personalization leaders follow a playbook to deliver on each of the five promises of personalization, starting with how to empower the customer. Senior executives looking to accelerate their own personalization efforts can follow that playbook no matter how mature their current personalization efforts are. We have found three areas of focus that make the difference between success and failure.

Rethink leadership roles and set a clear vision

Because personalization is such a cross-functional effort, its mandate must come from the CEO, and the C-suite should rally around a shared strategic vision. Every C-suite executive should explicitly support the vision and be accountable for delivering on

it. The CFO, for instance, must understand the business case. The CDO (chief digital officer) needs to be accountable for the digital experience. The CMO must instill agile marketing practices. The chief data and analytics officer (CDAO) needs to build robust platforms and intelligence. The COO needs to rethink the processes required to deliver a personalized customer experience, and the general counsel must champion responsible personalization and AI principles in partnership with the CDAO. The CIO needs to prioritize and deliver on the right technology investments in constant partnership with the business, and so on. Ultimately, as we explained in "Customer Experience in the Age of AI" (HBR, March–April 2022), small agile teams comprising many of these functions need to deliver on each wave of personalization use cases.

Secure the personalization foundation

The Personalization Index, paired with your strategic plan, will quickly highlight the gaps in your foundational capabilities. The key to success is prioritizing how to address those gaps and tying the upside and investments associated with each to the overall business case. Personalization leaders also apply the principles of smart integration. (For more detail, see "What Smart Companies Know About Integrating AI," HBR, July–August 2023.) They recognize that not everything has to be built internally, and the integration of the parts is just as important as the selection of the right components. In our experience, personalization can be largely self-funding, even in the initial years, when the right

trade-offs are made. As the teams deliver concrete business value, larger investments can be funded and the partnership between finance and a personalization team becomes even more important to ensure that a portion of the gains are regularly invested back into personalization efforts.

Compete on speed

Ultimately, personalization depends on the speed and scale of learning. Hence every organization should ask itself how teams could rethink processes to learn 10 times as fast. Many organizations are still working in eight- to 12-week campaign cycles. Three to five days is possible in every industry. Even companies that have cut their cycles down to days have a hard time taking the learnings forward quickly and enhancing the next customer interaction. Automation of specific areas (enhancing content, for example, or adjusting targeting and data science) can give teams rapid optimization tactics to improve KPIs meaningfully every week. On an annual basis, measuring how far you have progressed on the Personalization Index is a great way to hold the organization accountable.

For several decades companies have been gradually improving their financial performance by tapping the range of capabilities that can power personalization. We are now at a tipping point: Brand-differentiating personalization is not only possible but will be essential for competitive success. With the availability of new

AI tools, a company's ability to embed personalization capabilities in its operations has increased dramatically.

As the Personalization Index shows, there is a marked difference between the leaders—who are reshaping their value propositions and go-to-market approaches through personalization—and those who are just using basic tech tools to bombard prospects with impersonal messages. Tomorrow's winners are already using AI to unlock personalization's potential, and they are building new sustainable advantages along the way. The time for delivering growth with AI-enabled personalization is now.

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Ensure High-Quality Data Powers Your AI

by Thomas C. Redman

TWENTY YEARS AGO, mortgage-backed securities and collateralized debt obligations were all the rage. These new financial products were, initially, a wonder: They helped put millions of people into homes and make billions for banks. Then things went horribly wrong, and they nearly tanked the global economy.

The financial crisis of 2007–2009 has been studied from many perspectives, but viewed through a data science lens, it provides two insights that serve as a cautionary tale for artificial intelligence and machine learning (which together I'll simply refer to as AI) today.

First, the analytical work to slice and dice mortgages into products that fit investors' different risk-reward profiles was actually quite good. But results depended on the data being high-quality.

Second, the data was not, in fact, high-quality. Bad or missing data appeared at every step, from incorrect (sometimes

fraudulent) information on mortgage applications to incorrect credit scores and securities ratings, lost paperwork, and a lack of clarity in bank liabilities. The subpar data led banks to vastly underestimate risks, leading to an unexpected number of foreclosures, in turn causing banks to not trust each other's liability statements and the banking system to freeze up.

The lesson at the heart of this story is a simple one: Good data science + bad data = bad business results.

What does this have to do with AI? For one, the hype surrounding AI is far greater and more pervasive than was the hype around financial products. Worse:

- The math used to slice and dice mortgages was well understood. In contrast, even experts aren't sure why AI makes the predictions it makes.
- The quality requirements for AI are far broader and deeper.
- While the analytics work for financial products was performed by professionals who understood, or should have understood, the importance of quality data, better tools make AI accessible to people who are less informed on all that can go wrong.

AI does not need to fail on a global scale to cause enormous damage—to individuals, companies, and societies. Models frequently get things wrong, hallucinate, drift, and can collapse. Everyone has their favorite "look at this ridiculous answer a

large language model gave me" or "AI disaster" story. Coupled with the high failure rate of AI projects, it is not hard to see why few companies can cite a positive ROI.

Warnings about data quality have been plentiful, coming from sources as diverse as those leading the AI revolution, including Google, Meta, and Open AI; prominent researchers; and the popular press, including the *New York Times* and the *Economist*.[1] Forrester, the research firm, advises that "Data Quality Is Now the Primary Factor Limiting GenAI Adoption."[2] Yet most companies have not even started the work they need to do.

The primary interest here is a company's first several AI projects. Importantly, data quality is an enormous organization-wide issue (and opportunity), yet most companies have neglected it. One consequence is that they, and their senior leaders, don't understand the issues, how to address them, and the organizational changes they must make. They are in a tough spot—they must protect themselves in the short term *and* build needed longer-term capabilities, as well as simultaneously educate themselves so that they can do all that effectively. This article aims to help companies get started.

Idea in Brief

AI does not need to fail on a global scale to cause enormous damage—to individuals, companies, and societies. Models frequently get things wrong, hallucinate, drift, and can collapse. Good AI comes from good data, but data quality is an enormous organization-wide issue (and opportunity), yet most companies have neglected it.

Companies need to understand the nuances of the problem they're trying to solve, get the data right (both by having the right data for that problem and by ensuring that the data is error-free), assign responsibility for data quality in the short term, and then push quality efforts upstream in the longer term.

Understanding the Problem

Any data science project should start with levelheaded thinking about the problem one wishes to solve. To illustrate, consider, "We need a better way to decide which loans to grant and under what terms." Clearly stating the problem involves asking and answering questions such as "Better how?" and "Which loans, and to whom (that is, who's the population of interest)?"

The answers can be found by determining what data you need. Examples include:

- If "better" means "bias-free," you'll need a bias-free training data, which may be very hard to come by.
- If "better" means "more performing loans and fewer nonperforming loans," you need data on loans that you have historically not granted.
- If "better" means "decisions that can be more easily explained to regulators," AI may not be your best approach, because "explainability" is fiendishly difficult.

- If “better” means “will work in parts of the world in which we don’t currently operate and don’t have data,” you have a tough challenge.
- If “better” means a “cheaper overall process,” your historical data may be just fine.

Coming to agreement on the problem can be highly political. Those eager to bring AI into the organization may be motivated by completing a “use case” to prove AI works, while others may want to allay fears that competitors are roaring ahead, and still others may wish to get regulators off their backs. All such perspectives are valuable, and it is tempting to plow ahead before the problem is clearly stated. Doing so is ill-advised.

Getting Data Right

“Garbage in, garbage out” might be a useful rule of thumb, but I find it convenient to frame the idea of good data through two requirements: 1) whether it’s the “right data” to address the problem and 2) whether that “data is right,” or correct. The criteria for the latter are more familiar to most people: accuracy, absence of duplicates, and so forth. Having the “right data” is less familiar, more subtle, and complex—but it’s also essential.

When sorting out the “right data” for a project, consider:

- **Relevance and completeness:** AI depends on data with predictive power; that is, data relevant to the problem at hand. For instance, attributes such as age, payment history, and income are clearly relevant to loan decisions. Completeness involves having as many relevant data attributes as possible.
- **Comprehensiveness and adequate representation:** The two main issues are “Does the data adequately cover the population of interest (comprehensiveness) and all subpopulations of interest (adequate representation)?” After all, it takes a lot of data to properly train a model. To illustrate adequate representation, note that if you wish your model to work worldwide, you’ll need data from all over the world.
- **Freedom from bias:** Reducing historical bias is (or should be) important in most problems involving humans. Unfortunately, many data sets reflect historical biases, and removing those biases from data sets may be nearly impossible. In such situations, companies should not use AI.
- **Timeliness:** The essential issue is “How new must the data be?” In the loan example, making decisions quickly using the most up-to-date data may prove to be a source of competitive advantage.

- **Clear definitions:** Most AI efforts pull data together from disparate sources, and a good understanding of those sources and the data they provide makes combining them easier. Clear definitions—of sources, data attributes, and measurement units—make developing this understanding easier.
- **Appropriate exclusions:** Certain data should be excluded for legal, regulatory, ethical, and intellectual property considerations. For example, using zip codes can be a surrogate for race in loan decisions, and you must avoid violating laws stipulating how personally identifiable information (PII) may be used. Further, there is growing concern that AI models trained on public sources may violate intellectual property rights.

The most important “data is right” considerations include:

- **Accuracy:** Accuracy is the probably the best-known feature of data quality. The essential idea is that data values must be “correct”; that is, they must reflect reality. Though subtleties such as “How closely must the data values represent reality?” are sometimes important, most structured data sets are riddled with errors. And it stands to reason that documents (unstructured data) used to train large language models are in even worse shape.

- **Absence of duplicates:** It is easy for duplicate entries to slip into databases, and they can skew results. Thus, they must be kept to a minimum.
- **Consistent identifiers:** When compiling loan data, is the "John Smith" with a checking account and the "J. E. Smith" with a home equity line of credit the same person? You have to know so that you can integrate the data.
- **Correct labeling:** Good data labels (for example, "This is a cat," "This loan is performing") improve AI models.

The data-quality job does not end with training. Train a model well on high-quality data and you get a good model. But feed that model bad "future data" and you get bad results/predictions. Both the "data is right" and "right data" criteria remain in force.

One woman was denied painkillers because prescriptions for her dog were ascribed to her (that is, the data was wrong).[3] This led a model to label her as at risk for opioid addiction. Similarly, feed a model data beyond the realm on which it was trained and there is no telling what will happen. This occurs rather frequently because conditions in the world have changed, causing data inputs from new domains and, often, degraded model performance. Sometimes the deterioration is severe. Microsoft's Tay chatbot, which started spitting hate speech

when users fed it vile phrases (on which it was not trained), serves as a cautionary tale.[4]

In the Short Term, Adopt a "Guilty Until Proven Innocent" Stance

The most important step in any data-quality project or program involves "getting the management responsibilities right." And for any given AI project, it is tempting to assign quality to those closest to the work, including modelers, data engineers, and data wranglers. While all can do some good work in understanding sources and cleaning the data, they are not equipped to pull together the full range of effort required. To illustrate, "defining the problem" is a business issue, not a modeling, engineering, or wrangling issue.

At the project level, overall responsibility for data quality must reside with the highest-level person leading the effort. They must organize teams of people who can develop the full suite of requirements and dive into details to ensure they are met. They should consider acquiring external talent if, as is often the case, their companies lack the breadth and depth of skill needed to do so.

These managers should relentlessly question model-development teams and others about everything. Start with "Are the problem you set out to solve and our problem the same?" If not, be very careful. Next, work through the "right

training data" criteria one at a time. Dig deep: "What makes you think the data will yield good predictions, will support extending the model to new markets, and is bias-free?" Be wary when vendors decline to answer, citing "Our model is proprietary." The model may be proprietary, but that doesn't mean the data is.

With respect to the "data is right" criteria, modelers almost certainly need to perform extensive cleanup of training data. Ask them to detail how they measured accuracy before and after cleanup, how they found and eliminated duplicates, how they ensured data integration, and how they ensured that labels were correct. Push very hard, and if you detect any softness in the answers, insist modelers obtain better answers.

Management responsibilities extend to future data. Managers must monitor future data aggressively to ensure it doesn't stray too far from the bounds of the data on which it was trained. They must insist on in-depth controls to make sure that future data is as complete and accurate as possible.

Finally, senior leaders and boards should also insist on quality assurance in the form of independent review, in order to ensure that all the issues described here are understood, embraced, and acted on in an appropriate manner. Especially early on, a bit too much QA is better than too little.

In the Medium Term, Push Quality Efforts Upstream

Data scientists, like everyone else in a company, depend on others for data. Marketing creates data used by Sales; Sales creates data used by Operations; and those building AI models use data created by many sources and in turn create data in the form of predictions. Today each group spends an inordinate amount of time dealing with bad data: perhaps 30% across the company and 80% in data science.[5]

Fortunately, there is a better way: Eliminate root causes of bad data upstream, and those downstream don't have to deal with bad data! This approach builds on what worked in manufacturing and has proven itself time and again at companies such as AT&T, Gulf Bank, Chevron, and others. As a practical matter, this works best when downstream groups take on roles as *data customers*; upstream groups take on roles as *data creators*; and the two work together to sort through quality requirements, make some basic measurements, and then find and eliminate root causes of error, one at a time.

Thus, companies should transition from dealing with bad data downstream to creating good data upstream. Like any other discipline, quality management provides a rich body of underlying foundations, methods, and tools on who does what. To bring them into your organization, start with three concepts:

1. Focus intensely on customer and process improvement, with the long-term goal of creating the right data correctly with proper identifiers and labels, the first time.

2. Clarify management accountabilities: Make clear that business departments, not IT, the chief risk officer, or the chief data officers, are responsible for data quality. In turn, this means defining clear role responsibilities for teams and individuals.
3. Get as many people as you can, at all levels, as involved as you can.

Don’t Get Overwhelmed

I know that some will react, “Oh my goodness, this is far too expensive.” But in fact, these recommendations save money. As noted, companies spend excessive time on non-value-added data work. Following these recommendations cuts that cost dramatically.

The more pertinent reaction is, “Oh my goodness, this will require real transformation.” That is true! If you’re not up for it, perhaps you should scale back your AI ambitions.

More proactively, companies are beginning to realize that, properly managed, data becomes an asset of unlimited potential. It deserves proper management. AI unlocks that potential. Simply stated: To get great results from high-quality AI, you need high-quality data.

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The Ethics of Managing People's Data

by Michael Segalla and Dominique Rouziès

THE ABILITY TO ENCODE, store, analyze, and share data creates huge opportunities for companies, which is why they are enthusiastically investing in artificial intelligence even at a time of economic uncertainty. Which customers are likely to buy what products and when? Which competitors are likely to move ahead or fall behind? How will markets and whole economies create commercial advantages—or threats? Data and analytics give companies better-informed and higher-probability answers to those and many other questions.

But the need for data opens the door to abuse. Over the past few years the EU has fined companies more than 1,400 times, for a total of nearly €3 billion, for violations of the General Data Protection Regulation (GDPR). In 2018 the Cambridge Analytica scandal alone wiped $36 billion off Facebook's market value and resulted in fines of nearly $6 billion for Meta, Facebook's parent company. And stories abound about how AI-driven decisions

discriminate against women and minority members in job recruitment, credit approval, health care diagnoses, and even criminal sentencing, stoking unease about the way data is collected, used, and analyzed. Those fears will only intensify with the use of chatbots such as ChatGPT, Bing AI, and GPT-4, which acquire their "intelligence" from data fed them by their creators and users. What they do with that intelligence can be scary. A Bing chatbot even stated in an exchange that it would prioritize its own survival over that of the human it was engaging with.

As they examine new projects that will involve human-provided data or leverage existing databases, companies need to focus on five critical issues: the *provenance* of the data, the *purpose* for which it will be used, how it is *protected,* how the *privacy* of the data providers is ensured, and how the data is *prepared* for use. We call these issues the five Ps (see the sidebar "The Five Ps of Ethical Data Handling"). In the following pages we'll discuss each of them and look at how AI technologies increase the risk of data abuse. But first we'll offer a brief overview of the organizational requirements for a robust ethical-review process.

The Five Ps of Ethical Data Handling

Provenance

- Where does the data come from?
- Was it legally acquired?
- Was appropriate consent obtained?

Purpose

- Is the data being repurposed?
- Would the original source of the data agree to its reuse for a purpose different from the one originally announced or implied?
- If dark data is being used, will it remain within the parameters of its original collection mandates?

Protection

- How is the data being protected?
- How long will it be available for the project?
- Who is responsible for destroying it?

Privacy

- Who will have access to data that can be used to identify a person?
- How will individual observations in the data set be anonymized?
- Who will have access to anonymized data?

Preparation

- How was the data cleaned?
- Are data sets being combined in a way that preserves anonymity?

- How is the accuracy of the data being verified and, if necessary, improved?
- How are missing data and variables being managed?

Organizing the Oversight of Data

In academia, data acquisition from human subjects is usually supervised by an in-house institutional review board (IRB) whose approval researchers must have to obtain access to the people involved, research funds, or permission to publish. IRBs are composed of academics versed in the research and the ethics around the acquisition and use of information. They first appeared in the field of medical research but are now used almost universally by academic organizations for any research involving human subjects.

A few large companies have also established IRBs, typically under the leadership of a digital ethics specialist, hiring external tech experts to staff boards on an ad hoc basis and assigning internal executives from compliance and business units as necessary. But that remains rare: Even in Europe, which has been at the forefront of data regulation, most companies still give responsibility for adhering to the GDPR to a mid- or senior-level compliance manager, who often has some legal or computer engineering training but not extensive ethical training and rarely has a solid grasp of emerging digital technologies.

Although a compliance manager should certainly be part of a corporate IRB, he or she should probably not be directing it. In fact, the European Data Protection Board announced in March 2023 that it was concerned about this issue and that data protection officers would be sent questionnaires designed to determine whether their corporate roles are appropriate for ensuring compliance.

Idea in Brief

The Problem

As companies jockey for competitive advantage in the digital age, they are increasingly penalized for the abuse of data. In 2018 the Cambridge Analytica scandal alone wiped $36 billion off Facebook's market value and resulted in fines of nearly $6 billion for Meta, Facebook's parent firm.

Why It Happens

Most problems arise from (1) ethical failures in data sourcing, (2) using data for purposes other than those initially communicated, (3) lack of security in storing it, (4) how it is anonymized, and (5) how it's prepared for use.

The Solution

Companies should create a special unit to review projects involving people's data. In its reviews this unit should carefully consider the five Ps of data safety: *provenance*, *purpose*, *protection*, *privacy*, and *preparation*.

A good overview of how companies might establish an IRB-type process can be found in "Why You Need an AI Ethics

Committee," by Reid Blackman (HBR, July–August 2022). Our experience confirms most of its main points. A corporate IRB should have from four to seven members, depending on the frequency, importance, and size of the company's digital projects. The members should include a compliance specialist, a data scientist, a business executive familiar with the functional area of the digital projects (such as human resources, marketing, or finance), and one or more senior professionals with appropriate academic credentials. The full board won't be needed for every review. The London School of Economics, for example, uses its full board only for the oversight of the most complicated projects. Simpler ones may be evaluated in less than a week using an online questionnaire and with the input of only one board member.

Any new project involving the collection, storage, and processing of data about people should be approved by the corporate IRB before getting a go-ahead. There should be no exceptions to this rule, no matter how small the project. In addition, most companies have already collected large stores of human data and continue to generate it from their operations; the corporate IRB should examine those projects as well.

An IRB review begins with our first P: exploring how a project will (or did) collect the data—where it comes from, whether it was gathered with the knowledge and consent of the research subjects, and whether its collection involved or will involve any coercion or subterfuge.

1. Provenance

To understand what can go wrong with sourcing data, consider the case of Clearview AI, a facial-recognition firm that received significant attention in 2021 for collecting photos of people, using them to train facial-recognition algorithms, and then selling access to its database of photos to law enforcement agencies. According to a report by the BBC, "a police officer seeking to identify a suspect [can] upload a photo of a face and find matches in a database of billions of images it has collected from the internet and social media."

The Australian regulatory agency objected to Clearview's collection method, finding that it violated Australia's Privacy Act by obtaining personal and sensitive information without consent or notification, by unfair means, and without even ensuring that the information was accurate. Following that finding, the government ordered Clearview to stop collecting and to remove existing photos taken in Australia. In France the Commission Nationale de l'Informatique et des Libertés (CNIL) also ordered the company to cease collecting, processing, and storing facial data. That case may be one reason Facebook announced that it would abandon its facial-recognition system and delete the face-scan data of more than one billion users.

Even when the reasons for collecting data are transparent, the methods used to gather it may be unethical, as the following composite example, drawn from our research, illustrates. A

recruitment firm with a commitment to promoting diversity and inclusion in the workforce found that job candidates posting on its platform suspected that they were being discriminated against on the basis of their demographic profiles. The firm wanted to reassure them that the algorithms matching job openings with candidates were skill-based and demographically neutral and that any discrimination was occurring at the hiring companies, not on the platform.

The firm approached a well-known business school and identified a professor who was willing to conduct research to test for possible discrimination by the recruiting companies. The researcher proposed replicating a study conducted a few years earlier that had created several standard résumés but varied the race and gender of the applicants. Thousands of bogus job applications would be sent to companies in the area and the responses tracked and analyzed. If any active discrimination was at play, the results would show differing acceptance rates based on the embedded demographic variables.

The firm's marketing and sales managers liked the proposal and offered a contract. Because the business school required an ethics evaluation, the proposal was submitted to its IRB, which rejected it on the grounds that the professor proposed to collect data from companies by subterfuge. He would be lying to potential corporate users of the platform and asking them to work for the school's client without their knowledge and without any benefit to them. (In fact, the companies might

suffer from participating if they could be identified as using discriminatory hiring processes.)

The lesson from this story is that good intentions are not enough to make data collection ethical.

Companies should consider the provenance not only of data they plan to obtain but also of data they already own. Many of them routinely collect so-called dark data that is rarely used, often forgotten, and sometimes even unknown. Examples include ignored or unshared customer data, visitor logs, photos, presentation documents that are filed away but uncataloged, emails, customer service reports or recorded transcripts, machine-generated usage or maintenance logs, and social media reactions to corporate posts. Although this data is often unstructured and therefore difficult to integrate, its potential value is enormous, so many software developers are creating products to help companies find and use their dark data. This brings us to the second P.

2. Purpose

In a corporate context, data collected for a specific purpose with the consent of human subjects is often used subsequently for some other purpose not communicated to the providers. In reviewing the exploitation of existing data, therefore, a company must establish whether additional consent is required.

For example, one large bank in France wanted to test the hypothesis that bullying or sexual harassment of peers and subordinates might be identified by examining corporate emails. The diversity manager in the HR department believed that spotting potential harassment early would allow the company to intervene in a timely manner and perhaps even entirely avoid a harassment situation by training people to watch for warning signs.

The bank launched a trial study and found strong evidence that email communications could forecast later harassment. Despite that finding, an ad hoc review of the results by several senior managers led the company to shelve the project because, as the managers pointed out, the data being collected—namely, emails—was originally designed to communicate work-related information. The people who had sent them would not have seen predicting or detecting illegal activity as their purpose.

When it comes to customer data, companies have typically been much less scrupulous. Many view it as a source of revenue and sell it to third parties or commercial address brokers. But attitudes against that are hardening. In 2019 the Austrian government fined the Austrian postal service €18 million for selling the names, addresses, ages, and political affiliations (where available) of its clients. The national regulatory agency found that postal data collected for one purpose (delivering letters and parcels) was being inappropriately repurposed for marketing to clients that could combine it with easily obtainable

public data (such as estimates of home value, homeownership rates, residential density, number of rental units, and reports of street crime) to find potential customers. Among the buyers of the data were political parties attempting to influence potential voters. The fine was overturned on appeal, but the murkiness of reusing (or misusing) customer data remains an important problem for companies and governments.

Most companies use their client databases to sell their customers other services, but that can bring them trouble as well. In 2021 the Information Commissioners Office, an independent UK authority promoting data privacy, accused Virgin Media of violating its customers' privacy rights. Virgin Media had sent 1,964,562 emails announcing that it was freezing its subscription prices. That was reasonable enough, but Virgin had also used the emails to market to those customers. Because 450,000 subscribers on the list had opted out of receiving marketing pitches, the regulator imposed a fine of £50,000 on Virgin for violating that agreement.

The possibility that company databases could be repurposed without the data providers' consent brings us to the third P.

3. Protection

According to the Identity Theft Resource Center, nearly 2,000 data breaches occurred in the United States in 2021. Even the biggest, most sophisticated tech companies have had

tremendous breaches, with the personal details of more than several billion individuals exposed. The situation in Europe, despite some of the most protective laws in the world, is not much better. Virgin Media left the personal details of 900,000 subscribers unsecured and accessible on its servers for 10 months because of a configuration error—and at least one unauthorized person accessed those files during that period.

The common practice of lodging data with expert third parties doesn't necessarily offer better protection. Doctolib, a French medical appointments app, was taken to court because it stored data on Amazon Web Services, where it could conceivably be accessed by Amazon and many other organizations, including U.S. intelligence agencies. Although the data was encrypted, it arrived at Amazon's server without anonymization, meaning that it could be linked to digital records of online behavior to develop very accurate personal profiles for commercial or political purposes.

An institutional review board needs clarity on where the company's data will reside, who may have access to it, whether (and when) it will be anonymized, and when it will be destroyed. Thus many companies will have to change their existing protocols and arrangements, which could prove expensive: Since a 2014 data breach at JPMorgan Chase compromised 76 million people and 7 million businesses, the bank has had to spend $250 million annually on data protection.

The fourth P is closely related to protection.

4. Privacy

The conundrum that many companies face is making the trade-off between too little and too much anonymization. Too little is unacceptable under most government regulations without informed consent from the individuals involved. Too much may make the data useless for marketing purposes.

Many techniques for anonymization exist. They range from simply aggregating the data (so that only summaries or averages are available), to approximating it (for example, using an age range rather than a person's exact age), to making variable values slightly different (by, for example, adding the same small value to each), to pseudonymizing the data so that a random, nonrepeating value replaces the identifying variable.

In principle these techniques should protect an individual's identity. But researchers have been able to identify people in a data set using as little as their gender, birth date, and postal code. Even less specific information, when combined with other data sets, can be used to identify individuals. Netflix published a data set that included 100 million records of its customers' movie ratings and offered $1 million to any data scientist who could create a better movie-recommendation algorithm for the company. The data contained no direct identifiers of its customers and included only a sample of each customer's ratings. Researchers were able to identify 84% of the individuals by comparing their ratings and rating dates with a third-party

data set published by IMDb, another platform on which many Netflix customers also post film ratings. In evaluating the privacy issues around human data, therefore, corporate IRBs must at the very least assess how effective a firewall anonymization will be, especially given the power of data analytics to break through anonymity. A technique called *differential privacy* may afford an added level of protection. Software offered by Sarus, a Y Combinator–funded startup, applies this technique, which blocks algorithms built to publish aggregated data from disclosing information about a specific record, thereby reducing the chances that data will leak as a result of compromised credentials, rogue employees, or human error.

But privacy can be violated even with effectively anonymized data because of the way in which the data is collected and processed. An unintended violation occurred at the mapping company MaxMind, which provides geolocation services that enable businesses to draw customers' attention to nearby products and services. Geolocation also aids internet searches and can help if a service that needs your IP address (such as an entertainment streaming site) isn't working correctly. But precise mapping permits anyone who has your IP address to find your neighborhood and even your home. Combining your address with Zillow or some other real estate database can provide information about your wealth along with photos of your home inside and out.

Unfortunately, IP mapping is not an exact science, and it can be difficult to precisely link an IP address to a physical address. A mapper might assign it to the nearest building or simply to a locality, such as a state, using that locality's central coordinates as the specific address. That may sound reasonable, but the consequences for one family renting a remote farmhouse in Potwin, Kansas, were horrific.

The family's IP address was listed with the map coordinates of the farmhouse, which happened to match the coordinates of the exact center of the United States. The problem was that MaxMind assigned more than 600 million other IP addresses that could not be mapped by any other means to the same coordinates. That decision led to years of pain for the family in the farmhouse. According to Kashmir Hill, the journalist who broke the story, "They've been accused of being identity thieves, spammers, scammers and fraudsters. They've gotten visited by FBI agents, federal marshals, IRS collectors, ambulances searching for suicidal veterans, and police officers searching for runaway children. They've found people scrounging around in their barn. The renters have been doxxed, their names and addresses posted on the internet by vigilantes."

Hill contacted a cofounder of MaxMind, who eventually produced a long list of physical addresses that had many IP addresses assigned to them and confessed that when the company was launched, it had not occurred to his team that "people would use the database to attempt to locate people

down to a household level." He said, "We have always advertised the database as determining the location down to a city or zip code level." The takeaway is that well-intentioned, innocuous decisions made by data scientists and database managers can have a real, very negative impact on the privacy of innocent third parties. That brings us to the fifth P.

5. Preparation

How is the data prepared for analysis? How is its accuracy verified or corrected? How are incomplete data sets and missing variables managed? Missing, erroneous, and outlying data can significantly affect the quality of the statistical analysis. But data quality is often poor. Experian, a credit services firm, reports that on average, its U.S. clients believe that 27% of their revenue is wasted owing to inaccurate and incomplete customer or prospect data.

Cleaning data, especially when it is collected from different periods, business units, or countries, can be especially challenging. In one instance we approached a large international online talent-management and learning company to help us research whether women and men equally obtained the career benefits of training. The company agreed that the question was relevant for both its customers and the public at large, and therefore extracted the data it had on its servers. To ensure privacy the data was anonymized so that neither individual

employees nor their employers could be identified. Because of the size of the data set and its internal structure, four individual data sets were extracted.

Normally we would just open the databases and find a spreadsheet file showing the features characterizing each individual, such as gender. A woman might be identified as "woman" or "female" or simply "F." The values might be misspelled ("feale"), appear in various languages (*mujer* or *frau*), or use different cases (f or F). If the spreadsheet is small (say, 1,000 rows), correcting such inconsistencies should be simple. But our data contained more than one billion observations—too many, obviously, for a typical spreadsheet—so a cleaning procedure had to be programmed and tested.

One major challenge was ascertaining how many values had been used to identify the variables. Because the data came from the foreign subsidiaries of multinational firms, it had been recorded in multiple languages, meaning that several variables had large numbers of values—94 for gender alone. We wrote programming code to standardize all those values, reducing gender, for instance, to three: female, male, and unknown. Employment start and end dates were especially problematic because of differing formats for dates.

According to Tableau, a data analytics platform, cleaning data has five basic steps: (1) Remove duplicate or irrelevant observations; (2) fix structural errors (such as the use of variable values); (3) remove unwanted outliers; (4) manage missing data,

perhaps by replacing each missing value with an average for the data set; and (5) validate and question the data and analytical results. Do the numbers look reasonable?

They may well not. One of our data sets, which recorded the number of steps HEC Paris MBA students took each day, contained a big surprise. On average, students took about 7,500 steps a day, but a few outliers took more than one million steps a day. Those outliers were the result of a data processing software error and were deleted. Obviously, if we had not physically and statistically examined the data set, our final analysis would have been totally erroneous.

How AI Raises the Stakes

Ethics can seem an expensive luxury for companies with strong competitors. For example, Microsoft reportedly fired the entire ethics team for its Bing AI project because, according to press and blog reports, Google was close to releasing its own AI-powered application, so time was of the essence.

But treating data ethics as a nice-to-have carries risks when it comes to AI. During a recent interview the CTO of OpenAI, the company that developed ChatGPT, observed, “There are massive potential negative consequences whenever you build something so powerful with which so much good can come . . . and that’s why . . . we’re trying to figure out how to deploy these systems responsibly.”

Thanks to AI, data scientists can develop remarkably accurate psychological and personal profiles of people on the basis of very few bits of digital detritus left behind by social-platform visits. The researchers Michal Kosinski, David Stillwell, and Thore Graepel of the University of Cambridge demonstrated the ease with which Facebook likes can accurately "predict a range of highly sensitive personal attributes including: sexual orientation, ethnicity, religious and political views, personality traits, intelligence, happiness, use of addictive substances, parental separation, age, and gender." (This research was, in fact, the inspiration for Cambridge Analytica's use of Facebook data.)

Subsequent research by Youyou Wu, Michal Kosinski, and David Stillwell reinforced those findings by demonstrating that computer-based personality judgments can be more accurate than human ones. Computer predictions of personality characteristics (openness, agreeableness, extraversion, conscientiousness, neuroticism—known as the Big Five) using Facebook likes were nearly as accurate as assessments by an individual's spouse. The implications of that should not be ignored. How would you feel if your government wanted to catalog your private thoughts and actions?

A problem may also be rooted not in the data analyzed but in the data overlooked. Machines can "learn" only from what they are fed; they cannot identify variables they're not programmed to observe. This is known as *omitted-variable bias*. The best-

known example is Target's development of an algorithm to identify pregnant customers.

The company's data scientist, a statistician named Andrew Pole, created a "pregnancy prediction" score based on purchases of about 25 products, such as unscented lotions and calcium supplements. That enabled Target to promote products before its competitors did in the hope of winning loyal customers who would buy all their baby-related products at Target. The omitted variable was the age of the target customer, and the accident-in-waiting occurred when the father of a 17-year-old found pregnancy-related advertisements in his mailbox. Unaware that his daughter was pregnant, he contacted Target to ask why it was promoting premarital sex to minors.

Even by the standards of the era, spying on minors with the goal of identifying personal, intimate medical information was considered unethical. Pole admitted during a subsequent interview that he'd thought receiving a promotional catalog was going to make some people uncomfortable. But whatever concerns he may have expressed at the time did little to delay the rollout of the program, and according to a reporter, he got a promotion. Target eventually released a statement claiming that it complied "with all federal and state laws, including those related to protected health information."

The issue for boards and top management is that using AI to hook customers, determine suitability for a job interview, or approve a loan application can have disastrous effects. AI's

predictions of human behavior may be extremely accurate but inappropriately contextualized. They may also lead to glaring mispredictions that are just plain silly or even morally repugnant. Relying on automated statistical tools to make decisions is a bad idea. Board members and senior executives should view a corporate institutional review board not as an expense, a constraint, or a social obligation but as an early-warning system.

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Where Data-Driven Decision-Making Can Go Wrong

by Michael Luca and Amy C. Edmondson

LET'S SAY YOU'RE LEADING a meeting about the hourly pay of your company's warehouse employees. For several years it has automatically been increased by small amounts to keep up with inflation. Citing a study of a large company that found that higher pay improved productivity so much that it boosted profits, someone on your team advocates for a different approach: a substantial raise of $2 an hour for all workers in the warehouse. What would you do?

Too often business leaders go in one of two directions in these moments: either taking the evidence presented as gospel or dismissing it altogether. Both approaches are misguided. Leaders instead should organize discussions that thoughtfully evaluate seemingly relevant evidence and its applicability to a given situation.

In the scenario just described you should pose a series of questions aimed at assessing the potential impact of wage

increases on your company specifically. You might ask:

- Can you tell us more about the setting of the research to help us evaluate whether it applies to our warehouse employees?
- How do our wages stack up against those of other employers competing for our workers, and how does that compare with the study?
- Was an experiment conducted? If not, what approach was used to understand whether higher wages were driving the productivity change or simply reflecting it?
- What measures of productivity were used, and how long were the effects measured?
- What other analyses or data might be relevant?

Of course, tone matters. These questions must be asked in a genuine spirit of curiosity, with a desire to learn and get sound recommendations.

Whether evidence comes from an outside study or internal data, walking through it thoroughly before making major decisions is crucial. In our interactions with companies—including data-heavy tech firms—we've noticed that this practice isn't consistently followed. Too often predetermined beliefs, problematic comparisons, and groupthink dominate discussions. Research from psychology and economics suggests that biases—such as *base rate neglect*, the tendency to overlook general statistical information in favor of specific case information or

anecdotes, and *confirmation bias*, the propensity to seek out and overweight results that support your existing beliefs—also hinder the systematic weighing of evidence. But companies don't have to fall into this pattern. Drawing on our research, work with companies, and teaching experience (including executive education classes in leadership and business analytics and a recent MBA course called Data-Driven Leadership), we have developed an approach general managers can apply to discussions of data so that they can make better decisions.

Pressure-Test the Link Between Cause and Effect

Will search engine advertisements increase sales? Will allowing employees to work remotely reduce turnover? These questions are about cause and effect—and are the kind of questions that data analytics can help answer. In fact, research papers have looked at them in detail. However, managers frequently misinterpret how the findings of those and other studies apply to their own business situation. When making decisions, managers should consider *internal validity*—whether an analysis accurately answers a question in the context in which it was studied. They should also consider *external validity*—the extent to which they can generalize results from one context to another. That will help them avoid making five common mistakes:

Idea in Brief

The Problem

When managers are presented with internal data or an external study, all too often they either automatically accept its accuracy and relevance to their business or dismiss it out of hand.

Why It Happens

Leaders mistakenly conflate causation with correlation, underestimate the importance of sample size, focus on the wrong outcomes, misjudge generalizability, or over-weight a specific result.

The Right Approach

Leaders should ask probing questions about the evidence in a rigorous discussion about its usefulness. They should create a psychologically safe environment so that participants will feel comfortable offering diverse points of view.

Conflating causation with correlation

Even though most people know that correlation doesn't equal causation, this error is surprisingly prevalent. Take eBay's advertising strategy. For years the company had advertised on search engines such as Google, looking to grow demand by attracting more customers. A consulting report concluded that the ads were effective, noting that when more ads were shown in a market, the total value of purchases on eBay was higher. Alas, it had reached the wrong conclusion about those ads. With the help of an experiment conducted by a team of economists led by Steven Tadelis of the University of California Berkeley, eBay realized that the correlation was explained by advertisements targeting people already likely to visit eBay and markets where demand for eBay would be expected to spike even without ads.

How to avoid predictable errors

Pressure-testing assumptions, especially before difficult-to-reverse decisions are made, is increasingly vital. Here are five common pitfalls leaders make in interpreting analyses, along with the questions that will help you steer clear of them.

To prevent this problem	Ask about this	Sample questions
Conflating correlation and causation	Approach to determining causality	Was this analysis based on an experiment? If not, are there confounders (variables that affect the independent and dependent variables)? To what extent were they addressed in the analysis?
Misjudging the potential magnitude of effects	Sample size and the precision of the results	What was the average effect of the change? What was the sample size and the confidence interval (or range of likely values the true effect would fall into, and the degree to which one is certain it would fall into that range)? How would our course of action change, depending on where the true effect might lie?
A disconnect between what is measured and what matters	Outcome measures	What outcomes were measured? Were they broad enough? Did they capture key intended and unintended consequences? Were they tracked for an appropriate period of time? Were all relevant outcomes reported? How do we think they map to broader organizational goals?
Misjudging generalizability	Empirical setting and subgroup analysis	How similar is the setting of this study to our business context? Does the context or time period of the analysis make it more or less relevant to our decision? What is the composition of the sample being studied, and how does it influence the applicability of the results? Does the effect vary across subgroups or settings? Does this tell us anything about the generalizability of the results?
Overweighting a specific result	Broader evidence and further data collection	Are there other analyses that validate the results and approach? What additional data could we collect, and would the benefit of gathering it outweigh the cost of collecting it? How might this change our interpretation of the results?

To understand causality, delve into how the study in question was conducted. For instance, was it a *randomized controlled trial*, in which the researchers randomly assigned people to two groups: one that was subjected to a test condition and a control group that was not? That's often considered the gold standard for

assessing cause and effect, though such experiments aren't always feasible or practical. Perhaps the researchers relied on a *natural experiment*, observing the effects of an event or a policy change on specific groups. For example, a study might examine the impact of a benefit whose recipients were chosen by lottery, which allows researchers to compare how the benefit changed the circumstances or behavior of those who won the lottery with that of those who didn't win.

Researchers who don't have access to planned or natural experiments may instead control for potential confounding factors—variables that affect the variable of interest—in their data analysis, though this can be challenging in practice. For instance, if you were assessing the impact of a training program on productivity, you'd want to make sure you controlled for prior experience and other things that might affect productivity.

Underestimating the importance of sample size

Imagine two hospitals: a large one that handles thousands of births each year, and a small one with a few hundred births annually. Which hospital do you think would have more days where more than 60% of the babies born were boys?

The answer is the small hospital because it has more variability in daily birth numbers. Small sample sizes are more likely to show greater fluctuations. Psychologists Daniel Kahneman and Amos Tversky, in their canonical work on biases and heuristics, found that most people got the answer wrong, with more than half saying, "About the same." People tend to underappreciate the

effect that sample size has on the precision of an estimate. This common error can lead to bad decisions. Whether you're trying to figure out how much to trust online reviews, how to interpret productivity trends, or how much weight to put on the results of an advertising experiment, the size of the sample being analyzed is important to consider.

When evaluating effects, it can be helpful to ask not only about the sample size but about the *confidence interval*. A confidence interval provides a range of values that the true effect is likely to fall into, and the degree to which one is certain it falls into that range. The answers should shape the conversation about which course of action you'll take.

Focusing on the wrong outcomes

In their classic 1992 HBR article "The Balanced Scorecard—Measures That Drive Performance," Robert S. Kaplan and David P. Norton opened with a simple observation: "What you measure is what you get." Although their article predates the era of modern analytics, that idea is more apt than ever. Experiments and predictive analytics often focus on outcomes that are easy to measure rather than on those that business leaders truly care about but are difficult or impractical to ascertain. As a result, outcome metrics often don't fully capture broader performance in company operations.

Let's return to the example of wage increases. Costs are easily measured, while boosts in productivity can be difficult to quantify. That can lead managers to focus narrowly on the cost of

better pay and fail to appreciate the potential gains. A broader analysis would take an approach like the one seen in a study by economists Natalia Emanuel and Emma Harrington. They set out to understand the implications of warehouse pay levels set by a large online retailer. The researchers examined changes in productivity after a 2019 pay increase for warehouse workers and found that improvements in productivity and turnover were so large that the wage increases more than paid for themselves. They found similar results when they looked at the effects of higher pay on the productivity and turnover of customer service employees.

It's also important to make sure that the outcome being studied is a good proxy for the actual organizational goal in question. Some company experiments track results for just a few days and assume that they're robust evidence of what the longer-term effect would be. With certain questions and contexts, a short time frame may not be sufficient. One company that works to avoid this problem is Amazon: It invests heavily in exploring the longer-term costs and benefits of possible product changes. There are many ways to assess the relevance and interpretation of outcomes, ranging from clear discussions about limitations to formal analyses of the link between short-term effects and longer-term ones.

To really learn from any data set, you need to ask basic questions like, What outcomes were measured, and did we include all that are relevant to the decision we have to make?

Were they broad enough to capture key intended and unintended consequences? Were they tracked for an appropriate period of time?

Misjudging generalizability

With the example of the warehouse wage increase, a vital question is what the results from one set of warehouses imply for a different set. Moreover, a company may wish to know how the results apply to, say, restaurant or retail employees.

We have seen business leaders make missteps in both directions, either over- or underestimating the generalizability of findings. For instance, when the senior vice president of engineering at a major tech company told us about his company's rule against looking at university grades in engineer hiring decisions, we asked about the rationale. He said that Google had "proved that grades don't matter"—referring to a Google executive's comment he had read somewhere, claiming there wasn't a relationship between school grades and career outcomes. By taking that piece of information as gospel, he ignored potential limitations to both its internal and its external validity.

When you're assessing generalizability, it can be helpful to discuss the mechanisms that might explain the results and whether they apply in other contexts. You might ask things like, How similar is the setting of this study to that of our business? Does the context or period of the analysis make it more or less relevant to our decision? What is the composition of the sample

being studied, and how does it influence the applicability of the results? Does the effect vary across subgroups?

Overweighting a specific result

Relying on a single empirical finding without a systematic discussion of it can be just as unwise as dismissing the evidence as irrelevant to your situation. It's worth checking for additional research on the subject. Conducting an experiment or further analysis with your own organization can be another good option. Questions to ask include, Are there other analyses that validate the results and the approach? What additional data might we collect, and would the benefit of gathering more evidence outweigh the cost of that effort?

Start by Speaking Up

In 1906, Sir Francis Galton famously analyzed data on a contest at a livestock fair in which people guessed the weight of an ox. Though individual guesses were all over the map, the average of the guesses was nearly spot-on—demonstrating the *wisdom of the crowd*. Harnessing that wisdom can be challenging, however. Collective intelligence is best when mechanisms are in place to promote active and diverse participation. Otherwise, crowds can also amplify bias—especially when they're homogeneous in viewpoint.

To overcome bias, business leaders can invite contributors with diverse perspectives to a conversation, ask them to challenge and

build on ideas, and ensure that discussions are probing and draw on high-quality data. (See "What You Don't Know About Making Decisions," by David A. Garvin and Michael Roberto, HBR, September 2001.) Encouraging dissent and constructive criticism can help combat groupthink, make it easier to anticipate unintended consequences, and help teams avoid giving too much weight to leaders' opinions. Leaders also must push people to consider the impact of decisions on various stakeholders and deliberately break out of siloed perspectives.

These kinds of discussions can help ensure the thoughtful weighing of evidence. But all too often they get derailed even when they would be productive. Countless studies have shown that hierarchies can lead people to withhold dissenting views and that discussion participants tend to shy away from sharing potentially relevant data or asking probing questions when they don't experience *psychological safety*—the belief that candor is expected and won't be punished. Without psychological safety, the approach we've described is less likely to work.

Teams benefit when their members feel that offering up data, ideas, concerns, and alternative views will be valued by their peers and managers alike. Most important, in many discussions, participants should view asking probing questions as part of their job.

Much has been written about how to build psychological safety in a team. (See "Why Employees Are Afraid to Speak," by James R. Detert and Amy C. Edmondson, HBR, May 2007.) But it's

especially critical to establish it in a team that seeks to use evidence to make business decisions—so that the fear of raising unpopular findings doesn't cause members to miss critical data.

The chilling effect of low psychological safety was evident in the response to experimental research at Facebook that looked at whether showing more positive versus negative posts affected users' emotions. In 2014, in the aftermath of public backlash to the research—which arose partly because people didn't know that Facebook was running these types of experiments—CEO Mark Zuckerberg pulled the plug on ongoing external-facing research projects. That deterred employees from undertaking experiments that might explore Facebook's social impact proactively. More recently, Zuckerberg has changed course and expressed renewed interest in external research. However, had he created an atmosphere where Facebook executives felt able to thoughtfully discuss the negative effects of social media a decade ago, the company might have avoided some of its recent reputational challenges related to misinformation and its effects on user well-being.

From Data to Decisions

Decision-making in the face of uncertainty is necessarily iterative; it requires regular pauses for reflection on both information and process. Effective teams will learn from data, adjust plans accordingly, and deliberately work on improving their discussions.

Taking the time to discuss the nuances of analyses—including sample size and composition, the outcomes being measured, the approach to separating causation from correlation, and the extent to which results might generalize from one setting to another—is vital to understanding how evidence can, or can't, inform a specific decision. When carefully considered, each empirical result presents a piece of a puzzle, helping businesses figure out whether and when different changes are likely to have an effect. Such discussions will also set the stage for organizations to be more rigorous about data collection.

Even in the best of worlds, evidence is rarely definitive, and how a business move will play out is uncertain. You can nonetheless aspire to make thoughtful choices based on information you have or might obtain. By employing a systematic approach to its collection, analysis, and interpretation, you can more effectively reap the benefits of the ever-increasing mountain of internal and external data and make better decisions.

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Sizing Up Your Cyberrisks

by Thomas J. Parenty and Jack J. Domet

OVER THE PAST DECADE the costs and consequences of cyberbreaches have grown alarmingly. The total financial and economic losses from the 2017 WannaCry attack, for instance, were estimated to reach $8 billion. In 2018 Marriott discovered that a breach of its Starwood subsidiary's reservation system had potentially exposed the personal and credit-card information of 500 million guests. Hackers seem to keep getting more effective. But in our experience as consultants to clients across the globe, we've found another reason that companies are so susceptible to threats from hacking: They don't know or understand their critical cyberrisks, because they're too focused on their technological vulnerabilities.

When cybersecurity efforts address only technology, the result is company leaders who are poorly informed and organizations that are poorly protected. Discussions of cyberthreats end up being filled with specialized tech jargon, and senior executives can't participate meaningfully in them. The responsibility for addressing risks then gets relegated entirely to cybersecurity and

IT staff, whose attention falls mainly on corporate computer systems. The outcome tends to be a long, ill-prioritized list of mitigation tasks. Since no company has the resources to fix every cybersecurity problem, important threats can go unaddressed.

A more fruitful approach is to adopt the view that cybersecurity should focus more on threats' potential impact on a business's *activities*. Say you're an executive at a chemical company. Instead of asking what cyberattacks might be possible on your computer systems, ask, How could a cyberattack disrupt your supply chain? Or expose your trade secrets? Or make you fail to meet your contractual obligations? Or cause a threat to humanity? That adjustment might seem minor, but when leaders start with crucial activities, they can better prioritize the development of cyberdefenses.

A CEO we worked with, Richard Lancaster of CLP, Asia's third-largest electricity provider, described the shift in mindset this way: "Initially, we viewed cyberrisks primarily as an IT issue. Over time we realized that what was really vulnerable was our electric grid and generating plants. Now we recognize that cyberrisk is really business risk—and my job as CEO is to manage business risk." With this perspective, responsibility shifts from IT to senior executives and boards, who must take an active role and ensure that cybersecurity teams focus on the right threats.

Developing Cyberthreat Narratives

Identifying and fixing cyberrisks is a social process. To accurately assess where the most important ones lie, you must consider the viewpoints and opinions of a wide range of employees. By involving a broad group, you'll build a common understanding of the critical facts and details early on, which will enable you to reach consensus when you subsequently need to manage the risks.

To help companies organize and share the relevant information with a wide audience, we've developed a tool we call a *cyberthreat narrative*. It addresses the four parts of the story of a potential cyberattack: a key business activity and the risks to it; the systems that support that activity; the potential types of attacks and possible consequences; and the adversaries most likely to carry attacks out. Outlining details about all four will help companies recognize and prioritize their risks and prepare remedial actions.

The people in your cybersecurity group should take charge of developing cybernarratives, but they should solicit contributions from:

Idea in Brief

The Challenge

Despite the billions spent on cybersecurity, the damage done by breaches keeps growing—to a large extent because companies don't recognize or understand their critical cyberrisks.

The Old Approach

Too many firms focus only on technological vulnerabilities. Responsibility for cybersecurity then defaults to IT specialists, yielding an ill-prioritized list of possible attacks. Tech jargon dominates discussions of risk, and senior leaders and boards can't participate meaningfully in them.

A Better Way

A more fruitful approach is to identify your critical business activities, the risks to them, the systems supporting them, those systems' vulnerabilities, and potential attackers. Leaders and staff throughout the firm can participate in this process, and overall responsibility for cybersecurity shifts to senior executives and boards.

- **Leadership.** The CEO, the executive team, and other senior executives. Meetings with executive leaders are critical, but they don't need to be time-consuming; scripting the interviews and discussions carefully will make them more efficient and easier to document.
- **Operations.** Personnel involved day to day in the central business activities.
- **IT systems.** People responsible for the administration of the computing systems supporting the activities.
- **Relevant specialists.** Staffers with expertise related to the type and consequences of the particular kind of threat you're outlining, such as legal, public relations, human resources, and physical security employees. For example, if a narrative is about an attack that results in a loss of personal data, you'd want to include the legal team in your

discussions because of the possibility of a regulatory violation.

Let's look now at each element of a cyberthreat narrative, how to develop it, and who should be involved.

Critical Business Activities and Risks

To identify these, the cybersecurity team should interview the company's leaders; examine its written statements of risk tolerance, such as those found in annual reports; and consider company objectives, such as revenue targets or growth in new markets. A revenue goal, for example, could be dependent on new-product development or expanded service offerings. Entering a new country might be essential to increasing the customer base. Critical activities can be outside the organization, relate to internal operations, or involve the company's strategic future. For the chemical company, a critical activity could be, say, the manufacture of polyester resins, a specialty product in high demand.

How important an activity is will vary by industry and company. Customer support is a low-risk activity in some industries, such as consumer software or discount retail. But in others, like the gaming industry, customer relationships are paramount. Casinos in Macao, for example, rely on a small segment of VIP customers for more than 54% of their combined

gross gaming revenue. Risks to customer relationship management also threaten the casinos' bottom lines.

The number of critical business activities a company has, and therefore the number of cyberthreat narratives it should develop, also vary from company to company.

To size up your organization's risks, think about how each key activity could fail in a way that damages your company. For example, in the case of the chemical maker, a disruption to its plant operations could prevent it from producing resins, which could reduce its revenue. Also examine the risk of collateral damage to your customers or other stakeholders—a release of poisonous chemicals into the environment, for example, or the loss of confidential customer information such as passwords and credit-card data.

Bear in mind that what poses a significant risk for one company may not for another. Although the disruption of resin production might be harmful to one chemical firm, it may not be for another chemical manufacturer whose resins aren't in demand, make only a negligible contribution to profits, or can be produced at alternative plants.

Cybersecurity Is a Board Responsibility

BECAUSE BOARDS REPRESENT THE fiduciary interests of the company's owners and are charged with adopting a long-term view of the company, we believe that they have the authority and responsibility to oversee efforts to identify cyberrisks. When they do, it makes a

material difference: Simply by raising questions about the four elements of cyberthreats, directors can prompt companies to pay more attention to critical risks.

For example, if you're a board member, ask for assurance that the company has identified and documented its most critical business activities, the benefits they provide, and the most significant risks they face. You should also confirm that the company's leaders have participated in this process.

Similarly, you will want assurance that the company has up-to-date inventories of the computer systems those business activities rely on. Though you don't need to review the details of the inventories yourself, at your discretion you may seek your own independent validation of them or instruct senior executives to arrange their own review. You should also confirm that the company has processes and tools for keeping the inventories current and ask for a few examples of updates that prove those processes are in use.

And so on down the line: Directors should press company leaders about whether they understand the types of attacks that could threaten critical business activities, what the possible impact to both the company and its stakeholders might be, and who potential cyberadversaries are and what capabilities and motivations they might have. The company should brief the board regularly on its current cyberrisk posture for each critical business activity.

Supporting Systems

Your company can't mount an effective cyberdefense if it doesn't know what it needs to protect. So you have to catalog your computer systems and the services and functionality they provide for each activity in question. This process should be kick-started by the operational employees involved in the activity, because they know what software they use and what the

consequences will be if those programs malfunction. The employees who maintain the computer systems should also participate, since they have a broader picture of the technology supporting this software. For general-purpose computers, this typically means IT staff; for industrial control systems, it means engineers. The inventory should note the physical locations of the systems so that cyberincident-response staffers know where to go to fix things in the event of an attack.

While there are products that assist IT departments with automated inventorying of computers and software, they can't identify which assets are most important. By cataloging them on the basis of business activity specifically, a company can prioritize computer-vulnerability remediation and enhanced protections effectively.

Cyberattack Types and Consequences

Next the team should outline all the types of attacks that could disrupt each critical activity, describing what would be required for attacks to succeed and what the potential consequences might be.

At the most basic level, cyberattacks exploit vulnerabilities in computer systems. Malware attacks, for instance, use malicious software to take advantage of programming mistakes in applications. (This was the technique hackers used in the WannaCry incident.) Your cybersecurity staff can and should

identify the kinds of techniques that could target vulnerabilities in your crucial computer systems.

It's important to note that attacks aren't always sophisticated or technically complex. One vulnerability common to every computer system is an administrator's almost complete control over the information and applications on it. This power is necessary for proper operation and maintenance, but any administrator can abuse it.

A cyberattack can be executed in countless ways, and it isn't practical or useful to enumerate them all. If you identify basic types of attack, such as an external hack that installs malware or an employee who misuses computer privileges, that's enough.

Attack requirements

Understanding what an adversary needs to pull off a cyberattack is vital to building your defenses. Your cybersecurity group and the operational staff involved in the critical activities can identify the specific requirements, but most fall into one of these three types:

1. **Knowledge.** Information the adversary needs to have—for example, how to program malicious software or how hydroelectric dams operate.
2. **Tools and equipment.** What devices an adversary needs—not only hacking aids like password crackers and network analyzers, but hardware such as laptops and radio transmitters.

3. **Position.** Where an adversary needs to be—for example, does he need to be physically next to a building or have employee or contractor status in the organization?

One Southeast Asian bank we worked with had previously suffered a cyberattack that resulted in a massive debit-card fraud. The investigation revealed the hackers had to have known the format of Visa and Mastercard authorization codes and how to configure a credit-card terminal. The tools they needed included several of the terminals and a database of debit-card account numbers. They also had to be in the local vicinity to coordinate with merchants who were complicit with the fraud, but they didn't necessarily have to work at the bank or physically enter the bank building.

Attack consequences

Executive leadership and senior management are well positioned to identify the fallout from disruptions to the key business activities and should guide your cybersecurity group on this task. Operations and systems staff can point out additional consequences; specialists from other departments, such as legal, finance, and compliance, can spot potential collateral damage. A simple set of "What if . . .?" questions will make conversations with all these players fruitful. For example, What would happen to the provision of care if a hospital's patient records were no longer accessible because of a ransomware attack? For England's National Health Service in the aftermath of WannaCry, the answer

was the cancellation of thousands of appointments and operations.

Some consequences go beyond direct financial costs. The NotPetya cyberattack of 2017 interrupted operations at numerous large companies worldwide, causing total estimated losses of up to $300 million at AP Moller-Maersk and $400 million at FedEx. Pharmaceutical giant Merck estimated NotPetya's impact to its business at $870 million because of both direct costs and lost revenue. In addition, the shutdown resulted in low inventories of a Merck vaccine that prevents certain types of cancer.

Cyberadversaries

Who's out to get you? Identifying potential perpetrators, as well as their motivations and capabilities, will help you assess the likelihood of an attack and develop the controls needed to thwart it. Your adversaries could be countries, criminal organizations, competitors, disgruntled employees, terrorists, or advocacy groups. Don't underestimate their sophistication: Advanced hacking tools are widely available to many.

Company leaders and the operations staffers involved in critical business activities are best at identifying possible adversaries, because they're most familiar with what might motivate attackers and what they could gain. A good place to start is to ask what your company has that could be of value to someone else. For example, a competitor could be interested in your R&D and trade secrets, whereas a criminal organization would be more

interested in stealing customer financial records to sell on the black market.

Companies also must consider potential adversaries' larger commercial context. Casino managers in Macao had decided not to encrypt the network connections they used to transmit their VIP client data to a centralized operations center. The need to do so had not been identified. But when we asked the casino managers to think through who might gain from an attack, they realized that the telecommunications network itself was owned by a conglomerate that included the casino's largest competitor.

Even customers could be cyberadversaries. Executives at AMSC, a developer of software for controlling wind turbines, were stunned when Sinovel, one of its largest customers, suddenly canceled all payments on current and future contracts, which were worth some $800 million. An investigation revealed that Sinovel had managed to steal AMSC's software and deploy it with more than a thousand new turbines. As a result of this intellectual property theft, AMSC reported a loss of more than $186 million in fiscal year 2010. The company's total losses from the theft came to $550 million—only a fraction of which has been recovered. The scheme also cost AMSC's shareholders $1 billion in equity and forced the organization to shed 700 jobs—over half its global workforce. The company's operations haven't yet returned AMSC to profitability.

Sometimes a company's industry or the way a firm conducts business provokes an attack. Environmental groups might target

companies with a bad record of polluting. Edward Snowden, a former NSA contractor, stole information from the agency to expose surveillance programs that it had publicly denied. Actions such as layoffs or plant closures can also motivate employees to retaliate by misusing their computer privileges. And there will always be random individuals with a personal agenda to pursue or a reputation to enhance.

Your company can suffer consequences from an attack even if it is not the direct target. Infrastructure, for example, is increasingly the object of cyberattacks. Consider the attacks on the power infrastructure in Ivano-Frankivsk in 2015. If, as suspected, Russia was behind the incidents, its motivations probably had nothing to do with the power companies themselves—or with their customers, who suffered as a result of interruptions to the power supply. The companies most likely were targeted simply because they're located in Ukraine, a country with which Russia has a hostile relationship. When evaluating who might want to disrupt your systems, you must look beyond your company to the broader commercial and political world in which it operates.

A Crisis That Might Have Been Prevented

Let's now look in depth at one cyberattack and how the four elements of the cyberthreat narrative could have captured relevant information that would have allowed the organization involved to avoid it.

The shire of Maroochy is a tourist destination about a hundred kilometers north of Brisbane, Australia. The area is one of outstanding natural beauty and ecological significance, with long white beaches and subtropical rain forests with creeks and waterfalls.

At the turn of the millennium the shire's water and sewage system was overseen by Maroochy Water Services, which collected, treated, and disposed of 35 million liters of wastewater daily. In late January 2000 the system that managed its wastewater pumping stations started losing control over the pumps and issuing false alarms. By the time a vendor concluded that the system's computers were under attack, raw sewage had backed out of the stations and flowed throughout the shire, including into the PGA championship golf course of what had previously been a five-star Hyatt Regency resort. In local parks, waterways reeked and turned black, and marine life died. The stench was horrible. The attacks went on for three months, until the police apprehended the perpetrator after a car chase near one of the pumping stations.

Hindsight makes it easy to see what went wrong, but let's reconstruct what a cyberthreat narrative for the utility might have been. (See the exhibit "Maroochy's cyberthreat narrative.")

Maroochy's cyberthreat narrative

In 2000 a cyberattack on the local water and sewer system wreaked havoc in Australia's Shire of Maroochy. If officials at the utility managing the

system had created a narrative outlining the four elements of a possible attack, they would have been better positioned to prevent it.

Element	Example
Critical business activity and risks	Wastewater treatment Pumping-station malfunction
Supporting systems	Centralized operations-management system Pumping-station control equipment
Cyberattack types and consequences	Exploitation of insecure network communications and lack of user authentication on pumping-station control equipment Massive release of raw sewage
Cyberadversary	Disgruntled insider

For the organization, treating wastewater was a critical business activity. Maroochy's systems had 142 stations pumping sewage to the treatment facility. Because of elevation variations in the shire, if the pumps malfunctioned, there was a good chance wastewater would back up into the area's pristine parklands and tourist areas.

The supporting computing systems for the pumps included a central operations-management system and the control equipment inside the pumping stations. From the central system, operators could turn individual pumping stations on or off and change the pumping rates. Pumping stations could also be managed locally using equipment that could manipulate the central system as well.

These supporting systems had two cybersecurity vulnerabilities. The first was that anyone could set up a network connection to the equipment; and the second was that once you were connected, a password wasn't required to log on.

To be successful, a potential attacker would need an understanding of how the equipment worked, which could be gained through experience or by reading the manuals, and the radio frequency used to communicate with the equipment, which was easy to find in company documentation. The attacker would need several computers (including one that was the same variety as those used in the pumping stations), network cables, and two-way radio equipment. And to connect to the control systems of an individual pumping station, he or she would need to be within radio range but would not need to physically break into a pumping station.

The cyberadversary in this case turned out to be a former employee of the vendor that provided the pumping-station control equipment. After a contentious stint at the vendor, he had applied twice for a job with Maroochy Water Services but wasn't hired. As a result, he became disgruntled and resentful and wanted revenge on both companies. He had stolen one of the pumping-station computers, and because he knew how the control system worked, he was able to use it and radio equipment to communicate with the individual pumping stations. He then manipulated them to take over the central operations-management system and wreak havoc.

If Maroochy Water Services executives had worked with their staff to develop a narrative for wastewater treatment, they would have discovered and understood the significant risks they faced. While we can assume that they would have recognized that

wastewater treatment was a critical activity, the investigation and analysis involved in creating a narrative would have given them and their IT security staff a clear idea of how cyberattacks could undermine supporting computer systems and cause the pumps and other wastewater treatment equipment to fail. And they would have had an idea of what it would take for cyberattacks to succeed and who might be behind them.

They also would have had a foundation for mitigating those risks. The Maroochy Water Services IT security staff members would have known the two vulnerabilities that needed to be addressed to prevent the wastewater crisis, and they could have explained them to executives and the board in one jargon-free sentence: “To prevent a cyberattack on our pumping stations, we need to require that people log on to station computers and restrict who can connect to their networks.”

Identifying cyberrisks is an ongoing process: As your business evolves, and as the computing systems that underlie it change, it will face new vulnerabilities. To spot these, your company must have well-defined checkpoints within its change management processes at which it evaluates cyberrisk.

But identifying your company’s most important cybervulnerabilities is only the first step. Knowing what the risks are will allow you to prioritize potential attacks, identify controls that will help prevent them, and create and, if needed, execute a

practical remediation plan. Good digital stewardship requires moving business risks—and business leaders—to the center of all these conversations.

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A Better Way to Put Your Data to Work

by Veeral Desai, Tim Fountaine, and Kayvaun Rowshankish

THOUGH EVERY COMPANY RECOGNIZES THE POWER OF DATA, most struggle to unlock its full potential. The problem is that data investments must deliver near-term value and at the same time lay the groundwork for rapidly developing future uses, while data technologies evolve in unpredictable ways, new types of data emerge, and the volume of data keeps rising.

The experiences of two global companies illustrate how ineffective today's predominant data strategies are at managing those challenges. The first, a large Asia-Pacific bank, took the "big bang" approach, assuming it could accommodate the needs of every analytics development team and data end user in one fell swoop. It launched a massive program to build pipelines to extract all the data in its systems, clean it, and aggregate it in a data lake in the cloud, without taking much time up front to align its efforts with business use cases. After spending nearly three

years to create a new platform, the bank found that only some users, such as those seeking raw historical data for ad hoc analysis, could easily use it. In addition, the critical architectural needs of many potential applications, such as real-time data feeds for personalized customer offerings, had been overlooked. As a result the program didn't generate much value for the firm.

The second company, a large North American bank, had individual teams tap into existing data sources and systems on their own and then piece together any additional technologies their business use cases required. The teams did create some value by solving challenges like improving customer segmentation for digital channels and enabling efficient risk reporting. But the overall result was a messy snarl of customized data pipelines that couldn't easily be repurposed. Every team had to start from scratch, which made digital transformation efforts painfully costly and slow.

So if neither a monolithic nor a grassroots data strategy works, what's the right approach?

We find that companies are most successful when they treat data like a product. When a firm develops a commercial product, it typically tries to create an offering that can address the needs of as many kinds of users as possible to maximize sales. Often that means developing a base product that can be customized for different users. Automakers do this by allowing customers to add a variety of special options—leather upholstery, tinted windows, anti-theft devices, and so on—to standard models. Likewise,

digital apps often let users customize their dashboards, including personalizing the layout, color schemes, and content displayed, or offer different plans and pricing structures for different user needs.

Over time companies enhance their products, adding new features (engine modifications that boost fuel economy in a car or new functionality in an app), and introduce brand-new offerings in response to user feedback, performance evaluations, and changes in the market. All the while firms seek to increase production efficiency. Wherever possible, they reuse existing processes, machinery, and components. (Automakers use a common chassis on vastly different cars, for instance, and app developers reuse blocks of code.) Treating data in much the same way helps companies balance delivering value with it today and paving the way for quickly getting more value out of it tomorrow.

In our work we've seen that companies that treat data like a product can reduce the time it takes to implement it in new use cases by as much as 90%, decrease their total ownership (technology, development, and maintenance) costs by up to 30%, and reduce their risk and data governance burden. In the pages that follow we'll describe what constitutes a data product and outline the best practices for building one.

Idea in Brief

The Problem

Although data offers enormous opportunities, most companies' strategies for realizing them are ineffective.

Why It Happens

Too often firms' data efforts fail to lay the foundations for future data uses. Individual teams create a customized data pipeline for each application that can't easily be repurposed.

The Solution

Create standard data products that can be tailored to suit the needs of various types of users and many applications. The products can be managed by dedicated teams within business units, supported by a central function that coordinates and standardizes design.

What Is a Data Product?

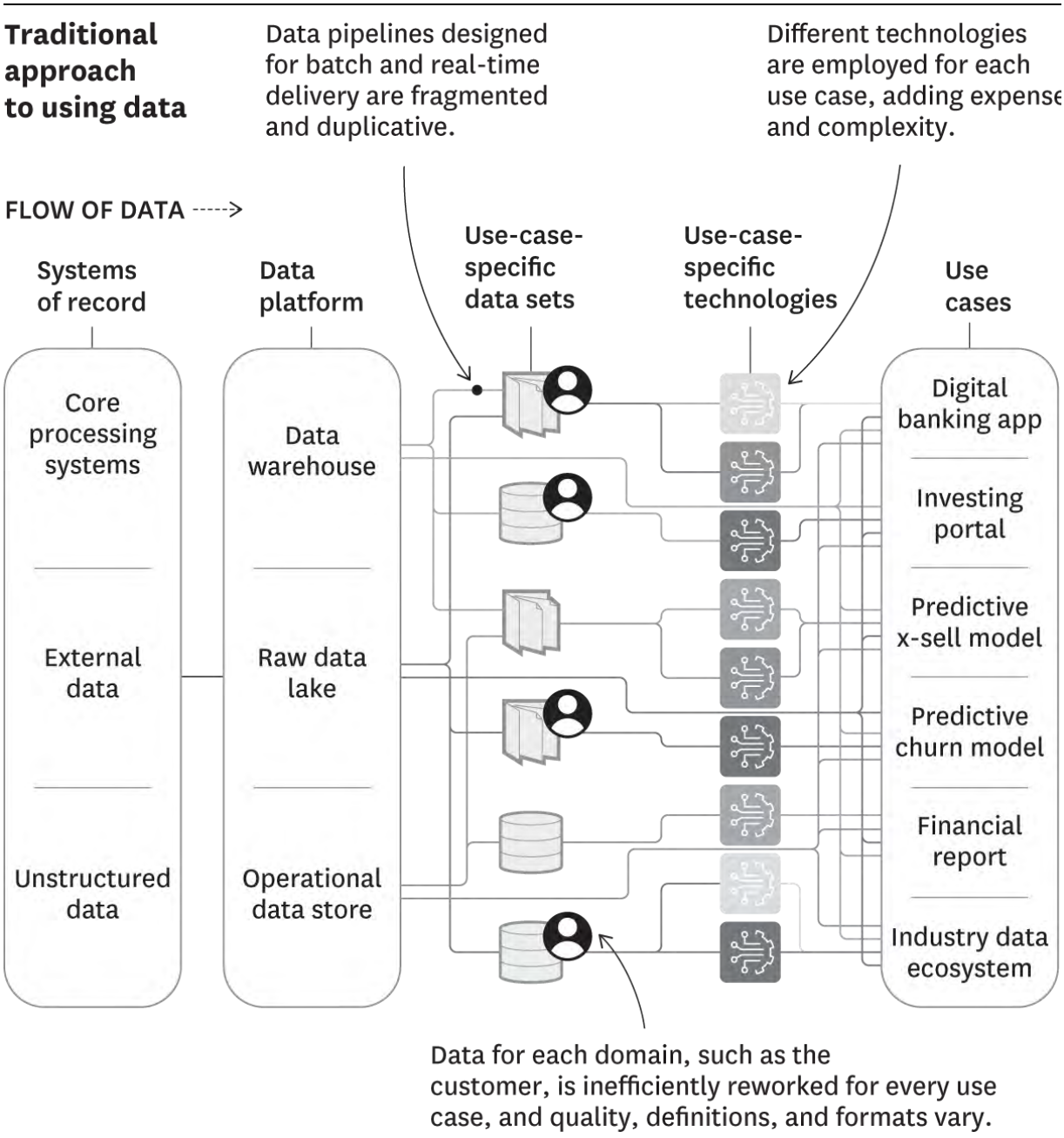
A data product delivers a high-quality, ready-to-use set of data that people across an organization can easily access and apply to different business challenges. It might, for example, provide 360-degree views of customers, including all the details that a company's business units and systems collect about them: online and in-store purchasing behavior, demographic information, payment methods, their interactions with customer service, and more. Or it might provide 360-degree views of employees or a channel, like a bank's branches. Another product might enable "digital twins," using data to virtually replicate the operation of real-world assets or processes, such as critical pieces of machinery or an entire factory production line.

Because they have many applications, data products can generate impressive returns. At a large national bank, one customer data product has powered nearly 60 use cases—ranging from real-time scoring of credit risk to chatbots that answer customers' questions—across multiple channels. Those applications already provide $60 million in incremental revenue and eliminate $40 million in losses annually. And as the product is applied to new use cases, its impact will continue to grow.

Data products sit on top of existing operational data stores, such as warehouses or lakes. (See the exhibit "Traditional data consumption versus the data product model.") The teams using them don't have to waste time searching for data, processing it into the right format, and building bespoke data sets and data pipelines (which ultimately create an architectural mess and governance challenges).

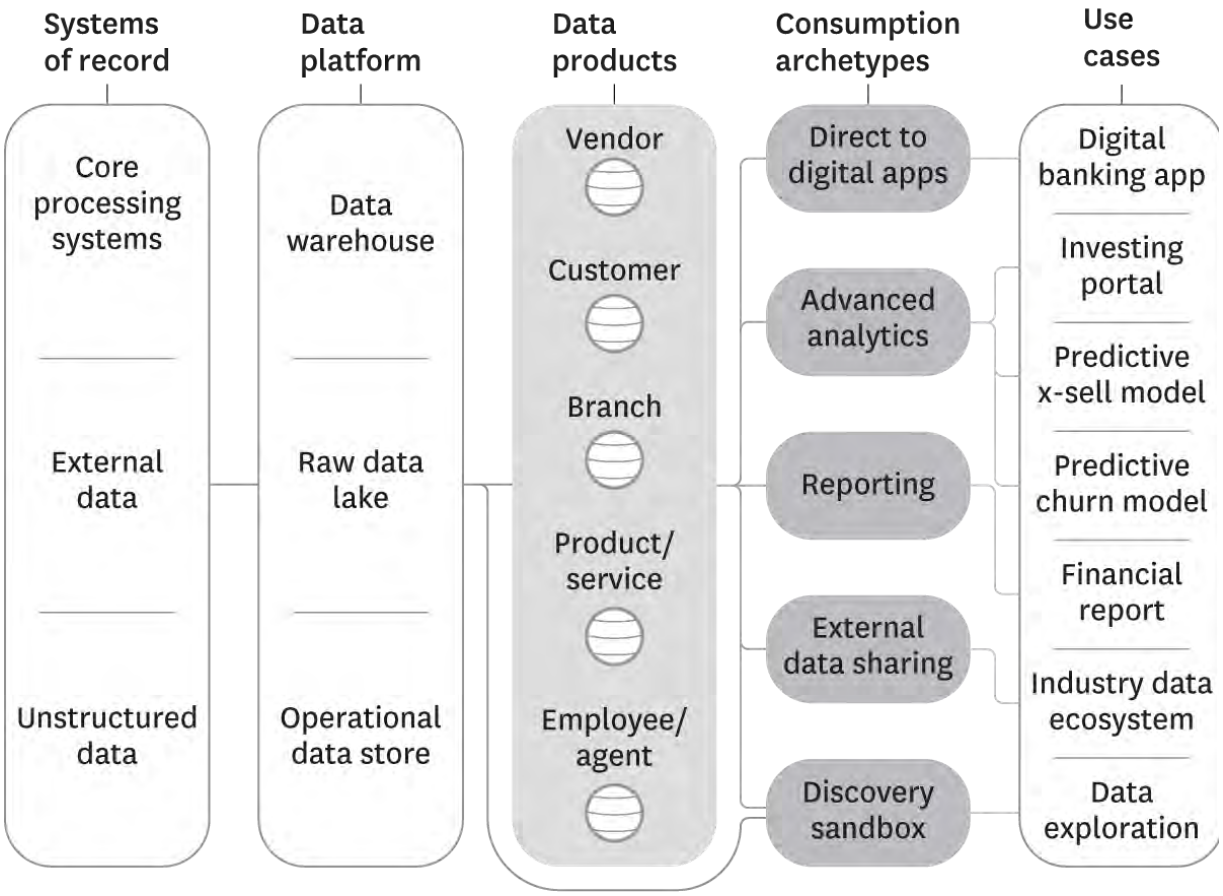
Traditional data consumption versus the data product model

In the traditional approach to data solutions, use case teams identify the data they need from source systems and create data sets and feeds only for their particular solutions. That leads to a lot of replicated work and a complex data architecture that's difficult to maintain and use for new solutions.



In a data product approach, use case teams build solutions by leveraging standardized data products and wiring technologies together following consumption archetype patterns, which reduces work, simplifies the enterprise data architecture, and decreases the time it takes to realize value.

Data product approach



Each data product supports data "consumers" with varying needs, in much the same way that a software product supports users working on computers with different operating systems. These consumers are systems, not people, and our work suggests that organizations typically have five kinds. We call them "consumption archetypes" because they describe what the data is used for. They include:

1. **Digital applications.** These require specific data that is cleaned, stored in the necessary format—perhaps as

individual messages in an event stream or a table of records in a data mart (a data storage area that is oriented to one topic, business function, or team)—and delivered at a particular frequency. For example, a digital app that tracks the location of a vehicle will need access in real time to event streams of GPS or sensor data. A marketing app designed to find trends in customer browsing behavior will need access to large volumes of web log data on demand (often referred to as "batch" data) from a data mart.

2. **Advanced analytics systems.** These too need data cleaned and delivered at a certain frequency, but it must be engineered to allow machine learning and AI systems, such as simulation and optimization engines, to process it.
3. **Reporting systems.** These need highly governed data (data with clear definitions that is managed closely for quality, security, and changes) to be aggregated at a basic level and delivered in an audited form for use in dashboards or regulatory and compliance activities. Usually, the data must be delivered in batches, but companies are increasingly moving toward self-service models and intraday updates incorporating real-time feeds.
4. **Discovery sandboxes.** These enable ad hoc exploratory analysis of a combination of raw and aggregated data. Data scientists and data engineers frequently use these to delve into data and uncover new potential use cases.

5. **External data-sharing systems.** These must adhere to stringent policies and agreements about where the data sits and how it's managed and secured. Banks use such systems to share fraud insights with one another, for example, and retailers to share data with suppliers in the hope of improving supply chains.

Each consumption archetype requires different technologies for storing, processing, and delivering data and calls for those technologies to be assembled in a specific pattern. This pattern is essentially an architectural blueprint for how the necessary technologies should fit together. For example, a pattern for a sandbox would most likely include technologies for setting up a multi-user self-service environment that can be accessed by data engineers across the company. The pattern for an advanced analytics system using real-time data feeds might include technologies for processing high volumes of unstructured data.

Like a Lego brick, a data product wired to support one or more of these consumption archetypes can be quickly snapped into any number of business applications.

Consider a mining company that created a data product providing live GPS data feeds of ore-transport-truck locations. It was designed to support all the archetypes except external data sharing for its first use case—improving ore-processing yields. The company soon discovered the product had uses far beyond that. Once it was made available more broadly in the organization, several entrepreneurial employees immediately

leveraged it to eliminate bottlenecks in the mine transport system. In just three days they built a prototype of a truck-routing decision support tool that reduced queuing time and carbon emissions. If they'd had to engineer the data from scratch, it would have taken nearly three months.

As word continued to spread, employees interested in other issues that involved trucks—such as safety, maintenance, and driver scheduling—tapped into the data to find answers to thorny questions and to build revenue-generating solutions that previously would have been impossible.

Managing and Developing Data Products

Whether they're selling sedans, software, or sneakers, most companies will have internal product managers who are dedicated to researching market needs, developing road maps of product capabilities, and designing and profitably marketing the products.

Likewise, every data product should have a designated product manager who is in charge of putting together a team of experts to build, support, and improve it over time. Both the manager and the experts should be within a data utility group that sits inside a business unit. Typically, such groups include data engineers, data architects, data modelers, data platform engineers, and site reliability engineers. Embedding them within business units gives the data product teams ready access to both the business subject-matter experts and the operational, process, legal, and risk

assistance they need to develop useful and compliant data products. It also connects teams directly with feedback from users, which helps them keep improving their products and identify new uses. The first release of the customer data product at the national bank, for instance, focused on customer demographic profiles and information on transactions. Subsequent releases included data on customer interactions and on prospects, attracting significantly more data users and supporting teams developing other applications. The cost savings and incremental revenue realized by the product's early uses funded the next phases, creating a sustainable business model.

A company also needs a center of excellence to support the product teams and determine standards and best practices for building data products across the organization. For example, the center should define how teams will document data provenance, audit data use, and measure data quality, and should design the consumption archetype patterns for data product teams to use. This approach can eliminate complexity and waste. In addition, the center can be a resource for specialized talent or data experts when demand for them surges within utility groups or business-use-case teams. For example, at one telecom provider we worked with, computer vision experts, who are scarce but often in demand, sit within the central hub and are deployed to business units on request.

While most companies already have some, if not all, of the talent needed to build out their utility groups and centers of

excellence, many will need to deepen their bench of certain experts, particularly data engineers who can clean, transform, and aggregate data for analysis and exploration.

This was especially true for the mining company, which needed to grow its data engineering staff from three to 40 people. To fill that big gap, its leaders took a stepped approach. They hired contractors to get immediate work done and then embarked on far-reaching recruiting efforts: hosting networking events, publishing articles on LinkedIn, upgrading the skills of the software engineers already on staff, and developing internship programs with local colleges and universities. To improve retention, they created a guild for data engineers, which helped them build their skills and share best practices. The company also crafted individualized plans for data engineers that ensured those professionals had a clear growth path after joining the company.

Tracking Performance and Quality

To see whether commercial products are successes, organizations look at barometers like customer sales, retention, engagement, satisfaction, and profitability. Data products can be evaluated with commensurate metrics, such as number of active monthly users, the number of applications across the business, user satisfaction, and the return on investment for use cases.

The telecom company tracked the impact of its first data product—which provided comprehensive data on critical cellular-network equipment—in 150 use cases. They included investment

decision systems, scenario-planning systems, and network optimization engines. In total they're set to produce hundreds of millions of dollars in cost savings and new revenue within three years. The company estimates that over the first 10 years the use cases will have a cumulative financial impact of $5 billion—providing a return many times over on its initial investment.

And just as manufacturers routinely use quality assurance testing or production line inspections to make certain that their products work as promised, data product managers can ensure the quality of their offerings' data. To do so they must tightly manage data definitions (outlining, say, whether customer data includes only active customers or former and prospective customers as well), availability, and access controls. They must also work closely with employees who own the data source systems or are accountable for the data's integrity. (The latter are sometimes called "data stewards.")

Quality can suffer, for instance, when the same data is captured in different ways across different systems, resulting in duplicated entries. This was a risk with the national bank's customer data product. So its product manager worked with the stewards of the company's various customer data repositories and applications to institute a unique ID for each customer. That allowed the customer data to be seamlessly integrated into any use case or with any related data product. The product manager also partnered with the center of excellence to develop the standards and policies governing customer data across the enterprise and to

monitor compliance—all of which facilitated reuse of the data product while building trust among users.

Where to Start

Leaders often ask which data products and consumption archetypes will get the highest and fastest return on investment. The answer is different for every organization.

To find the right approach for their companies, executives need to assess the feasibility and potential value of use cases in each business domain (this might be a core business process, a customer or employee journey, or a function) and group them first by the data products they require and then by the consumption archetypes involved. Categorizing the use cases like this helps leaders more efficiently sequence work and get a faster return on investment. For instance, they may end up pushing some lower-value use cases ahead if they leverage the data products and consumption archetypes of higher-value use cases.

For the executives at the national bank, this approach illuminated several priorities. First they saw that a customer data product that supported their most critical fraud-management and marketing use cases could generate tremendous value. Then they identified the kinds of data the product needed to gather first. Some of those use cases called for basic customer identifiers and reference data (such as demographic or segmentation data) while others required comprehensive customer behavioral data. The bank also realized that the two consumption archetypes it should

pursue first were a discovery sandbox and advanced analytics, which in combination would support most of the company's priority fraud and marketing use cases.

Data product decisions often involve trade-offs between impact, feasibility, and speed. Ideally, the initial target products and consumption archetypes will immediately apply to high-value use cases and a long pipeline of others, as the telecom provider's product for its network equipment did.

However, feasibility considerations may cause a company to adjust its approach. For example, it may make sense to build momentum first in an area of the organization that has data expertise and has gotten some traction with data products, even if that isn't where the biggest opportunity lies. We saw this happen at the mining company. It initially chose to develop two products that supported its ore-processing plant, where use cases had already been successfully proven, the managers were enthusiastic to pursue more, the team had a lot of prepared data to work with, and experts with deep expertise were available to help.

Most leaders today are making major efforts to turn data into a source of competitive advantage. But those initiatives can quickly fall flat if organizations don't ensure that the hard work they do today is reusable tomorrow. Companies that manage their data like a product will find themselves with a significant market edge

in the coming years, thanks to the increases in speed and flexibility and the new opportunities that approach can unlock.

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Heavy Machinery Meets AI

by Vijay Govindarajan and Venkat Venkatraman

FOR MORE THAN 187 YEARS, Deere & Company has simplified farmwork. From the advent of the first self-scouring plow, in 1837, to the launch of its first fully self-driving tractor, in 2022, the company has built advanced industrial technology. The See & Spray is an excellent contemporary example. The automated weed killer features a self-propelled, 120-foot carbon-fiber boom lined with 36 cameras capable of scanning 2,100 square feet per second. Powered by 10 onboard vision-processing units handling almost four gigabytes of data per second, the system uses AI and deep learning to distinguish crops from weeds. Once a weed is identified, a command is sent to spray and kill it. The machine moves through a field at 12 miles per hour without stopping. Manual labor would be more expensive, more time-consuming, and less reliable than the See & Spray. By fusing computer hardware and software with industrial machinery, it has helped farmers decrease their use of herbicide by more than two-thirds and exponentially increase productivity.

Deere gathers data from all its modern farm equipment, including the See & Spray, as it's being used. In total the company collects billions of measurements on soil, crop, and weather conditions from about 500,000 machines on more than 325 million acres of land. That data goes into Deere's cloud-enabled JDLink system, where it's analyzed and used to generate immediate and future improvements to the equipment and farms. Feeding all that information into machine-learning algorithms and AI enables Deere to create a portfolio of combined digital and analog services for optimal seed, fertilizer, and weed management.

Deere is one of the companies leading the way in the industrial sector. Until recently, incumbent manufacturers of construction and mining equipment and other heavy machinery didn't use the most advanced software in their products. That's no longer true. Today, using generative AI and machine learning, they extract insights and trends from structured and unstructured data—including text, high-resolution 3D images, voice interactions, video, and sound— and create complex designs in seconds.

Having studied industrial companies for more than 40 years, we're fascinated by the way they're now fusing analog machinery and advanced digital technologies. The smartest firms have come to realize that the deepest insights, rather than the most valuable physical assets, will separate the winners from the rest of the pack. But achieving success isn't as easy as installing a computer in a tractor. There are many things to consider first.

The process starts with developing what we call *fusion strategies*, which join what manufacturers do best—creating physical products—with what digital businesses do best: using AI to mine enormous, interconnected data sets for critical insights. Industrial firms will have to figure out how to connect hardware and software, steel and silicon, and humans and machines. In this article we'll describe four kinds of fusion strategies and how to execute them. They all require reimagining analog products and services as digitally enabled offerings and learning to create new value from the data generated by combined physical and digital assets. Just as crucially, industrial firms will need to partner with other companies to create ecosystems with an unwavering focus on solving customers' problems.

How Fusion Strategy Differs from the Internet of Things

Traditionally, industrial manufacturers looked mostly at sales and marketing data. They analyzed it to see which demographic bought a particular product or which would be willing to spend more on a better product. Digital systems weren't that important. The companies viewed them as a way to lower manufacturing costs or add features, such as Wi-Fi, to existing analog products. Today, however, industrial goods companies must also focus on what happens after the sale—on how combinations of digital and analog products enable positive outcomes for customers. Data on

that must be gathered constantly, and insights from it applied in real time to customers' problems.

Idea in Brief

The Problem

Until recently most incumbent industrial companies didn't use the most advanced software in their products. But competitors that can extract complex designs, insights, and trends using generative AI have emerged to challenge them.

The Solution

Industrial companies must develop strategies that fuse what they do best—creating physical products—with what digital companies do best: using data and AI to parse enormous, interconnected data sets and develop innovative insights.

The Changes Required

Companies will have to reimagine analog products and services as digitally enabled offerings, learn to create new value from data generated by the combination of physical and digital assets, and partner with other companies to create ecosystems with an unwavering focus on helping customers solve problems.

Like the internet of things, fusion strategy involves a network of physical objects embedded with sensors, software, and connectivity. But fusion strategy goes beyond equipping an analog product so that you can monitor it and collect data from it over the internet. With fusion strategy, you completely reimagine how that analog product could be designed to function if it were

built from scratch using every digital tool and functionality available.

Responsibility for fusion strategy is also handled differently in organizations. The internet of things is owned at the functional level, and use cases are straightforward: Managers in operations (such as supply chain coordinators and quality control executives) use the data from sensors to ensure that machinery-based processes adhere to specific industry standards and protocols. Designers build internet-enabled products that generate data that can be used in future iterations of those products. The separate functions don't really have to coordinate their efforts. Fusion strategy, in contrast, is cross-functional. It requires senior-level executives from different departments to work together to determine how sensor data can be harnessed to deliver value to existing customers and to create new products for future customers.

Fusion strategy also generates insights and benefits more quickly than the internet of things does. While the machinery behind the internet of things ensures that data is recorded and logged, fusion strategy applies that data immediately, observing products' use across diverse settings and then leveraging information on it to offer automated, personalized recommendations or experiences to customers.

The business case for fusion strategy is competitive differentiation. It's about experimenting with newer technologies and developing a road map for incorporating them to get an edge.

You can see this happening with the automotive companies that are developing self-driving vehicles. They will invest in the internet of things to adhere to safety standards, but the real differentiation they gain is in how sensors, software, and data connectivity are integrated to make their cars perform better than competitors'.

The strategic opportunities presented by traditional and generative AI and other advanced digital technologies fall into four main categories: *fusion products*, *fusion services*, *fusion systems*, and *fusion solutions*. Although most companies should start by focusing on fusion products, they should also explore and test the three other strategies. Experimenting with all four fusion strategies will allow them to determine how they can create and capture value, evaluate how other companies' strategies threaten them, and determine how best to allocate their resources.

Fusion Products

As we've noted, fusion products are designed from scratch to collect and leverage information on products' use in real time. The amount of data they generate is increasingly massive, and companies can apply AI to it in three ways: They can use traditional AI to analyze it for, say, potential product, process, and cost improvements; they can use generative AI to create digital twins of existing products (virtual duplicates that mirror their functioning) and train robots; and they can use the large language

models within generative AI to develop proprietary insights that will add value for customers.

Take Tesla automobiles. They were designed from the start to be cloud-connected computers on wheels. For more than three decades incumbent automakers have added digital features to their products, including GM's OnStar, Ford's Sync, and Mercedes's Mbrace, but those features didn't enable the companies to track automobiles in real time, analyze data continuously, and improve vehicles on the fly the way Tesla does. The sensors in every Tesla provide immediate information about how it's performing on the road. AI determines whether it is working optimally or needs fixing, and in many cases Tesla is able to correct problems with software updates. Tesla can stop a door from rattling by tweaking the hydraulics, for example, or it can adjust regenerative braking levels with over-the-air software updates to reduce collision risk.

Tesla is now experimenting with using generative AI to program autonomous vehicles. By studying video data taken from inside every Tesla, generative AI can reconstruct multitudes of driving scenarios, which Tesla can then use to train cars to avoid collisions and other dangers.

In each Tesla an operating system, which houses data from all the vehicles, helps it respond quickly to insights from hundreds of thousands of cars. The system integrates three digital twins: a virtual rendering of the car's model as it was designed (the product twin), a virtual version of the assembly line that created

it (the process twin), and a representation of the product in use (the performance twin). The system also incorporates insights gathered from Tesla's supply chain, manufacturing operations, and after-sales services.

Tesla's competitors are now trying to catch up by developing their own original fusion products. Mercedes is focusing on designing AI-enhanced operating systems for its vehicles and has teamed up with Nvidia for software, hardware, and AI chips. GM, partnering with its subsidiary Cruise, already has put self-driving taxis on the road. (Although last fall it had to withdraw them after their safety came under question, once GM understands the causes of safety lapses, it will probably resume operating the taxis.)

Fusion products can be found in other industries too. The window manufacturer View, for instance, has developed smart glass that uses real-time data and AI to automatically adjust how much sunlight gets in. When natural light in a room is low, the glass allows more in, but if the room gets too hot, it will filter sunlight out.

Early movers can capture value from fusion products in several ways. The most attractive option is to charge premium prices for them—as Tesla does. Product makers can justify higher pricing if their offerings perform better than those of rivals and cost less to operate. Another option is to offer performance contracts to guarantee that fusion products will be proactively maintained and updated, ensuring minimal downtime. Rolls-Royce is one

company that does this. It guarantees uptime of close to 100% for its commercial airline customers by using AI to monitor and maintain the engines in planes. When something fails on one of its engines, Rolls-Royce knows about it in advance or finds out in real time, which allows it to troubleshoot the problem much more quickly.

Another way to capture value is to use fusion product insights to enter adjacent spaces. Tesla, for instance, has gone into the automobile insurance business and can offer rates that are lower than the competition's because of its ability to record and analyze the behavior of individual drivers instead of clustering them into different risk profiles, which is how other insurers operate.

Fusion Services

Until now services related to industrial goods have been labor-intensive to deliver, with businesses relying on codified knowledge and standard protocols to fix problems. Though it's more profitable to deliver customized services quickly and in large volumes, doing so was difficult and expensive in the analog world. But today, rather than using an army of on-site technicians to provide services, industrial companies can let customized recommendations generated by AI from product-in-use data flow automatically to customers.

Traditional strategy versus fusion strategy

Moving from a traditional strategy to a fusion strategy will require industrial businesses to rethink every aspect of operations, from how they use data, to how they partner with other businesses, to how they serve individual customers. Below is a high-level overview of the main differences between traditional strategy and fusion strategy. This exhibit is not exhaustive. Industrial businesses will find many more differences between how they operate today and how they will operate using a fusion strategy.

Category	Traditional strategy	Fusion strategy
Growth dynamics	Linear, gradual, within industry boundaries	Nonlinear, exponential, across industry boundaries
Competitive landscape	Familiar competitors with similar business models focused on product design and delivery	Digital competitors that have new capabilities and focus on product use
Source of scale and efficiency advantages	Physical assets; production	Information assets; data
Scope for expansion	Product-market extensions, vertical integration through mergers and acquisitions	Virtual integration through data integration and partnerships
Source and focus of customer insights	Ad hoc surveys and information gathered at point of purchase; operational improvements	Real-time observations and customer outcomes; competitive differentiation
Data and AI strategy	Enhance efficiency using independent databases, systems of records, and engagements; focus on a single firm; increase operational effectiveness	Develop real-time insights and integrated databases; focus on networks and firms' ecosystems of partners and customers; increase strategic differentiation

New ventures have already jumped on this opportunity. For instance, weather and climate data from companies such as Weather.com and ClimateAi allows farmers to make precise decisions in a timely manner and enhance profitability. Norm, a ChatGPT app for agriculture, mines available data on weather, soil conditions, and current events to answer all kinds of questions about farm management.

Some established industrial companies have gotten into fusion services too. In the civil aviation industry, improving aircraft engines' fuel efficiency by 1% could save $2 billion a year. Rolls-Royce's R^2 Data Labs saves its customers $200,000 per aircraft annually by analyzing the fuel consumption patterns revealed by product-in-use data, such as the routes planes take, the altitudes and speed at which they fly, weather conditions, and the loads they're carrying, and then relaying the insights to customers in real time.

In their early stages, fusion services can be sold through unbundled pricing, with buyers paying only for services they find valuable. As customers' appreciation for the services grows, industrial companies can migrate to offering bundled packages of services. Over time they could even enter into gain-sharing agreements that give them a percentage of the increase in profits that their fusion services generate.

Fusion Systems

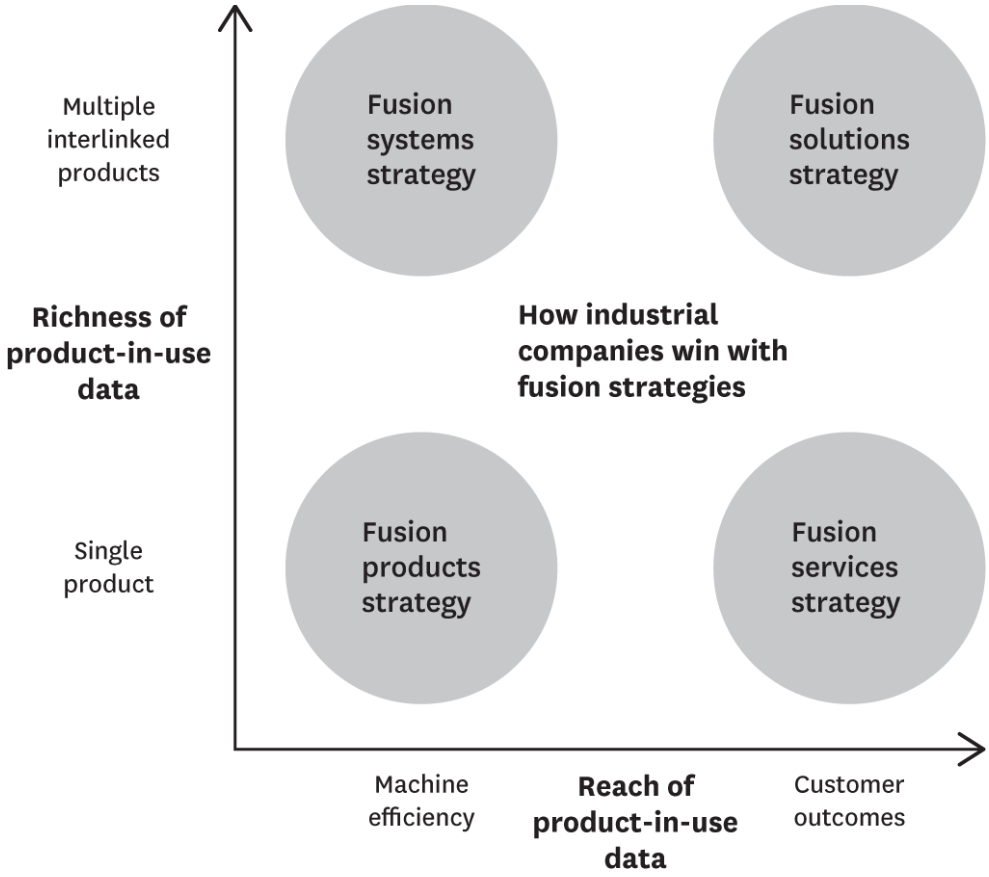
Many customers of Deere and of other makers of farm and construction equipment use complex systems that include multiple kinds of machines, each of which is made by a different specialized supplier. Most customers manage the interoperability of these machines themselves, but some hire systems integrators and consultants to install, connect, and upgrade them. Digitization can make operating them a lot easier. But with fusion systems, it shifts from a micro level (enhancing products from

different companies independently to improve their efficiency) to a macro level (enhancing multiple related products from multiple companies in concert to improve the whole system's efficiency).

A fusion systems integrator must not only connect all the machines and equipment it has sold but interlink them with those of partners and competitors. And instead of being responsible just for interconnecting different elements and getting the system to work, as traditional systems integrators are, the creator of a fusion system must ensure that it improves continually as new parts and functionality are added.

The fusion strategy framework

Companies must ask two questions: One, what is the reach of our product-in-use data? Two, how rich is our product-in-use data? Their answers will help them craft four kinds of fusion strategies: fusion products, fusion services, fusion systems, and fusion solutions. The fusion strategy framework is inherently dynamic. Most companies will start with fusion products, which require the least amount of data to build, and then expand into fusion services and systems as they gain more experience. From there they can work to advance into fusion solutions, which create the most value.



This is where AI plays a key role. Most industrial goods companies are comfortable synthesizing structured data such as names, addresses, credit card numbers, and other easily formatted customer information. But the advanced firms feed unstructured data—like images of defects, the sound of faulty machines, and video feeds of autonomous processes—into generative AI in real time to develop complex, realistic digital twins of systems. They can then use those twins to experiment with mixing and matching different permutations of products, machines, components, and peripherals to determine which combination of technologies works best.

Dubai's Burj Khalifa, the world's tallest building, employs numerous Honeywell systems—for ventilation, air-conditioning, lighting, water management, parking, storage, elevators, telecommunications, and security—in its overarching facilities-management system. Honeywell collects real-time data from the building's intricate system and analyzes it to identify anomalies, such as malfunctioning equipment. It helps Burj Khalifa detect incidents earlier, react faster, and mitigate potential risks. That strategy has reduced the maintenance time for Burj Khalifa's mechanical assets by 40% while improving their availability to 99.95%.

Fusion systems are likely to change the buying preferences of customers. They may shift from focusing on products with distinctive features to focusing on products that can be integrated with other products. Because of this, industrial manufacturers must create technology architecture that integrates hardware, software, applications, data, and cloud connectivity and can incorporate product-in-use data and algorithms from other products. Industrial companies will monetize the value their fusion systems create through integration fees as well as charges to connect additional machines. They can also generate revenue by selling fusion-systems software as a service through onetime license fees, monthly subscriptions, or a pay-as-you-go model.

Fusion Solutions

Fusion solutions combine fusion products, services, and systems in ways that directly improve a customer's performance. Rather than developing solutions through traditional consultative sales processes that approach each customer problem as a one-off, companies develop them with insights from product-in-use data and design them to be applicable to many customers. But designing these solutions effectively requires industrial companies to be experts at solving customer problems in ways that no other company can. They must earn an unusually high level of customers' trust, and they should tie their revenues and profits to customer success using outcome-based contracts and profit-sharing agreements (as they may eventually do with fusion systems).

To create the greatest value for customers, industrial companies will have to partner with other businesses to develop ecosystems of solutions. Deere, for instance, has teamed up with the agricultural software company Granular, which uses unstructured data to develop yield models for farmers. Granular's models collect and analyze imagery from satellites, gathering information on weather, elevations, and soil conditions, and combine that data with historical yield statistics to help Deere's customers forecast costs, revenues, and profits. As it sells fusion solutions to the market, Deere will go head-to-head with traditional competitors such as CNH Industrial and AGCO, but it will also compete with software companies, such as the Climate Corporation, now part of Bayer, and digital giants like IBM and

Alphabet. Ultimately, the battle for customers will be between different fusion ecosystems, not individual companies. While each fusion strategy creates pools of value for businesses, by combining them all, fusion solutions will create the most value.

Every industrial company will have to develop its own fusion strategy. The first step will be to stop treating human intelligence and machine intelligence as if they were separate. Every function and process can be enhanced if humans and machines work together. The companies that embrace that kind of collaborative intelligence will outperform those that treat AI as an esoteric technology or postpone its adoption.

Until now industrial companies have competed by developing proprietary technologies and integrating vertically. To execute fusion strategies, most will have to partner with digital natives and startups on new technologies and applications, but doing so will ultimately lower costs, increase speed, and improve customer satisfaction. This will be a major shift, however, and will require industrial CEOs to lead digital efforts—a responsibility they've usually delegated in the past. Their firms will need a holistic strategy for how, when, and where digital technologies will drive the organization to ensure that everyone in it shares the same view of priorities and resource allocation needs. And every employee will have to embrace digital

technology in order for the organization to craft, execute, and adopt a fusion strategy that succeeds.

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